SEMIGROUPS OF OPERATORS, COSINE OPERATOR FUNCTIONS, AND LINEAR DIFFERENTIAL EQUATIONS

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This survey presents a systematic exposition of the elements of the theory of operator semigroups (OS's) in Banach space from Hille–Yosida to the end of 1989. There is a parallel exposition of the theory of cosine operator functions (COF's). The paper contains the following divisions: Linear differential equations in Banach space, reduction of the Cauchy problem for second order equations to the Cauchy problem for first order equations, one-parameter OS's and COF's, differentiable OS's, analytic OS's, Fredholm OS's, positive OS's, stable OS's, spectral properties of OS's and COF's, compactness properties of OS's and COF's, uniformly continuous OS's and COF's, almost periodic OS's and COF's, uniformly bounded OS's and COF's, the theory of perturbations for OS's and COF's, adjoint OS's and COF's.

INTRODUCTION

Individual results obtained prior to 1983 in the theory of linear and nonlinear semigroups of operators and its applications were surveyed by S. G. Krein and M. I. Khazan in [60]. In that survey the authors announced a proposal for a second part that would consider other classes of differential equations in Banach spaces. However, in studying the literature that appeared between 1983 and 1989, it was discovered that there was a large group of publications on various problems in the theory of linear semigroups; of these publications, several monographs were particularly illuminating: [126, 162, 211, 213, 229, 341, 390]. Unfortunately, practically nothing on these problems was published in Russian. As a result, a large part of the present survey is devoted to presenting anew the theory of linear semigroups of the class C_0 (OS's). For continuity in exposition we sometimes restate results that have already been presented in [60].

Another object of study in the present paper is cosine operator functions (COF's) associated with second-order differential equations of hyperbolic type in Banach space, the theory of which has developed vigorously in the past twenty years. There is, for example, the monograph [213], which is devoted entirely to this theory. In Russian there are neither monographs nor surveys. It should be noted that the theory of COF's runs parallel to the theory of semigroups. Thus, in each chapter of the present survey we first present material on semigroup theory, and then the corresponding COF theory. In particular, this shows how the two complement each other.

A priori constraints on the size of the survey made it impossible to include classical problems such as, for example, application of fractional powers and interpolation spaces, the theory of boundary-value problems for equations of second and higher order in Banach space, asymptotic methods for investigating equations with a small parameter, coercivity inequalities, problems on discrete approximation of differential equations, or many applications. In a word, our intention is to provide a springboard for discussion of the modern theory of differential equations in Banach space, and we assume that these problems will be considered later. To conclude our advertisement for the theory of OS's and COF's, we are deeply gratified to note that the monograph [229] is being published by "Vysshaya Shkola," and Mir will be publishing a translation of [178] in 1991.

The paper is structured as follows: There are 9 sections, each of which contains divisions that are further divided into paragraphs. A paragraph contains a complete proposition. Accordingly, a tripartite numbering system is used: for example, 1.3.2 is the second proposition of the third division of the first section. Within each section, the section numbers are omitted. Thus, formulas within a section have only two-position numbering, although references from out-

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side the section in which the formula is defined contain the section number. Bibliographic references are not necessarily primary, since we frequently refer to publications in which a given problem is discussed in sufficient detail.

The bulk of our notation is in common use. For the reader's convenience the survey contains a glossary of notation. Equations used to define some quantity are indicated with a := sign.

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1. LINEAR DIFFERENTIAL EQUATIONS IN BANACH SPACE

1.1. Preliminary Remarks. Let E be a Banach space over the field of complex numbers with norm $\|\cdot\|$. We denote the set of linear operators mapping $D(A) \subset E$ into E by L(E), and the continuous linear operators by B(E). We denote the domain and range of an operator A by D(A) and $\mathscr{R}(A)$, respectively. Equipped with the norm $|||x||| := \|x\| + \|Ax\|$, the linear manifold D(A) becomes a Banach space, which we denote by $\mathscr{D}(A)$. We use traditional notation for the resolvent set $\rho(A)$ and spectrum $\sigma(A)$ of the operator A. We denote by $(\overline{D(A)} = E)$ the set containing the operators that are closed in E, have dense domains $\mathscr{C}(E) \subset L(E)$, and have nonempty resolvent sets ($\rho(A) \neq \varnothing$). Let $J \subseteq \mathbf{R} := (-\infty, +\infty)$ and $N = \{1,2,\ldots\}$.

Definition 1. A function $u(t) : J \to E$ is said to be continuous at a point $t_0 \in J$ if

$$\lim_{t \to t_0} \| u(t) - u(t_0) \| = 0.$$

A functions is said to be continuous on J if it is continuous at every point $t \in J$; in this case we write $u(t) \in C(J;E)$.

Definition 2. The derivative of a function u(t) at a point $t_0 \in J$ is an element $u'(t_0) \in E$ such that

$$\lim_{\Delta t \to 0} \| (u(t_0 + \Delta t) - u(t_0))/\Delta t - u'(t_0) \| = 0.$$

If a function u(t) has a continuous derivative u'(t) at every point $t \in J$, it is said to be continuously differentiable on J. In this case we write $u \in C^1(J;E)$.

The definition of the set of k-times continuously differentiable functions $C^k(J;E)$ is analogous to Def. 2.

1.2. Cauchy Problem for n-th-order Differential Equations. In a Banach space E, we consider the Cauchy problem for an n-th order equation of the form

$$u^{(n)}(t) = Au(t), \quad t \in \bar{\mathbf{R}}_+ := [0, \infty), \tag{2.1}$$

$$u(0) = u^0, \; u'(0) = u^1, \; \ldots, \; u^{(n-1)}(0) = u^{n-1}, \tag{2.2}$$

where $A \in \mathscr{C}(E)$, and $u^{(n)}(t)$ denotes the n-th derivative (i.e., $(d/dt)^n u(t)$) of the function u(t).

Definition 1. A function $u(t) : \mathbf{R}_+ \to E$ is said to be a solution of Eq. (2.1) if $u(t) \in C^n(\bar{\mathbf{R}}_+; E)$, $u(t) \in D(A)$ for any $t \in \mathbf{R}_+$ and (2.1) is satisfied.

Definition 2. A Cauchy problem (2.1)-(2.2) is said to be uniformly well-posed on $\bar{\mathbf{R}}_+$ if:

(i) there exists a dense subset $D \subseteq E$ such that for $u^0, u^1, \ldots, u^{(n-1)} \in D$ there exists a unique solution on $\mathbf{R}_+$ of Eq. (2.1) with the property that $u^{(k)}(0) := \lim_{h \to 0} u^{(k)}(h) = u^k$; $k = \overline{0, n-1}$ (i.e., k runs through all integers from 0 to n — 1);

(ii) the solution of (2.1)-(2.2) is uniformly stable with respect to t on any compact $\mathscr{K} \subset \bar{\mathbf{R}}_+$, i.e., the conditions $u_p^k \to 0$ ($u_p^k \in D$; $k = \overline{0, n-1}$) as $p \to \infty$, imply convergence of the solutions $u_p(t) \to 0$ of (2.1)-(2.2) uniformly with respect to $t \in \mathscr{K}$, where $u_p^{(k)}(0) = u_p^k$; $k = \overline{0, n-1}$.

To study the properties of solutions to uniformly well-posed Cauchy problem (2.1)-(2.2) we introduce the propagators $\mathscr{F}_j(t)$, $j = \overline{0, n-1}$, which provide a solution of (2.1)-(2.2) with initial conditions $u_j^{(k)}(0) = \delta_{jk} x$ (δ_{ik} is the Kronecker delta), i.e., $u_j(t) = \mathscr{F}_j(t)x$.

1.2.1. [209, 381] The operators $\mathscr{F}_j(t)$, $j = \overline{0, n-1}$ are densely defined, strongly continuous in t, and for any t may be extended continuously to all of E.

1.2.2. [209, 381] If u(t) is a solution of (2.1)-(2.2), then

$$u(t)=\sum_{j=0}^{n-1}\mathcal{F}_j(t)u^j, \quad t\in\bar{\mathbf{R}}_+. \tag{2.3}$$

THEOREM 1 ([209, 53]). The Cauchy problem (2.1)-(2.2) with $n \geq 3$ is uniformly well-posed if and only if $A \in B(E)$, i.e., A is a continuous operator with $D(A) = E$. In this case we have

$$\mathcal{F}_j(t)=\sum_{k=0}^{\infty}t^{kn+j}A^k/(nk+j)!. \tag{2.4}$$

1.3. Cauchy Problem for First Order Equations and C_0-semigroups. In order to consider the case n = 1, we present the following definition.

Definition 1. A one-parameter family of operators $U(t) \in B(E)$, $t \in \mathbf{R}_+$, is said to be a semigroup of operators on E if

$$\begin{aligned}&\text{(i) } U(t+s)=U(t)U(s) \quad \text{for all} \quad t,\ s\geqslant 0;\\ &\text{(ii) } U(0)=I \text{ is the identity operator on E.}\end{aligned} \tag{3.1}$$

A semigroup is said to be strongly continuous if $\lim_{t\to s}U(t)x=U(s)x$ $(s>0,\ x\in E)$. Finally, a semigroup belongs to the class C_0 (and we will call it an OS) if

(iii) $\lim_{t\to 0+}U(t)x=x$, for all $x \in E$.

Definition 2. A linear operator A with domain

$$D(A):=\{x\in E:\ \lim_{h\to 0+}((U(h)-I)/h)x \text{ exists}\} \tag{3.2}$$

and defined so that

$$Ax:=\lim_{h\to 0+}((U(h)-I)/h)x \quad \text{for } x\in D(A) \tag{3.3}$$

is called the generating operator (infinitesimal generator) of the OS U(t). In the sequel we will usually denote a semigroup with generating operator A by exp(tA).

1.3.1. [288] The generating operator of an OS belongs to $\mathscr{C}(E)$.

THEOREM 1 ([59]). A Cauchy problem (2.1)-(2.2) with n = 1 is uniformly well-posed if and only if the operator A is the generating operator of an OS.

1.3.2. The solution of a uniformly well-posed problem (2.1)-(2.2) is given by the formula

$$u(t)=\exp(tA)u^0,\ t\geqslant 0. \tag{3.4}$$

1.4. Cauchy Problem for Second Order Differential Equations and Cosine Operator Functions.

Definition 1. Let $\mathscr{B}$ be a Banach algebra with identity $\bar{e}$. A mapping $C(t): \mathbf{R}\to\mathscr{B}$ is said to be a cosine function if

(i) $C(t + s) + C(t - s) = 2C(t)C(s)$ for all $t, s \in \mathbf{R}$ (the d'Alembert equation);

(ii) $C(0) = \bar{e}$.

When $\mathcal{B}=B(E)$, where E is a Banach space, $\bar{e}$ = I (the identity operator), it is usually required that

(iii) the operator function C(t) is strongly continuous in E.

In what follows we will be interested primarily in cosine operator functions with properties (i)-(iii), which we will call COF.

A COF C(t) is associated with a sine operator function (SOF)

$$S(t)x:=\int_0^t C(s)x\,ds,\quad t\in\mathbf{R},\quad x\in E \tag{4.2}$$

and linear manifolds

$$E^k:=\{x\in E: C(t)x\in C^k(\mathbf{R};E)\};\ k=1,2.$$

Definition 2. A linear operator A with domain D(A) that consists of all x for which the limit

$$\lim_{h\to 0}((C(2h)-I)/2h)x:=Ax \tag{4.3}$$

exists is said to be the (infinitesimal) generating operator for the COF C(t).

1.4.1. [381] The generating operator for a COF is densly defined, closed, and belongs to $\mathcal{C}(E)$.

We denote the COF and SOF with generating operator A by C(t,A) and S(t,A), respectively.

1.4.2. [396] The domain D(A) of the generating operator A for the COF C(t,A) coincides with E^2.

THEOREM 1 [210]. A Cauchy problem (2.1)-(2.2) with n = 2 is uniformly well-posed if and only if the operator A is a generating operator for a COF.

1.4.3. [210] A solution of a uniformly well-posed problem (2.1)-(2.2) with n = 2 for all $u^0\in D(A)$ and $u^1\in E^1$ exists and may be written in terms of a COF and an SOF as follows:

$$u(t)=C(t,A)u^0+S(t,A)u^1,\ t\in\mathbf{R}. \tag{4.4}$$

1.5. Representation of Cosine Operator Functions in Terms of Groups of Operators. Assume we are given a one-parameter group of operators exp(tA) (for the definition, see 2.2) with values in a Banach algebra with identity $\bar{e}$. Then the function

$$C(t):=(\exp(tA)+\exp(-tA))/2,\ t\in\mathbf{R}, \tag{5.1}$$

has the properties (4.1).

Definition 1. When $\mathcal{B}=B(E)$, is a cosine function for which representation (5.1) is possible, we say that it is an exponential cosine operator function.

1.5.1. [398] The generating operator B for an exponential COF of the form (5.1) may be represented in the form $B=A^2$.

1.5.2. [7, 272, 273, 327, 402] Examples of COF's that are not exponential have been constructed.

1.5.3. [141] If a cosine function C(t,A) with values in $\mathcal{B}$ and properties (4.1) is measurable, the Banach algebra $\mathcal{B}$ can be extended to a Banach algebra $\tilde{\mathcal{B}}$, in which C(t,A) is exponential.

1.5.4. [141] If, for some $t\in R$, the spectrum of a COF C(t,A) is contained in a singly connected open subset of the complex plane that does not contain unity, then C(t,A) is an exponential COF.

1.5.5. [141] It follows from 5.4 that if $\|C(t_0,A)\|<1$ for some t_0, then the COF C(t,A) is exponential.

1.5.6. [141] Let ξ_0 be a nilpotent element of $\mathcal{B}$ of order 2, i.e., $\xi_0\neq 0$ and $\xi_0^2=0$, and let $a(t,\tau):\mathbf{R}\times\mathbf{R}\to\mathbf{R}$ be a biadditive symmetric function. Then

$$C(t,A):=\bar{e}+a(t,t)\xi_0 \text{ is a COF.}$$

1.5.7. [141] For a COF C(t,A) to be exponential it is necessary that there exist a function $\hat{K} : R \to \tilde{R}_+$ such that

$$|a(t, \tau)| \leqslant \hat{K}(t)\hat{K}(\tau) \quad \text{for all} \quad t, \tau \in \bar{R}_+. \tag{5.2}$$

1.5.8. [190] There exists a symmetric biadditive $a(t,\tau)$ such that for it there is no function $\hat{K}$ with property (5.2).

1.5.9. [141] Thus, the study of COF's cannot be reduced by means of representation (5.1) to the study of OS's even by extension of the Banach algebra $\mathscr{B}$. The requirement of measurability in Proposition 5.3 is essential.

1.6. Supplementary Comments. As we can see from Theorems 3.1 and 4.1, uniformly well-posed Cauchy problems are closely related to OS's and COF's. Generalization of the notion of well-posedness to the solution to a Cauchy problem leads to broader classes of semigroups and cosine operators. We note briefly possible approaches for generalization, leaving detailed discussion for a subsequent survey. Other (non-semigroup) approaches to the study of abstract equations of the form $Au'(t) + Bu(t) = f(t)$, where A and B are self-adjoint and $A \geq 0$, may be found in [49]-[51].

Cauchy problems that are well-posed in the sense of distributions, and problems associated with them, are considered in [166, 379, 404].

The publications [192, 319, 392, 393] are devoted to the study of so-called C-semigroups, which are used to study both first- and second-order problems, as well as semigroups with a singularity at zero.

Concerning n-parameter semigroups, see [178] and [241].

There are a number of publications (see, for example, [204, 214, 216]) that deal with the so-called incomplete Cauchy problem for Eq. (2.1) on the half-axis $\tilde{R}_+$ (i.e., part of the conditions is given at zero, and part at infinity). The solvability of such problems is related to the nature of the spectrum of the operator A that appears in (2.1) and the stability of certain semigroups.

2. SEMIGROUPS OF OPERATORS OF THE CLASS C_0 (OS'S)

2.1. Fundamental Properties of OS's.

2.1.1. ([288]) For any OS exp(tA) there exist constants $M \geq 1$ and $\omega \in \mathbf{R}$ such that

$$\|\exp(tA)\| \leqslant M \exp(\omega t) \quad \text{for} \quad t \geqslant 0. \tag{1.1}$$

Definition 1. An OS exp(tA) is said to be uniformly bounded if $\|\exp(tA)\| \leq M$ for $t \geq 0$ and some constant M. An OS is said to be a contaction semigroup if $M = 1$.

Note that if an OS exp(tA) satisfies bound (1.1), then $e^{-\alpha t}\exp(tA)$ is an OS satisfying (1.1) with $\omega_1 = \omega - \alpha$.

2.1.2. [126] For an OS exp(tA) the limit

$$\omega(A) := \lim_{t \to \infty} t^{-1} \ln \|\exp(tA)\| = \inf_{t>0} t^{-1} \ln \|\exp(tA)\|$$

exists and is equal to $\inf\{\omega \in R$: for every $\delta > \omega$ there exists a constant M_δ with the property that $\|\exp(tA)\| \leq M_\delta \exp(\delta t)$, $t \geq 0\}$.

Definition 2. The number $\omega(A)$ is called the type of the OS exp(tA).

2.1.3. [126] The type of an OS may be either finite or equal to $-\infty$.

2.1.4. [178] Let E = H be a Hilbert space. Then for an OS exp(tA) we have

$\omega(A) = \inf\{\alpha \in \mathbf{R} : \alpha + i\beta \in \rho(A)$ for all $\beta \in \mathbf{R}$ and the family of resolvents $R(\alpha + i\beta, A)$ with $\beta \in \mathbf{R}$ is uniformly bounded$\}$.

2.1.5. [288] An operator $A \in L(E)$ with $\overline{D(A)} = E$ may be a generating operator for only one OS.

2.1.6. [35] If we are given an OS exp(tA) and $\operatorname{Re}\lambda > \omega(A)$, then $\lambda \in \rho(A)$ and

$$R(\lambda, A)x = \int_0^\infty e^{-\lambda t} \exp(tA)x\,dt \quad \text{for all} \quad x \in E. \tag{1.2}$$

The problem of whether an operator A generates an OS is the subject of the so-called Hille—Yosida—Feller—Phillips—Miyadera Theorem.

THEOREM 1 ([35]). Operator $A \in L(EP)$ is a generating operator for OS exp(tA) if and only if $A \in \mathscr{G}(E)$ and there exist constants M and ω such that $\{\lambda \in \mathbf{C} : \operatorname{Re} \lambda > \omega\} \subset \rho(A)$ and

$$\| R(\lambda, A)^n \| \leqslant \frac{\mathbf{M}}{(\operatorname{Re}\lambda - \omega)^n} \quad \text{for} \quad \operatorname{Re}\lambda > \omega, \quad n \in \mathbf{N}. \tag{1.3}$$

We will indicate that operator $A \in \mathscr{G}(E)$ generates OS exp(tA) satisfying estimate (1.1) by writing $A \in \mathscr{G}(\mathbf{M}, \omega)$.

2.1.7. [35] If the OS in the statement of Theorem 1 is a contraction OS, then condition (1.3) can be replaced by $\{\lambda \in \mathbf{C}, \operatorname{Re}\lambda > 0\} \subset \rho(A)$ and

$$\| R(\lambda, A) \| \leqslant 1/(\operatorname{Re}\lambda) \quad \text{for} \quad \operatorname{Re}\lambda > 0. \tag{1.4}$$

2.1.8. [35] If some half-plane $\operatorname{Re}\lambda > \omega$ is included in $\rho(A)$ and $\| R(\lambda, A) \| \leq (\operatorname{Re}\lambda - \omega)^{-1}$ with $\operatorname{Re}\lambda > \omega$, then the operator $A \in \mathscr{G}(E)$ is a generating operator for an OS.

THEOREM 2 ([44], [101]). For an OS exp(tA) we have

(i) if $x \in D(A)$ then the function $t \to \exp(tA)x$ is differentiable with respect to x and

$$\frac{d}{dt} \exp(tA)x = A \exp(tA)x = \exp(tA)Ax \quad \text{for} \quad t \geqslant 0;$$

(ii) for any $x \in E$ and $t > 0$ the element $\int_0^t \exp(sA)x\,ds$ belongs to D(A) and

$$A \int_0^t \exp(sA)x\,ds = \exp(tA)x - x;$$

(iii) if $x \in D(A)$, then $\exp(tA)x - \exp(\tau A)x = A \int_\tau^t \exp(sA)x\,ds$ for $t, \tau \geq 0$;

(iv) the domain of any natural power of a generating operator of an OS is invariant with respect to the semigroup, i.e., for any $t \geq 0$ we have $\exp(tA)x \in D(A^n)$ for $x \in D(A^n)$, $n \in N$.

2.1.9. [41] Equation (1.2) can be inverted for $x \in E$ and $t \geq 0$ in several ways:

(i) (the Yosida approximation)

$$\exp(tA)x = \lim_{\lambda \to \infty} \exp(t\lambda AR(\lambda, A))x;$$

(ii) (the Widder—Post approximation formula)

$$\exp(tA)x = \lim_{k \to \infty} \left[\frac{k}{t} R\left(\frac{k}{t}, A\right)\right]^k x = \lim_{k \to \infty} (-1)^k \frac{1}{k!} \left(\frac{k}{t}\right)^{k+1} R\left(\frac{k}{t}, A\right)^k x.$$

2.1.10. [333] Assume that we are given an OS exp(tA) and some closed subspace $F \subset E$. The following conditions are equivalent:

(i) the subspace F is invariant with respect to the operators of exp(tA) for $t \geq 0$;

(ii) there exist numbers $s_0 \in \mathbf{C}$, $r \in \mathbf{R}$, and $n_0 \in N$ such that F is an invariant subspace of the operators $R(s_0 + nr, A)$ for all $n \geq n_0$.

2.1.11. [158] For an element $x \in E$ to be invariant with respect to an OS exp(tA) and for an OS exp(tA) to be

contracting or positive, it is necessary and sufficient for the resolvent $R(\lambda,A)$ to have these same properties, respectively, for $\lambda > \omega(A)$.

2.1.12. [35] The set of elements of the form $\int_0^t \exp(sA)x\,ds$ $(t>0, x\in E)$ in D(A) is dense in E, and, clearly,

$$\lim_{t\to 0+} t^{-1}\int_0^t \exp(sA)x\,ds = x \quad \text{for all} \quad x\in E. \tag{1.5}$$

2.1.13. [341] For any $x \in E$, $\lambda \in \mathbf{C}$, we have

$$(\lambda I - A)\int_0^t \exp(\lambda(t-s))\exp(sA)x\,ds = (\exp(\lambda t)I - \exp(tA))x. \tag{1.6}$$

Definition 3. A subset $\mathscr{D}\subset D(A)$ is said to be a core of an arbitrary operator $A\in\mathscr{C}(E)$, if the closure of its restriction $A|_{\mathscr{D}}$ is A, or, equivalently, if $\mathscr{D}$ is dense in D(A) in the graphic norm of A.

2.1.14. [44] If a linear manifold $\tilde{\mathscr{D}}\subseteq D(A)$ is dense in E and is $\{\exp(tA), t \geq 0\}$-invariant, then $\tilde{\mathscr{D}}$ is a core of A.

2.1.15. [35, 101] For an OS exp(tA) set $D(A^k)$ is dense in E for all $k \in \mathbf{N}$. Moreover, set $D(A^\infty) := \bigcap_{k=0}^{\infty} D(A^k)$ (sometimes called the set of infinitely differentiable vectors of A) is also dense in E and is a core of A.

2.1.16. [101] We define the set $\tilde{E}_\alpha := \mathscr{R}(\exp(\alpha A))$ for each $\alpha > 0$ and $\tilde{E}_0 := \bigcup_{\alpha>0} \tilde{E}_\alpha$. Then $\tilde{E}_\beta \subseteq \tilde{E}_\alpha$ for $0 \leq \alpha \leq \beta$.

2.1.17. [126]. For OS's exp(tA) and exp(tB) the following conditions are equivalent:

(i) exp(tA)exp(tB) = exp(tB)exp(tA) for $t \geq 0$;

(ii) $R(\lambda,A) \cdot R(\lambda,B) \cdot R(\lambda,A)$ for some $\lambda \in \rho(A) \cap \rho(B)$;

(iii) condition (ii) holds for all $\lambda \in \rho(A) \cap \rho(B)$. When these conditions are satisfied, the following OS is defined: exp(tC) := exp(tA) × exp(tB), $t \in \mathbf{R}_+$. The set $D_0 := D(A) \cap D(B)$ is a core of the operator C and Cx = Ax +Bx for all $x \in D_0$.

2.1.18. [158, 333] For an OS exp(tA) we have ($x \in E$)

$$\exp(tA)x = \lim_{\lambda\to\infty} e^{-\lambda t}\sum_{k=0}^{\infty}\frac{(\lambda t)^k}{k!}[\lambda R(\lambda, A)]^k x; \tag{1.7}$$

$$\exp(tA)x = \lim_{\lambda\to\infty} \lambda\sum_{k=1}^{\infty}(-1)^k e^{k\lambda t}\frac{1}{(k-1)!}R(\lambda k, A)x; \tag{1.8}$$

$$\exp(tA)x = re^{s_0 t}\lim_{k\to\infty} k\sum_{p=1}^{\infty}(-1)^{p-1}e^{pkrt}\frac{R(s_0+pkr, A)}{(p-1)!}x \tag{1.9}$$

uniformly on any finite segment [0,T].

2.1.19. [158] If $x \in D(A^m)$, then Taylor's formula

$$\exp(tA)x = \sum_{k=0}^{m-1}\frac{t^k}{k!}A^k x + \frac{1}{(m-1)!}\int_0^1 (t-s)^{m-1}\exp(sA)A^m x\,ds \tag{1.10}$$

is valid and we have

$$[\exp(tA)-I]^m x=\int_0^t \dots \int_0^t \exp((s_1+\dots+s_m)A)\,A^m x\,ds_1\dots ds_m.$$

We define the modulus of continuity of an OS exp(tA) on [0,T] to be

$$\omega_T(\delta,\ \exp(\cdot A)x)=\sup_{0\leqslant t,s\leqslant T}\{\|\exp(tA)x-\exp(sA)x\|:|t-s|\leqslant\delta\}.$$

When T = 1, we will simply write $\omega(\delta,\exp(\cdot A)x)$.

We also set

$$\mathscr{A}_\tau := (\exp(\tau A)-I)/\tau. \tag{1.11}$$

2.1.20. [128] For any $x \in E$ we have

$$\exp(tA)x=\lim_{\tau\to 0}\exp(t\mathscr{A}_\tau)x,$$

where convergence is uniform on any finite segment [0,b] and

$$\|\exp(tA)x-\exp(t\mathscr{A}_\tau)x\|\leqslant\omega_{b+1}(\tau^{1/3},\ \exp(\cdot A)x)+K\tau^{1/3}\|x\|,$$

where $K := Mb(M^b+4Me^{b(M-1)})$ and does not depend on τ, while $M := \max_{t\in[0,b]}\|\exp(tA)\|$.

2.1.21. [158] Assume that we are given an OS exp(tA) and

$$\Delta_\tau^m\exp(tA):=\sum_{k=0}^m(-1)^{m-k}C_m^k\exp((t+k\tau)A).$$

Then

$$\frac{1}{\tau^m}\Delta_\tau^m\exp(tA)=\left[\frac{\exp(\tau A)-I}{\tau}\right]^m\exp(tA)=\mathscr{A}_\tau^m\exp(tA)$$

and

$$\exp(tA)x=\lim_{\tau\to 0}\sum_{k=0}^\infty\frac{t^k}{k!\,\tau^k}\Delta_\tau^k\exp(\zeta A)x|_{\zeta=0},$$

where convergence is uniform with respect to $t \in [0,T)$ for all $T < \infty$.

2.1.22. [158] Assume that we have an OS exp(tA). Then

$$\exp(tA)x=\lim_{k\to\infty}\left[(1-t)I-t\exp\left(\frac{1}{k}A\right)\right]^k x$$

uniformly with respect to $t \in [0,1]$, where

$$\left\| \sum_{k=0}^{m} C_m{}^k t^k (1-t)^{m-k} \exp\left(\frac{k}{m} A\right) x - \exp(tA)x \right\| \leqslant$$
$$\leqslant \frac{3}{2} \omega\left(\frac{1}{\sqrt{m}}, \exp(\cdot A)x\right)$$

and

$$\left\| \int_0^1 \left\{ \sum_{k=0}^{m} C_m{}^k t^k (1-t)^{m-k} \exp\left(\frac{k}{m} A\right) \right\} x dt - \int_0^1 \exp(tA)x dt \right\| \leqslant$$
$$\leqslant \omega\left(\frac{1}{m+1}, \exp(\cdot A)x\right).$$

A more general approximation theorem is given in [158].

2.1.23. [158] Assume that for $t \geq 0$ and $x \in D(A^{m-1})$

$$\tilde{B}_t{}^m := \frac{m!}{t^m}\left(\exp(tA)x - \sum_{k=0}^{m-1} \frac{t^k}{k!} A^k x\right) \tag{1.12}$$

(the expression in parentheses is the residual of the Taylor series —see Prop. 1.19). Then the limit $\tilde{B}_t{}^m x$ exists if and only if $x \in D(A^m)$ and $\lim_{t \to 0+} \tilde{B}_t{}^m x = A^m x$.

2.1.24. [158] Assume that for $t \geq 0$, and $x, g_{i,m} \in E$

$$\tilde{P}_t{}^m(g_{0,m}, \ldots, g_{m-1,m})x := \frac{m!}{t^m}\left(\exp(tA)x - \sum_{k=0}^{m-1} \frac{t^k}{k!} g_{k,m}\right) \tag{1.13}$$

(note that operator (1.12) is a special case of (1.13)). Then the limit of expression (1.13) as $t \to 0$ exists if and only if $x \in D(A^m)$ and $g_{i,m} = A^i x$. In this case the limit is equal to $A^m x$.

2.1.25. [158] Assume that for $t > 0$

$$\tilde{C}_t{}^m x := \frac{1}{t^m}\left\{ \sum_{k=0}^{m} (-1)^{m-k} C_m{}^k \exp(ktA)x \right\}. \tag{1.14}$$

Then the limit of expression (1.14) as $t \to 0$ exists if and only if $x \in D(A^m)$ and is equal to $A^m x$.

2.1.26. [158] Set $T_m(t) := I - (I - \exp(tA))^m$. Then

(i) if $x \in E$ and $\| T_m(t)x - x \| = o(t^m)$ as $t \to 0+$, then

$$T_m(t)x = x \quad \text{for all} \quad t \in \bar{\mathbf{R}}_+;$$

(ii) if $x \in D(A^m)$, then

$$\| T_m(t)x - x \| = O(t^m) \quad \text{for} \quad t \to 0+;$$

(iii) in the case of a reflexive E (ii) is invertible.

2.1.27. [158] Let A be a generating operator for an OS. Then for fixed m ∈ N we have

$$\text{(i) } x\in E\Rightarrow\lim_{\lambda\to\infty}\|(\lambda R(\lambda,\ A))^m x-x\|=0;$$

$$\text{(ii) } x\in D(A)\Rightarrow\lim_{\lambda\to\infty}\left\|\frac{\lambda}{m}\{[\lambda R(\lambda,\ A)]^m x-x\}-Ax\right\|=0;$$

$$\text{(iii) } x\in D(A^m)\Rightarrow\lim_{\lambda\to\infty}\left\|\lambda^m\left\{\lambda R(\lambda,\ A)x-\sum_{k=0}^{m-1}A^k x/k!\right\}-A^m x\right\|=0.$$

2.1.28. [158] Let $x \in D(A^{m-1})$, $y \in E$, and

$$\lim_{\lambda\to\infty}\left\|\lambda^m\left\{\lambda R(\lambda,\ A)x-\sum_{k=0}^{m}\lambda^{-k}A^k x\right\}-y\right\|=0.$$

Then $x \in D(A^m)$ and $A^m x = y$. If, in this case, $y = 0$, then $\lambda R(\lambda,\ A)x=\sum_{k=0}^{m-1}A^k x/\lambda^k$ for $\lambda > \max(0,\omega(A))$.

2.1.29. [158] The following assertions are equivalent ($x \in E$):

(i) $\psi(t,x) := \|(\exp(tA) - I)x\| = O(t)$ as $t \to 0$;

(ii) $\|[\lambda R(\lambda,A)]^m x - x\| = O(\lambda^{-1})$ with fixed $m \in N$ as $\lambda \to \infty$.

If E is reflexive, each of conditions (i) and (ii) is equivalent to the condition $x \in D(A)$.

2.1.30. [158] For $x \in D(A^{m-1})$ the following conditions are equivalent:

$$\text{(i)}\left\|\exp(tA)x-\sum_{k=0}^{m-1}(tA)^k x/k!\right\|=O(t^m/m!)\ \text{ as }\ t\to 0+;$$

$$\text{(ii)}\left\|\lambda R(\lambda,\ A)x-\sum_{k=0}^{m-1}A^k x/k!\right\|=O(\lambda^{-m})\ \text{ as }\ \lambda\to\infty.$$

If E is reflexive, each of condtions (i) and (ii) is equivalent to the condtion $x \in D(A^m)$.

2.1.31. [158] For a fixed $x \in E$ the following conditions are equivalent:

$$\text{(i) } \|(\exp(tA)-I)^m x\|=O(t^m)\ \text{ as }\ t\to 0+;$$

$$\text{(ii) } \|[\lambda R(\lambda,\ A)-I]^m x\|=O(\lambda^{-m})\ \text{ as }\ \lambda\to\infty;$$

$$\text{(iii)}\left\|\left\{\prod_{k=1}^{m}\left[\frac{\lambda}{k}R\left(\frac{\lambda}{k},\ A\right)-I\right]\right\}x\right\|=O(m!\ \lambda^{-m})\ \text{ as }\ \lambda\to\infty.$$

When $x \in D(A^{m-1})$ each of these assertions is equivalent to the assertions of 1.30.

2.1.32. [158] If x is such that

$$\|(\exp(tA)-\exp(sA))x\|\leqslant L(x)\frac{(t-s)^\gamma}{(t+s)^{\gamma/2}},\ s,\ t\in\overline{\mathbf{R}}_+,$$

then $\|[\exp(t\mathscr{A}_\tau)-\exp(tA)]x\|\leqslant L(x)\tau^{\gamma/2}\cdot$ for $t \geq 0$.

2.1.33. [158]. Let H be an infinite-dimensional Hilbert space, let $P \in B(H)$ be a projector of finite rank, and let exp(tA) be such that $\sup_{t\in\overline{\mathbf{R}}_+}\|\exp(tA)\|=\infty$ and $\sup_{t\in\overline{\mathbf{R}}_-}\|\exp(tA)\exp(-tP)\|<\infty$. Then closure of set $\bigcup_{t\geqslant 0}\mathscr{R}(\exp(tA)\cdot(I-P))$ is a proper exp(tA)-invariant subspace of H.

2.1.34. [92] If two operators A and B are self-adjoint and the operator $\exp(-B/2)\exp(-A)\exp(-B/2)$ is bounded, then the operator A + B is bounded below and

$$\|\exp(-(A+B))\| \leqslant \|\exp(-B/2)\exp(-A)\exp(-B/2)\|.$$

Definition 4. A sequence $\{x_k\}_{k\in\mathbf{Z}} \subset E$ is said to be Cesaro summable if the limit

$$C_1-\sum_{k\in\mathbf{Z}} x_k := \lim_{n\to\infty}\sum_{k=-n}^{n}(1-|k|/(n+1))x_k$$

exists.

2.1.35. [420] Let $A\in\mathscr{G}(\mathbf{M}, \omega)$, $x\in E$, $\zeta_0\in\mathbf{C}$ and $\{\zeta_0 + ik : k \in \mathbf{Z}\} \subset \rho(A)$. Then the following conditions are equivalent:

(i) $x\in\mathscr{R}(\exp(2\pi\zeta_0)I-\exp(2\pi A))$;

(ii) the limit $C_1-\sum_{k\in\mathbf{Z}} R(\zeta_0+ik, A)x$ exists;

(iii) the limit $C_1-\sum_{k\in\mathbf{Z}} R(\zeta_0+ik, A)Q_k(\zeta_0)x$ exists, where $Q_k(\xi) := (2\pi)^{-1}\int_0^{2\pi}\exp(-ikt)\exp(-t\xi)\exp(tA)\times$ $x\,dt$. If there exists an $r > 0$ such that $\zeta\to C_1-\sum_{k\in\mathbf{Z}} R(\zeta+ik, A)x$ is analytic in $\Delta(r) := \{z \in \mathbf{C} : |z - \zeta_0| < r\}$, then the function $\zeta \to (\exp(2\pi\zeta)I - \exp(2\pi A))^{-1}x$ is analytic in $\Delta(r)$.

2.2. One-parameter Groups of Operators.

Definition 1. A one-parameter group of operators, or simply a group, is a family of operators satisfying the conditions of Definition 1.3.1 with $\bar{\mathbf{R}}_+$ replaced by $\mathbf{R}$.

2.2.1. [35] If an OS exp(tA) can be extended to a group (sometimes we say can be imbedded in a group), the extension is unique. We then use the same notation exp(tA) for the group.

THEOREM 1. [35] For an operator $A \in L(E)$ to be a generating operator of a group exp(tA), it is necessary and sufficient for $A\in\mathscr{G}(E)$ and there exist constants M and ω such that $\{\lambda \in \mathbf{C} : \operatorname{Re}\lambda > \omega\} \subset \rho(A)$ and

$$\|R(\lambda, A)^k\| \leqslant \frac{M}{(\operatorname{Re}\lambda-\omega)^k} \quad \text{for } |\operatorname{Re}\lambda| > \omega, \quad k\in\mathbf{N}.$$

2.2.2. If $A \in B(E)$, then a simple example of a group of operators is given by the formula

$$\exp(tA)=\sum_{k=0}^{\infty}(tA)^k/k!, \quad t\in\mathbf{R}.$$

2.2.3. [288] If, for an OS exp(tA) and every $t > 0$, the operator $\hat{\Phi}(t) := [\exp(tA)]^{-1} \in B(E)$ exists, then $\hat{\Phi}(t)$ is an OS with generating operator —A that can be imbedded in $\Phi(t)$, $t \in \mathbf{R}$, in the following manner:

$$\Phi(t)=\begin{cases}\exp(tA) & \text{for } t\geqslant 0,\\ \hat{\Phi}(t) & \text{for } t<0.\end{cases}$$

We will denote the set of operators generating a group with bound (1.1) for $t \geq 0$ by $\mathscr{GR}(\mathbf{M}, \omega)$.

2.2.4. [288] If $0 \in\rho(\exp(t_0A))$ for some $t_0 > 0$, then $0 \in \rho(\exp(tA))$ for all $t > 0$ and the OS exp(tA) can be imbedded in a group.

2.2.5. [126] Consider an operator $\exp(t_0A) - I$ that is completely continuous for some $t_0 > 0$. Then the OS exp(tA) is invertible for some $t > 0$ and it can be imbedded in a group.

2.2.6. [181] Let exp(tA) be a group of operators with unbounded A. Then there exists a number μ such that for any α, β $(0 \le \alpha,\beta < \infty)$ we have

$$\sup_{t\in[\alpha,\beta]}(\|\exp(tA)-I\|-1-e^{\mu t})>0.$$

2.2.7. [153] Let exp(tA) and exp(tG) be two groups of operators that commute with each other, and let $\tilde{x}$ be such that $\tilde{x}\in D(A^2)\cap D(G^2)$; $A\tilde{x}$, $G\tilde{x}\in D(A)\cap D(G)$; $AG\tilde{x}=GA\tilde{x}$. Then $\|A\tilde{x}-iG\tilde{x}\|\le 2\|\tilde{x}\|\, m(\rho)/\rho+2\sqrt{2}\times h(\tilde{x})m(1/\rho)/\rho+4\|A\tilde{x}+iG\tilde{x}\|m(\rho)\cdot\ln\rho$ for all $\rho\ge 1$, where $h(x):=\|A(Ax+iGx)\|+\|B(Ax+iGx)\|$, $m(r):=\sup\{\|\exp(tA)\exp(sG)\|: t,\ s\in\mathbb{R};\ |t+is|\le r$ for $r>0\}$.

2.2.8. [153] Under the conditions of 2.7 we have

$$\max(\|A\tilde{x}\|,\|G\tilde{x}\|)\le\|\tilde{x}\|\, m(\rho)/\rho+\sqrt{2}\,h(\tilde{x})\,m(1/\rho)/\rho+$$
$$+\|A\tilde{x}+iG\tilde{x}\|(1/2+2m(\rho)\ln\rho),\quad \rho\ge 1.$$

2.2.9. [153] Given the conditions of 2.7 and additionally $m(\rho)/\rho\to 0$ as $\rho\to\infty$, then for all L, $\varepsilon>0$ there exists a $\delta>0$ such that for $\|\tilde{x}\|$, $h(\tilde{x})<L$, the inequality $\|A\tilde{x}+iG\tilde{x}\|<\delta$ implies $\|A\tilde{x}-iG\tilde{x}\|<\varepsilon$. If, in addition to the conditions of 2.7, we have $m(\rho)\le m$ for $\rho>0$, then for $\rho\ge 1$ we have

$$\|A\tilde{x}-iG\tilde{x}\|\le$$
$$\le\begin{cases}4m\cdot\|A\tilde{x}+iG\tilde{x}\|, & \text{if } \|\tilde{x}\|+\sqrt{2}h(x)\le 2\cdot\|A\tilde{x}+iG\tilde{x}\|;\\ 4m\cdot\|A\tilde{x}+iG\tilde{x}\|(1+\ln[\|\tilde{x}\|+\sqrt{2}h(x))/(2\|A\tilde{x}+iG\tilde{x}\|)]) & \text{otherwise.}\end{cases}$$

Definition 2. The set

$$\mathfrak{A}(A):=\{x\in D(A^\infty):\exists\alpha,\ c>0:\|A^kx\|\le c\alpha^k n^k,\ k\in\mathbf{N}\}$$

is called the set of analytic vectors of the operator A, and

$$\mathfrak{A}_c(A):=\{x\in D(A^\infty):\forall\alpha>0\,\exists c>0:\|A^kx\|\le c\alpha^k n^k,\ k\in\mathbf{N}\}$$

is called the set of integral vectors of the operator A.

It is clear that for any $A\in\mathscr{B}(E)$ we have $\mathfrak{A}_c(A)\subseteq\mathfrak{A}(A)\subseteq D(A^\infty)$.

2.2.10. [24] If $B\in B(E)$, then $\mathfrak{A}(A)=D(A^\infty)=\mathfrak{A}_c(A)=E$.

2.2.11. When A is unbounded, the set $D(A^\infty)$ may contain only the null element. However, if $A\in\mathscr{G}(\mathbf{M},\omega)$, then $\overline{D(A^\infty)}=E$.

2.2.12. [86] If $A\in\mathscr{GR}(\mathbf{M},0)$, then $\mathfrak{A}_c(A)$ is dense in E.

2.2.13. [22] There is an example of an operator $A\in\mathscr{G}(\mathbf{M},0)$, such that $\mathfrak{A}(A)=\{0\}$.

We define

$$D(A,\eta):=\{x\in D(A^\infty):\exists c=c(x,\eta)>0,\ \|A^kx\|\le c\eta^k,\ k\in\mathbf{N}\}$$

and

$$\mathrm{EXP}(A):=\{x\in D(A^\infty):\exists\alpha,\ c>0:\|A^kx\|\le c\alpha^k,\ k\in\mathbf{N}\}.$$

The set EXP(A) is called the set of vectors of exponential type.

2.2.14. [22] The set $D(A,\eta)$, provided with the norm $\|x\|_{D(A,\eta)}:=\sup_{k\in\mathbf{N}}\eta^{-k}\|A^kx\|$, is a Banach space.

2.2.15. [22] The following propositions are true:

(i) $D(A,\eta_1) \subseteq D(A,\eta_2)$ and $\|x\|_{D(A,\eta_2)} \leqslant \|x\|_{D(A,\eta_1)}$ for $x \in D(A,\eta_1)$ and $\eta_1 \leq \eta_2$;

(ii) the set $D(A,\eta)$ is invariant under A and $\|Ax\|_{D(A,\eta)} \leqslant \eta \|x\|_{D(A,\eta)}$ for $x \in D(A,\eta)$;

(iii) $\bigcup_{k=1}^{\infty} \sigma(A|_{D(A,k)}) \subseteq \sigma(A)$ and $\|R(\lambda, A)x\|_{D(A,\eta)} \leqslant \|R(\lambda, A)\| \times \|x\|_{D(A,\eta)}$ for $\lambda \in \rho(A)$ and $x \in D(A,\eta)$;

(iv) if $\overline{EXP(A)} = E$, any element $x \in E$ can be represented in form $x = \sum_{k=1}^{\infty} x_k$, where $x_k \in D(A, k)$, $\sum_{k=1}^{\infty} \|x_k\|_{D(A,k)} < \infty$, and $\|x\| = \inf \sum_{k=1}^{\infty} \|x_k\|_{D(A,k)}$, where inf is taken over all possible representations of $x \in E$.

Moreover,

$$D(A) = \left\{ x \in E : x = \sum_{k=1}^{\infty} x_k,\ x_k \in D(A, k),\ \sum_{k=1}^{\infty} k \|x_k\|_{D(A,k)} < \infty \right\}$$

and

$$Ax = \sum_{k=1}^{\infty} A|_{D(A,k)} x_k, \quad x \in D(A).$$

2.2.16. [22] If $A \in \mathcal{GR}(M, 0)$, then $\overline{EXP}(A) = E$.

2.3. Differentiable Semigroups of Operators.

Definition 1. An OS exp(tA) is said to be differentiable beginning with t_0 (or to become differentiable) if for any $x \in E$ the function $t \to \exp(tA)x$ is differentiable on (t_0, ∞). An OS is said to be differentiable if $t_0 = 0$.

Concerning the properties of differentiable OS's, as well as criteria for whether an OS becomes differentiable, see [60, 341].

2.3.1. [341] If there exist constants $C > 0$ and $\delta_c > 0$ such that

$$\|\exp(tA) - I\| \leqslant 2 + C \cdot t \cdot \ln t, \quad 0 < t < \delta_c, \tag{3.1}$$

then the OS exp(tA) becomes differentiable at 3M/C, where the constant M is determined by (1.1).

2.3.2. [419] If an OS exp(tA) becomes differentiable, then the operator exp(tA) is surjective for all $t \in \mathbf{R}_+$.

2.3.3. [419] Assume that $A \in \mathcal{G}(M, \omega)$ and the operator $\exp(t_0 A)$ is not surjective for some $t_0 > 0$. Then the OS exp(tA) is not surjective for any $t > 0$, and in this case the OS exp(tA) does not become differentiable.

2.4. Analytic Semigroups. By $\Sigma(\theta) := \{z \in \mathbf{C} : \arg z| < \theta\}$ we denote a sector of the complex plane **C**.

Definition 1. An OS exp(tA) is said to be an analytic semigroup with angle $\theta \in (0, \pi/2)$, or simply analytic, if it can be extended to an operator function exp(zA) that is analytic in $\Sigma(\theta)$ and strongly continuous at zero on any ray inside $\Sigma(\theta)$. If for any positive $\psi < \theta$ we have $\|\exp(zA)\| \leq M_\psi < \infty$, $z \in \Sigma(\psi)$, the semigroup is said to be bounded analytic.

2.4.1. An operator function exp(zA) preserves, in $\Sigma(\theta)$, the semigroup property

$$\exp((z_1 + z_2)A) = \exp(z_1 A)\exp(z_2 A), \quad z_1,\ z_2 \in \Sigma(\theta).$$

In $\Sigma(\theta)$ with $\omega > \omega(A)$ the bound

$$\|\exp(zA)\| \leqslant M \exp(\omega \cdot \operatorname{Re} z) \quad (|\arg z| \leqslant \theta_1 < \theta < \pi/2). \tag{4.1}$$

remains valid.

We will denote by $\mathcal{H}(\theta, \omega)$ the set of generating operators of analytic semigroups with angle θ that satisfy (4.1).

THEOREM 1 ([59, 60]). For an OS exp(tA) the following assertions are equivalent:

(i) the OS exp(tA) is analytic;

(ii) there exist M and $\omega \in \mathbf{R}$ such that for $\operatorname{Re}\lambda \geq \omega > \omega(A)$ the resolvent $R(\lambda,A)$ is defined and

$$\|R(\lambda, A)\| \leqslant \mathbf{M} \cdot |\lambda - \omega|^{-1}. \tag{4.2}$$

(iii) the OS exp(tA) is differentiable and for its derivative we have the bound

$$\left\| \frac{d}{dt} \exp(tA) \right\| \leqslant \mathbf{M} \cdot t^{-1} \cdot \exp(\omega t), \quad t \in \bar{\mathbf{R}}_+, \tag{4.3}$$

for some M, $\omega \in \mathbf{R}$;.

(iv) the generating operator of exp(tA) has, for some λ_0 and $\varepsilon > 0$, the bound

$$\|A(I - \lambda A)^{-k}\| \leqslant C/(k\lambda), \quad (0 < \lambda < \lambda_0,\ k \in \mathbf{N},\ k\lambda \leqslant 1);$$

(v) for an OS exp(tA) the generating operator has, for some M, ω, $\rho > 0$, the bound

$$\left\| \frac{d^{m+k}}{d\lambda^{m+k}} (\lambda^k R(\lambda, A)) \right\| \leqslant \mathbf{M} m! k! \rho^k / (\lambda - \omega)^{m+1}, \quad (m, k \in \mathbf{N},\ \lambda > \omega(A)).$$

2.4.2. [158] For an analytic OS exp(tA) conditions $\bar{E}_t \subseteq D(A)$ and $\| tA\exp(tA) \| \leq M$ for $t \geq 0$ are equivalent to the statement that OS exp(tA) can be analytically continued to sector $\Sigma(1/me)$ and condition $\lim_{\zeta \to 0} \exp(\zeta A)x = x (\zeta \in \Sigma(\theta))$ is satisfied. Also, when $0 < t \leq 1$ and $\alpha \in (0, 1/(me))$ we have

$$A \exp(tA) = \frac{1}{2\pi i} \int_{|\zeta - t| = t \sin\alpha} \exp(\zeta A)/(\zeta - t)^2 \, d\zeta.$$

2.4.3. [158] If an analytic OS exp(tA) has the bound

$$\|\exp(\zeta A)\| \leqslant \mathbf{M}_1 \quad \text{for} \quad \zeta \in \left\{ \zeta \in \mathbf{C} : 0 \leqslant \operatorname{Re}\zeta \leqslant 2,\ |\arg\zeta| \leqslant \frac{\varepsilon}{e\mathbf{M}} \right\},$$

where M is given by 4.2, then for the α given in 4.2 we have

$$\|A \exp(tA)\| \leqslant \mathbf{M}_1/(t \cdot \sin\alpha).$$

2.4.4. [44] For an analytic OS exp(tA) we have

$$\text{(i)} \left\| \frac{d^k}{dt^k} (\exp(tA)) \right\| = \| A^k \exp(tA) \| \leqslant \mathbf{M}_k e^{\omega t}/t^k, \ t \geqslant 0;$$
$$\text{(ii)} \ \| A^k (\exp(tA) - \exp(sA)) \| \leqslant \mathbf{M}_k (t^k - s^k) e^{\omega t}/[k(ts)^k], \ 0 < s \leqslant t$$

where $\mathbf{M}_k = \mathbf{M}^k \cdot k^k$ and $\omega > \omega(A)$.

2.4.5. [193] A generating operator A generates a bounded analytic OS exp(tA) with angle θ if and only if $\Sigma(\theta + \pi/2) \subset \rho(A)$ and for any $\psi \in (0, \theta)$ there exists a constant $\mathbf{M}_\psi$ such that $\| R(\lambda, A) \| \leq \mathbf{M}_\psi / |\lambda|$ for $\lambda \in \Sigma(\psi + \pi/2)$.

2.4.6. [44] An OS exp(tA) is analytic if and only if there exist an $\omega_1 > \omega(A)$ and an $\mathbf{M} \geq 0$ with the property

$$\|R(\omega_1 + i\eta, A)\| \leqslant \mathbf{M}/|\eta| \quad \text{for all} \quad \eta \in \mathbf{R}.$$

2.4.7. [293] If A generates a bounded group of operators exp(tA), then $-A^2$ generates a bounded analytic OS with angle $\pi/2$. The statement remains true if the word "bounded" is omitted.

2.4.8. Assume we are given an OS exp(tA) that is analytic in $\Sigma(\theta)$. Then the operator function $U(t) = \exp(t\exp(i\varphi)A)$ is an OS for all $\varphi \in (-\theta,\theta)$.

2.4.9. [293] Assume that $-A$ generates a bounded analytic OS exp(tA) with angle θ and $n(\pi/2 - \theta) < \pi/2$. Then the operator $-A^n$ generates a bounded analytic OS with angle $\pi/2 - n(\pi/2 - \theta)$ (for the case $n = 2$ see [231]).

2.4.10. [293] If A generates a bounded analytic OS with angle $\pi/2$, then $-A^n$ generates a bounded analytic OS for all $n \in \mathbf{N}$.

2.4.11. [292] Assume that we are given a bounded analytic OS exp(tA) with angle $\theta \in (0,\pi/2]$. If $\mathscr{R}(A)$ is dense in E, then $\mathscr{N}(A) = \{0\}$ and the operator A^{-1} generates a bounded analytic OS U(t) with angle θ that is given by an integral that converges in the uniform operator topology:

$$U(t) = \frac{1}{2\pi i}\int_{\Gamma} e^{-z}A(zA + tI)^{-1}\,dz,$$

where $\Gamma = \partial[\{x \in \mathbf{C} : |x| = r > 0\} \cup \Sigma(\theta)]$.

2.4.12. [292] If E is reflexive, then in 4.11 condition of denseness may be replaced by condition $\mathscr{N}(A) = \{0\}$.

2.4.13. [162] The following statements are equivalent:

(i) the OS exp(tA) is analytic;

(ii) there exists a number $\zeta \in \mathbf{C}$, $|\zeta| = 1$, such that

$$\varliminf_{t\to 0+}\inf_{x}\{\|\zeta x - \exp(tA)x\| : x \in E,\ \|x\| = 1\} > 0;$$

(iii)

$$\varlimsup_{t\to 0+}\sup\{t\,\|A\exp(tA)x\| : x \in D(A),\ \|x\| = 1\} < \infty;$$

(iv) the operator $\begin{pmatrix} A & O \\ A & A \end{pmatrix}$ generates an OS on $E \times E$;

(v) there exist a constant C and a number $b > 0$ such that $n\|(\lambda R(\lambda, A)^n - (\lambda R(\lambda, A))^{n-1}\| \leqslant C$ for $\lambda > nb$, $n \in \mathbf{N}$;

(vi) there exists a polynomial q such that

$$\varlimsup_{t\to 0+}\|q(\exp(tA))\| < \sup\{|q(z)| : z \in \mathbf{C},\ |z| = 1\};$$

(vii) the strict inequality

$$\lim_{h\to\infty}\sup_{s,t\in\mathscr{K}}\left\|\left\{q\left(\exp\left(\frac{st}{n}A\right)\right)\right\}^n\exp(tA)\right\| < \\ < L(q) := \sup\{|q(z)| : z \in \mathbf{C},\ |z| = 1\}$$

is satisfied for all compact $\mathscr{K} \subset \mathbf{R}$ and all polynomials q(z) such that $|q(1)| < L(q)$;

(viii) for any $\zeta \in \mathbf{C}$ such that $|\zeta| = 1$ and $\zeta \neq 1$, for sufficiently small t the operator $\zeta I - \exp(tA)$ is invertible.

2.4.14. [162] If we have, for an OS exp(tA), the inequality

$$\varlimsup_{t\to 0+}\|\exp(tA) - I\| < 2,$$

then the OS exp(tA) is analytic. The converse is generally not true, but does hold, for example, in the case of a uniformly convex E or a contraction OS.

2.4.15. [59] If an OS exp(tA) is analytic with angle θ, then $\tilde{E}_\alpha \subseteq D(A^\infty)$ for all complex $\alpha \in \Sigma(\theta)$.

2.4.16. [59] Let exp(tA) be a bounded analytic OS and let $\delta = \pi/2 + \psi$, with ψ as in 4.5. Then

$$\exp(tA)=\frac{1}{2\pi i}\int_{\Gamma_\varphi} e^{\lambda z}R(\lambda, A)\,d\lambda, \tag{4.4}$$

where $\Gamma_\varphi=\Gamma_{\varphi_+}\cup\Gamma_{\varphi_-}$, $\Gamma_{\varphi_\pm}=\rho e^{\pm i\varphi}$ $(\pi/2<\varphi<\delta,\ 0<\rho<\infty)$.

2.4.17. [158] Assume we are given an analytic OS exp(tA). Then

(i) $\psi(t,x) = O(t^\alpha)$ for $t \to 0+$ and $0 < \alpha < 1$ if and only if $\| A\exp(tA) \| = O(t^{\alpha-1})$ for $t \to 0+$, which, in turn, occurs if and only if $\| A^2\exp(tA) \| = O(t^{\alpha-2})$ for $t \to 0+$;

(ii) if $\psi(t,x) = O(t)$, then $|A\exp(tA)| = O(\log(1/t))$ and $\| A^2\exp(tA) \| = O(t^{-1})$ for $t \to 0+$.

2.4.18. [28] If A generates a bounded analytic OS, the set of integral vectors $\mathfrak{A}_c(A)$ is dense in E.

2.4.19. [28] Assume that A generates an analytic contraction OS exp(tA). Then $\mathfrak{A}_c(A)=\bigcap_{t>0}\mathcal{R}(\exp(tA))$, there exist inverse operators for the OS exp(tA) that are defined on $\mathfrak{A}_c(A)$, , and $[\exp(tA)]^{-1}x=\sum_{k=0}^{\infty}t^k(-A)^k x/k!$ for all $x\in\mathfrak{A}_c(A)$.

2.4.20. [169] Let $A\in\mathcal{H}(\pi/2,0)$ and $B^2 = A$. Then for any $\eta > 0$ the Banach spaces $D(B,\eta)$ and $D(A,\eta^2)$ coincide. In this case EXP(B) = EXP(A).

2.5. Fredholm Semigroups of Operators. Given an operator $A\in\mathcal{C}(E)$, we denote the dimension of the kernel $\mathcal{N}(A)$ by $\operatorname{nul}(A):=\dim\mathcal{N}(A)$. Defect of an operator A is codimension of $\mathcal{R}(A)$, i.e., $\operatorname{def}(A)=\dim(E/\mathcal{R}(A))$. If nul(A) or def(A) is finite, then the index of the operator can be defined: ind(A) := nul(A) — def(A).

Definition 1. An operator $A\in\mathcal{C}(E)$ is said to be Fredholm (Noetherian) if $\mathcal{R}(A)$ is closed and nul(A) and def(A) are finite. If, however, only one of the numbers nul(A) or def(A) is finite, the operator is said to be semi-Fredholm.

Definition 2. An OS exp(tA) is said to be semi-Fredholm if the operator exp(tA) is semi-Fredholm for all $t \geq 0$.

2.5.1. [219] Assume $t_0 > 0$. The following statements are true:

(i) if $\operatorname{nul}(\exp(t_0A)) < \infty$, then $\operatorname{nul}(\exp(tA)) = 0$ for any $t \geq 0$;

(ii) if $\operatorname{def}(\exp(t_0A)) < \infty$, then $\operatorname{def}(\exp(tA)) = 0$ for any $t \geq 0$;

(iii) if an operator $\exp(t_0A)$ is Fredholm, then the operators exp(tA) are bijective for all $t \in \mathbf{R}_+$;

(iv) if the operator $\exp(t_0A)$ is semi-Fredholm, then for all $t > 0$ the operators exp(tA) are semi-Fredholm and $\operatorname{nul}(\exp(tA)) = \operatorname{nul}(\exp(t_0A))$ and $\operatorname{def}(\exp(tA)) = \operatorname{def}(\exp(t_0A))$.

THEOREM 1. [219] Let E be a reflexive space. Then an OS exp(tA) is semi-Fredholm if and only if for all $\mu > \omega(A)$ and $1 \leq \rho < \infty$

$$\inf_{\|x\|=1}\int_0^\infty e^{-\mu t}\|\exp(tA)x\|^\rho\,dt+\inf_{\|y^*\|=1}\int_0^\infty e^{-\mu t}\|\exp(tA)^*y^*\|^\rho\,dt>0,$$

where $\exp(tA)^*$ is the operator adjoint to exp(tA).

2.5.2. [219] A differentiable OS with an unbounded generator cannot be semi-Fredholm.

2.5.3. [219] If an OS exp(tA) is semi-Fredholm and has an unbounded generator, then $\overline{\lim_{t\to0+}}\|\exp(tA)-I\|\geqslant2$.

We define reduced minimal modulus of an operator $A \in L(E)$ as $\gamma(A):=\inf\{\|Ax\|/\operatorname{dist}(x,\mathcal{N}(A)) : x\notin\mathcal{N}(A)\}$, where $\operatorname{dist}(x,\mathcal{N}(A))$ is the distance from the element x to the kernel $\mathcal{N}(A)$.

2.5.4. [44] The range of an operator $A\in\mathcal{C}(E)$ is closed if and only if $\gamma(A) > 0$.

2.5.5. [219] If an OS exp(tA) is semi-Fredholm, then

(i) $\sup_{t>0} t^{-1} \ln [\gamma(\exp(tA))] = \lim_{t\to\infty} t^{-1} \ln [\gamma(\exp(tA))] \leqslant \omega(A)$;

(ii) there exist constants $\delta > 0$ and α such that

$$\delta \cdot \exp(\alpha \cdot t) \leqslant \gamma(\exp(tA) \leqslant \exp(t\omega(A)), \quad t \geqslant 0.$$

2.5.6. [219] Assume the operator $\exp(t_0 A)$ is semi-Fredholm for some $t_0 > 0$ (and consequently for all $t \geq 0$ — see 5.1). Then, in the half-plane

$$\mathbf{SF}(A) := \{z \in \mathbf{C} : \operatorname{Re} z < \sup_{t>0} t^{-1} \ln [\gamma(\exp(tA))]\}$$

we have

(i) $\lambda I - A$ is semi-Fredholm for all $\lambda \in \mathbf{SF}(A)$;

(ii) $\operatorname{nul}(\lambda I - A)$ and $\operatorname{def}(\lambda I - A)$ are constant for $\lambda \in \mathbf{SF}(A)$, where

$$\begin{aligned} &\operatorname{nul}(\lambda I - A) = 0, \quad \text{if} \quad \operatorname{nul}(\exp(t_0 A)) < \infty, \\ &\operatorname{nul}(\lambda I - A) > 0, \quad \text{if} \quad \operatorname{nul}(\exp(t_0 A)) = \infty, \\ &\operatorname{def}(\lambda I - A) = 0, \quad \text{if} \quad \operatorname{def}(\exp(t_0 A)) < \infty, \\ &\operatorname{def}(\lambda I - A) > 0, \quad \text{if} \quad \operatorname{def}(\exp(t_0 A)) = \infty; \end{aligned}$$

(iii) $|\gamma(\lambda I - A) - \gamma(\lambda_1 I - A)| \leqslant |\lambda - \lambda_1|$ for $\lambda, \lambda_1 \in \mathbf{SF}(A)$.

2.5.7. [219] If an OS exp(tA) is semi-Fredholm and the resolvent $R(\lambda, A)$ is compact, then exp(tA) is bijective for $t \geq 0$.

2.5.8. [219] The following statements are equivalent:

(i) $\| \exp(tA)x \| \geq \exp(\rho t) \| x \|$ for all $x \in E$ and $t \geq 0$;

(ii) $\| (\lambda I - A)x \| \geq (\rho - \lambda) \| x \|$ for all $\lambda < \rho$ and $x \in d(A)$;

(iii) for all $x \in D(A)$ with unit norm there exists an element $x^* \in \theta(x)$ (see Chap. 7) such that $\operatorname{Re}\langle Ax, x^* \rangle \geq \rho$.

2.5.9. [219] For $A \in \mathscr{G}(1,0)$ the following statements are equivalent:

(i) the operator exp(tA) is isometric for every $t > 0$;

(ii) $\| (\lambda I - A)x \| \geq -\lambda \| x \|$ for all $\lambda \leq 0$ and $x \in D(A)$;

(iii) for every $x \in E$ there exists an $x^* \in \theta(x)$ such that $\operatorname{Re}\langle Ax, x^* \rangle = 0$.

When $E = H$ the assertions of 5.8 and 5.9 may be restated in terms of the numerical range of the operator

$$W(A) := \{(Ax, x) : \|x\| = 1, \ x \in D(A)\}.$$

2.5.10. [219] The following statements are equivalent:

(i) the OS exp(tA) can be imbedded in a group and $\gamma(\exp(tA)) \geq e^{\rho t}$;

(ii) $\operatorname{Re} W(A) \geq \rho$;

(iii) $\| (\lambda I - A)x \| \geq (\rho - \lambda) \| x \|$ for all $\lambda < \rho$ and $x \in D(A)$.

When these statements are true we have

$$\lim_{t\to 0} \gamma(\exp(tA)) = 1.$$

2.6. Spectral Properties of Operator Semigroups. Given an operator $A \in \mathscr{G}(E)$ we will, as ususal, separate its spectrum into three sections, the point spectrum $P\sigma(A)$, the continuous spectrum $C\sigma(A)$, and the residual $R\sigma(A)$.

Definition 1. By the Fredholm domain of an operator $A \in \mathscr{G}(A)$ we mean the set $\Omega\mathbf{F}(A)$ of all points $z \in \mathbf{C}$ for which the operator $zI - A$ is Fredholm.

Definition 2. The complement of $\Omega\mathbf{F}(A)$ is called the essential spectrum of the operator $A \in \mathscr{G}(E)$ and is denoted by $E\sigma(A)$.

2.6.1. [178] It is clear that $E\sigma(A) \subseteq \sigma(A)$ for all $A \in \mathscr{B}(E)$.

Definition 3. We denote be $EF\sigma(A)$ the set of all $z \in \mathbf{C}$ for which the operator $zI - A$ is not semi-Fredholm.

Definition 4. The set of all points $\lambda \in \mathbf{C}$ for which the operator $\lambda I - A$ is not injective or $\mathscr{R}(\lambda I - A)$ is not closed in E is called the approximative spectrum of the operator $A \in \mathscr{B}(E)$ and denoted by $A\sigma(A)$.

2.6.2. [126] For any $A \in \mathscr{B}(E)$ we have

$$P\sigma(A) \subseteq A\sigma(A) \text{ and } \sigma(A) = A\sigma(A) \cup R\sigma(A).$$

2.6.3. [126] For any $A \in \mathscr{B}(E)$ we have

(i) the boundary $\partial\sigma(A)$ of the set $\sigma(A)$ is contained in $A\sigma(A)$;

(ii) $R\sigma(A) = P\sigma(A^*)$, where A^* is the adjoint of A.

As usual, for an operator $B \in B(E)$ we define the spectral radius

$$r(B) := \sup\{|\lambda| : \lambda \in \sigma(B)\},$$

and the essential spectral radius

$$Er(B) := \sup\{|\lambda| : \lambda \in E\sigma(B)\}.$$

Definition 5. The measure of noncompactness of a Kuratowski set $\Omega \subset E$ is the quantity $\alpha(\Omega) := \inf\{d > 0$: there exists a finite number of subsets $\{\Omega_j\}_{j=1}^k \subset E$, such that $\Omega \subset \bigcup_{j=1}^k \Omega_j$, $\sup\{\|x - y\|: x, y \in \Omega_i\} \leqslant d$, $i = \overline{1, k}\}$. The measure of noncompactness of the empty set is taken to be zero.

For detailed information on the measure of noncompactness see, for example, [2].

Definition 6. Let $B \in B(E)$. The quantity $|||B|||_\alpha := \inf\{d > 0 : \alpha(B\Omega) \leqslant d \cdot \alpha(\Omega)$ for any bounded $\Omega \in E$ is called the measure of noncompactness of the operator B.

2.6.4.[126] For any $B \in B(E)$ we have

$$Er(B) = \lim_{k\to\infty} |||B^k|||_\alpha^{1/k} = \lim_{k\to\infty} \|B^k\|_*^{1/k}, \tag{6.1}$$

where $\|B\|_* = \inf\{\|B - K\| : K \in B_0(E)\}$.

2.6.5. [126] Let $A \in \mathscr{G}(\mathbf{M}, \omega)$. Then

$$S(A) \leqslant \omega(A). \tag{6.2}$$

2.6.6. [126, 424] Inequality (6.2) may be strict.

2.6.7. [126, 401] The equation $\omega(A) = S(A)$ holds in each of the following cases:

(i) the OS exp(tA) becomes compact;

(ii) the OS exp(tA) becomes differentiable;

(iii) the OS exp(tA) is analytic.

2.6.8. [126, 178] For $A \in \mathscr{G}(\mathbf{M}, \omega)$ we have

$$r(\exp(tA)) = \exp(\omega(A)t), \quad t \geqslant 0.$$

Definition 7. The essential growth index of an OS exp(tA) is the quantity

$$E\omega(A) := \inf_{t>0} t^{-1} \ln |||\exp(tA)|||_\alpha = \lim_{t\to\infty} t^{-1} \cdot \ln |||\exp(tA)|||_\alpha.$$

2.6.9. [178] Let $A \in \mathcal{G}(\mathbf{M}, \omega)$. Then $E\omega(A) \leq \omega(A)$.

2.6.10. [178] Let $A \in \mathcal{G}(\mathbf{M}, \omega)$. Then (see 6.8)

$$Er(\exp(tA)) = \exp(E\omega(A)t) \text{ при всех } t \geqslant 0.$$

2.6.11. [178] We set $S_1(A) := \sup\{\operatorname{Re}\lambda : \lambda \in \sigma(A) \setminus E\sigma(A)\}$. Then

(i) $\sup\{\operatorname{Re}\lambda : \lambda \in E\sigma(A)\} \leqslant E\omega(A)$;
(ii) $\omega(A) = \max\{S(A), E\omega(A)\} = \max\{S_1(A), E\omega(A)\}$.

2.6.12. [101] Let $A \in \mathcal{G}(\mathbf{M}, \omega)$. If infinity is an isolated sigular point of the resolvent $R(\lambda, A)$, then it is an essential singularity.

If $A \in B(E)$, then we have

$$\exp(t\sigma(A)) = \sigma(\exp(tA)) \setminus \{0\}, \tag{6.3}$$

which is called the spectral mapping theorem for operator semigroups. Here $\exp(t\Omega) := \{\lambda \in \mathbf{C} : z \in \Omega : \lambda = \exp(tz)\}$.

2.6.13. [126] For an OS exp(tA) Eq. (6.3) holds if, for example,

(i) the OS exp(tA) is compact;

(ii) the OS exp(tA) becomes differentiable;

(iii) the OS exp(tA) is analytic;

(iv) the OS exp(tA) becomes uniformly continuous.

2.6.14. [126] If $\|\exp(tA)\| \leq C \cdot t^k$ for a group exp(tA) for some $C > 0$, $k \in \mathbf{N}$, and all $t \in \mathbf{R}$, then we have the following relationship, which is weaker than (6.3):

$$\sigma(\exp(tA)) = \overline{\exp(t\sigma(A))} \text{ for all } t \in \mathbf{R}.$$

2.6.15. [126] For an OS exp(tA) we have the inclusion

$$\exp(t\sigma(A)) \subseteq \sigma(\exp(tA)) \text{ for } t \geqslant 0.$$

In more detail,

(i) $\exp(tP\sigma(A)) = P\sigma(\exp(tA)) \setminus \{0\}$;
(ii) $\exp(tR\sigma(A)) = R\sigma(\exp(tA)) \setminus \{0\}$;
(iii) $\exp(tA\sigma(A)) \subseteq A\sigma(\exp(tA))$;
(iv) $\exp(tE\sigma(A)) \subseteq E\sigma(\exp(tA))$;
(v) $\exp(tEF\sigma(A)) \subseteq EF\sigma(\exp(tA))$.

2.6.16. [126] Let $A \in \mathcal{G}(\mathbf{M}, \omega)$. Then for all $t \in \overline{\mathbf{R}}_+$

(i) if $\lambda \in C\sigma(A)$ and none of the numbers $\lambda_k = \lambda + 2\pi ik/t$, $k \in \mathbf{Z}$, belongs to $P\sigma(A) \cup R\sigma(A)$, then $\exp(t\lambda) \in C\sigma(\exp(tA))$;

(ii) if $\lambda \in R\sigma(A)$ and none of the numbers λ_k, $k \in \mathbf{Z}$, belongs to $P\sigma(A)$, the $\exp(t\lambda) \in R\sigma(\exp(tA))$;

(iii) if $\exp(t\lambda) \in R\sigma(\exp(tA))$, then none of the λ_k, $k \in \mathbf{Z}$, belongs to $P\sigma(A)$ and there exists a $\tilde{k} \in \mathbf{N}$ such that $\lambda_{\tilde{k}} \in R\sigma(A)$;

(iv) if $\lambda \in P\sigma(A)$, then $\mu := \exp(t\lambda) \in \sigma(\exp(tA))$. If, however, $\mu \in P\sigma(\exp(tA))$, then there exists a $\tilde{k} \in \mathbf{Z}$ such that $\lambda_{\tilde{k}} \in P\sigma(A)$ and $\mathcal{N}((\mu I - \exp(tA))^l) = \bigvee_{k=-\infty}^{\infty} \mathcal{N}((\lambda_k I - A)^l)$ for all integers $l \geq 1$. Here $\bigvee_{k=-\infty}^{\infty} \Omega_k$ is the smallest closed subspace containing all of the subsets Ω_k.

2.6.17. [178] Assertion (6.16) (i) cannot be reversed, or, more accurately, it is possible that $\exp(t\lambda) \in C\sigma(\exp(tA))$ at the same time as $\lambda + 2\pi ik/t \in \rho(A)$, $k \in \mathbf{Z}$.

2.6.18. [126] Let E = H be a Hilbert space and $A \in \mathcal{G}(\mathbf{M}, \omega)$. Then for each $t \in \tilde{\mathbf{R}}_+$ the set $\sigma(\exp(tA))\backslash\{0\} =$ $\{\exp(t\lambda)$: either $\lambda_k = \lambda + \frac{2\pi i k}{t} \in \sigma(A)$ for some $k \in \mathbf{Z}$ or the sequence $\| R(\lambda_k, A) \|_{k \in \mathbf{Z}}$ is unbounded$\}$.

2.6.19. [126, 178] For an OS exp(tA) we have

(i) $\sigma(\exp(tA)) = \sigma(\exp(tA)^\circ) = \sigma(\exp(tA)^*)$ for all $t \geq 0$;

(ii) $R\sigma(\exp(tA)) = P\sigma(\exp(tA)^\circ) = P\sigma(\exp(tA)^*) \subseteq R\sigma(\exp(tA)) \cup P\sigma(\exp(tA))$ for all $t \in \tilde{\mathbf{R}}_+$;

(iii) $\sigma(A) = \sigma(A^\circ) = \sigma(A^*)$;

(iv) $R\sigma(A) = P\sigma(A^\circ) = P\sigma(A^*) \subseteq R\sigma(A) \cup P\sigma(A)$;

(v) $S(A) = S(A^*)$ and $\omega(A) = \omega(A^*)$.

2.7. Stable Operator Semigroups.

2.7.1. [101] There is a well-known classical example in which $\sigma(A) = \varnothing$ and $\| \exp(tA) \| = \exp(\pi t/2)$, $t \in \tilde{\mathbf{R}}_+$.

2.7.2. [424] When E = H is a Hilbert space, for any real $a < b$ there exists an OS exp(tA) such that $S(A) = a$ and $\| \exp(tA) \| = \exp(bt)$, $t \geq 0$.

These examples indicate that there is a difference between the finite and infinite dimensional cases when the operator A is represented as a matrix. We set

$$\omega(x, A) := \inf \{\omega \in \mathbf{R} : \|\exp(tA)x\| \leqslant \mathbf{M} \exp(\omega t) \quad \text{for some} \quad \mathbf{M} \text{ and } t \geqslant 0\}$$

is the exponential growth bound for the function exp(tA)x;

$$\omega_1(A) := \sup \{\omega(x, A) : x \in D(A)\}$$

is the exponential growth bound for the solution of Cauchy problem (1.2.1) with n = 1.

It is clear that $\omega(A) = \sup\{\omega(x,A) : x \in E\}$.

2.7.3. [126] For any $x \in E$ and any OS exp(tA) we have

$$\omega(x, A) = \lim_{t \to \infty} t^{-1} \ln \| \exp(tA)x \|;$$

$$\omega(x, A) = \inf \left\{ \operatorname{Re}\lambda : \int_0^\infty \exp(-\lambda t) \| \exp(tA)x \| \, dt \quad \text{exists} \right\}$$

$$\omega_1(A) = \inf \left\{ \operatorname{Re}\lambda : \int_0^\infty \exp(-\lambda t) \exp(tA)x \, dt \quad \text{exists as an improper integral for any } x \in E \right\} =$$

$$= \inf \left\{ \operatorname{Re}\lambda : \int_0^\infty e^{-\lambda t} \| \exp(tA)x \| \, dt \quad \text{exists for any} \quad x \in D(A) \right\}.$$

If $\mathcal{N}(A) = \{0\}$, then, for $x \in D(A)$ we have $\omega(x, A) = \inf \{\operatorname{Re}\lambda : \int_0^\infty e^{-\lambda t} \exp(tA) Ax \, dt$ exists as an improper integral$\}$.

2.7.4. [126] It is clear that $S(A) \leq \omega_1(A) \leq \omega(A)$, and there are examples in which both inequalities are simultaneously strict.

Definition 1. An Os exp(tA) is said to be

a) stable if $\| \exp(tA)x \| \to 0$ as $t \to \infty$ for all $x \in D(A)$;

b) uniformly stable if a) is satisfied for all $x \in E$;

c) exponentially stable if $\omega_1(A) < 0$;

d) uniformly exponentially stable if $\omega(A) < 0$.

2.7.5. [126] Let $A \in B(E)$ and $S(A) < 0$. Then an OS exp(tB) is uniformly exponentially stable. In the case of an unbounded generating operator this is generally false.

2.7.6. [126] The following statements are equivalent:

(i) the OS exp(tA) is stable;

(ii) the subspace $\mathcal{N}(A)=\{0\}$ and the integral $\int_0^\infty \exp(tA)x\,dt$ exist for all $x \in D(A)$.

2.7.7. [126]. The following assertions are equivalent:

(i) the OS exp(tA) is stable and $A \in \mathcal{G}(\mathbf{M}, 0)$;

(ii) the OS exp(tA) is uniformly stable;

(iii) $A \in \mathcal{G}(\mathbf{M}, 0)$, and there exists a subset $\tilde{D} \subset E$ that is dense in E and such that integral $\int_0^\infty \exp(tA)x\,dt$ exists for all $x \in \tilde{D}$.

We write

$$b(x) := \inf\left\{\operatorname{Re}\lambda : \int_0^\infty e^{-\lambda t}\exp(tA)x\,dt \quad \text{exists} \right\}.$$

2.7.8. [332]. Given an OS exp(tA) and $x \in D(A)$, we have

(i) $\omega(x,A) = \max(b(x),b(Ax))$;

(ii) if $b(Ax) \geq 0$, then $0 \leqslant \omega(x, A) \leqslant b(Ax) = \inf\{\omega \in \mathbf{R} : \|\exp(tA)x - x\| \leqslant \mathbf{M}e^{\omega t}$ for some M and all $t \in \tilde{\mathbf{R}}_+\}$.

(iii) if $\mathcal{N}(A) = \{0\}$, then $\omega(x,A) = b(Ax)$;

(iv) if $b(Ax) < 0$ then $b(Ax) = \inf\{\omega \in \mathbf{R} : \|\exp(tA)x - y\| \leqslant \mathbf{M}e^{\omega t}$ for some M and $t \in \tilde{\mathbf{R}}_+\}$, where $y \in \mathcal{N}(A)$. Moreover, if $\mathcal{N}(A) = \{0\}$, then $\omega(x,A) < 0$.

Definition 2. For $x \in E$ we introduce the following notation:

$\gamma(x,A) := \inf\{a \in \mathbf{R}$: the function $\lambda \to R(\lambda,A)\chi$ has a holomorphic continuation to the half-plane $\{\lambda \in \mathbf{C} : \operatorname{Re}\lambda \geq a\}\}$,

$\gamma_0(x,A) := \inf\{a \in \mathbf{R} : a > \gamma(x,A)$ and the function $R(\lambda,A)x$ is bounded in the half-plane $\{\lambda \in \mathbf{C} : \operatorname{Re}\lambda \geq a\}\}$;

$\gamma_1(x,A) := \inf\{a \in \mathbf{R} : a > \gamma(x,A)$ and the function $R(\lambda,A)x$ uniformly converges to zero with respect to $\operatorname{Re}\lambda$ as $|\operatorname{Im}\lambda| \to \infty$ throughout the half-plane $\{\lambda \in \mathbf{C} : \operatorname{Re}\lambda \geq a + \varepsilon\}, \varepsilon > 0\}$.

2.7.9. [332] For any $x \in E$ we have $\gamma(x,A) \leq \gamma_0(x,A) \leq \gamma_1(x,A) \leq \omega(A)$. If, however, $x \in D(A^2)$, then $\gamma_1(x,A) \leq \omega(x,A) \leq \max\{\gamma(x,A),\gamma_0(A^2x,A)\}$.

2.7.10. [332] Assume that we are given a positive uniformly bounded OS exp(tA) in a space E with a normal reproducing cone. If $\lim_{\lambda \to 0^+} R(\lambda, A)x$ exists for all $x \in D(A)$, then the OS is uniformly stable.

2.7.11. [126] An OS exp(tB) is uniformly exponentially stable if and only if $S(A) < 0$ and there exists a t_0 such that $A\sigma(\exp(t_0A))$ is inside the unit circle.

2.7.12. [126] The following statements are equivalent:

(i) $\omega(A) < 0$;

(ii) $\lim_{t \to \infty} \|\exp(tA)\| = 0$;

(iii) for some $t_0 > 0$ we have $\|\exp(t_0A)\| < 1$;

(iv) for any (some) $p \geq 1$ the integral $\int_0^\infty \|\exp(tA)x\|^p dt$ exits for any $x \in E$.

2.7.13. [126] Given an OS exp(tA) that is stable. Then

(i) $S(A) \leq 0$ and $\operatorname{Re}\lambda < 0$ for any $\lambda \in P\sigma(A) \cup R\sigma(A)$;

(ii) the limit $\lim_{\lambda \to 0+} \lambda R(\lambda, A)x$ exists for any $x \in D(A)$.

2.7.14. [126] Assume that exp(tA) is a uniformly bounded analytic OS. Then the following statements are equivalent:

(i) $0 \notin P\sigma(A) \cup R\sigma(A)$;

(ii) the OS exp(tA) is uniformly stable.

2.7.15. [377] Let $S(A) \le -\delta < 0$. Then

(i) if $\|R(\lambda,A)\| = O(|\lambda|^k)$ as $|\mathrm{Im}\ \lambda| \to \infty$, $\mathrm{Re}\ \lambda \ge -\delta$, and $k \ge -2$, then there exist $0 < \mu < \delta$ and $g(\mu) > 0$ such that $\|A^{-(k+2)}\exp(tA)x\| \le g(\mu)e^{-\mu t}\|x\|$ for any $x \in E$, $t \ge 0$;

(ii) if (i) is satisfied for $k \ge -1$, then there exists a $\mu \in (0,\delta)$ such that

$$\int_0^\infty |e^{\mu t} \langle A^{-(k+1)} \exp(tA)x, x^* \rangle|^2 dt \le K(\mu, x, x^*)$$

for all $x \in E$, $x^* \in E^*$.

2.7.16. [126] Assume we are given a positive OS exp(tA) and $D(A)_- := -D(A)_+$. Then $\omega_1(A) < 0$ if and only if $E_+ \subset AD(A)_-$.

2.7.17. [126] Assume we are given an OS exp(tA) that is positive and bounded. If a subset $\mathcal{F} \subset E_+$ that is total in E is such that $\lim_{\lambda \to 0+} R(\lambda, A)x$ for all $x \in \mathcal{F}$, then the OS exp(tA) is uniformly stable.

2.7.18. [377] Assume that $S(A) \le -\delta < 0$ and exp(tA) becomes differentiable. Then there exist constants $\mu \in (-\delta,0)$ and $K = K(\mu)$ such that $\|(\exp(tA)\| \le K \exp(-\mu t)$ for $t \ge 0$.

2.7.19. [124] Assume that E is reflexive, $A \in \mathcal{G}(\mathbf{M}, 0)$, $P\sigma(A) \cap i\mathbf{R} = \varnothing$, and the set $\sigma(A) \cap i\mathbf{R}$ is countable. Then the OS exp(tA) is uniformly stable.

2.7.20. [124] There are examples in which an OS exp(tA) is uniformly stable, $P\sigma(A) \cap i\mathbf{R} = \varnothing$, but $\sigma(A) \cap i\mathbf{R} \ne \varnothing$.

2.7.21. [124] There is an example of an OS that is not uniformly stable in a reflexive E but is such that $P\sigma(A) = \varnothing$ and $\sigma(A) \subseteq i\mathbf{R}$.

2.7.22. [124] Assume $A \in \mathcal{G}(\mathbf{M}, 0)$, $R\sigma(A) \cap i\mathbf{R} = \varnothing$, and the set $\sigma(A) \cap i\mathbf{R}$ is countable. Then the OS exp(tA) is uniformly stable.

2.7.23. [124] 7.21 is precise in the following sense:

(i) there exists a unitary group exp(tA) such that $\sigma(A) \subseteq i\mathbf{R}$, $R\sigma(A) = \varnothing$ and at the same time this group is unstable; more precisely, the condition $\exp(tA)x \to O$ as $t \to \infty$ implies $x = O$;

(ii) there exists an OS exp(tA) such that $\omega(A) = 0$ and $\mathrm{Re}\ \lambda < 0$ for $\lambda \in \sigma(A)$, but the exp(tA) is not uniformly stable;

(iii) let $E = [C,\infty)$ and $(\exp(tA)f)(x) = f(t + x)$. Then $\sigma(A) = \{\lambda \in \mathbf{C} : \mathrm{Re}\ \lambda \le 0\}$ and the OS exp(tA) is uniformly stable.

2.7.24. [124] Assume that E is reflexive, $A \in \mathcal{G}(\mathbf{M}, 0)$, and the set $\sigma(A) \cap i\mathbf{R}$ is countable. Then E is the direct sum of invariant subspaces E_{st} and E_{ap}, where $E_{st} := \{x \in E : \lim_{t\to\infty} \exp(tA)x = O\}$, $E_{ap} := \mathrm{span}\{x \in D(A): Ax = \lambda x \ \ \lambda \in i\mathbf{R}\}$. Moreover, the restriction of the OS exp(tA) to E_{ap} can be imbedded in a group on E_{ap}.

2.7.25. [124] Given an OS exp(tA) that becomes continuous with respect to norm, if $\mathrm{Re}\ \lambda < 0$ for all $\lambda \in \sigma(A)$, then the OS is uniformly stable.

2.7.26. [126] Assume that a positive OS exp(tA) is uniformly bounded and becomes continuous with respect to norm. Then the following conditions are equivalent:

(i) the OS exp(tA) is uniformly stable;

(ii) $0 \notin R\sigma(A)$, i.e., $\mathcal{N}(A^*) = \{O\}$;.

When E is reflexive, conditions (i) and (ii) are equivalent to

(iii) $0 \notin P\sigma(A)$, i.e., $\mathcal{N}(A) = \{O\}$.

2.7.27. [124] If, in 7.22, condition $A \in \mathcal{G}(\mathbf{M}, 0)$ is replaced by assumptions $S(A) \le 0$ and $\sup_{t\ge 0} \|\exp(tA)B\| < \infty$, where $B \in B(E)$ is an operator that commutes with OS exp(tA), then the assertion is replaced by $\lim_{t\to\infty} \|\exp(tA) \cdot Bx\| = 0$ for any $x \in E$.

2.7.28. [124] Let $\operatorname{Re}\lambda < 0$ for all $\lambda \in \sigma(A)$ and $\sup_{t\geqslant 0} \| \exp(tA)Ax \| < \infty$ for some $x \in D(A)$. Then the OS exp(tA) is stable.

2.7.29. [311] Assume we are given an OS exp(tA) that is uniformly bounded, the set $\sigma(A) \cap i\mathbf{R}$ is countable, and the adjoint A^* is such that $P\sigma(A^*) \cap iR = \varnothing$. Then the OS exp(tA) is uniformly stable.

2.7.30. [311] Let exp(tA) be a uniformly bounded group and $\sigma(A) \cap iR = \varnothing$. Then it is uniformly stable.

Let $A \in \mathscr{G}(E)$, $m \in \mathbf{N} \cup \{0\}$. For $\alpha \in \rho(A)$ we set

$$\omega_m(A) := \inf\{\omega \in \mathbf{R} : \sup_{t\geqslant 0} \| e^{-\omega t} \exp(tA) R(\alpha, A)^m \| < \infty\}$$

and

$$\Xi_m(A) := \inf\{s > \omega(A) : \|R(a+ib, A)\| = O(|b|^m) \quad \text{for } b \to \infty \text{ and } a \geqslant s\}.$$

2.7.31. [378, 407] Given an OS exp(tA) in a complex Banach space E. Then for any $m \in N \cup \{0\}$ (see 7.15)

(i) $\omega_{m+2}(A) \leq \Xi_m(A)$;

(ii) if E = H is a Hilbert space, then $\omega_m(A) \leq \Xi_m(A)$;

(iii) the relation $\omega_{m+1}(A) \leq \Xi_m(A)$ is satisfied for a broad class of spaces E including, in particular, Lebesgue, Besov, and Sobolev spaces.

2.7.32. [378] There are examples in which

(i) $\omega_{m+1}(A) < \omega_m(A)$ and $S(A) < \Xi_{m+1}(A) < \Xi_m(A)$, $m \in N \cup \{0\}$.

(ii) E is uniformly convex and $\omega_1(A) < \Xi_0(A) < \omega_0(A)$.

Definition 3. An OS exp(tA) in a Hilbert space H is said to be weakly L^p stable if for any $x, y \in H$ we have

$$\int_0^\infty |(\exp(tA)x, y)|^p \, dt < \infty.$$

2.7.33. [418] Weak L^p stability of an OS for some $p \in [1,\infty)$ implies weak stability for the OS for any $p \in [1,\infty)$.

2.7.34. [418] An OS exp(tA) defined in H is weakly L^p stable for some $p \in [1,\infty)$ if and only if it is uniformly exponentially stable.

2.7.35. [418] If there exists a $p \in [1,\infty)$ such that for all $x \in E$

$$\int_0^\infty \| \exp(tA)x \|^p \, dt < \infty, \tag{7.1}$$

then the OS exp(tA) is uniformly exponentially stable.

2.7.36. [418] An OS exp(tA) defined in H is uniformly exponentially stable if and only if for some $p \in [1,\infty)$ inequality (7.1) is satisfied.

2.7.37. [418] In general, weak L^p stability does not imply uniform exponential stability.

2.7.38. [126] Given an OS exp(tA) that is positive. Then, if either

(i) the operator $(\lambda I - A)$ is invertible for some $\lambda \geq 0$, or

(ii) the operator A is invertible and $A^{-1} \leq O$;

then the OS is exponentially stable.

2.7.39. [418] With $1 \leq p \leq \infty$ we have, for all $x \in E$ and $y^* \in E^*$,

$$\left(\int_0^\infty |\langle \exp(tA)x, y^* \rangle|^p \, dt \right)^{1/p} \leqslant \mathbf{M}_0 \| x \| \cdot \| y^* \| \tag{7.2}$$

with some constant $M_0 \geq 0$ if the integral on the left exists.

2.7.40. [418] Given the conditions and notation of 7.38 with $p^{-1} + q^{-1} = 1$, then for all $\varphi(t) \in L^q[0,\infty)$ and $x \in E$ the limit

$$\xi(\varphi) := \lim_{T\to\infty} \int_0^T \varphi(t)\exp(tA)x\,dt$$

exists and $\|\xi(\varphi)\| \leqslant M_0\|x\| \cdot \|\varphi\|_{L^q[0,\infty)}$. This statement holds when the space E is weakly sequentially complete (in particular, when it does not contain c_0).

2.7.41. [418] Assume that (7.2) is satisfied and $1 \leq p \leq \infty$. Then, for any $z \in \mathbf{C}$ with $\operatorname{Re} z > 0$ we have $z \in \rho(A)$ and for any $x \in E$ we have

$$R(z, A)x = \lim_{T\to\infty} \int_0^T e^{-zt}\exp(tA)x\,dt.$$

2.7.42. [418] Under the conditions of 7.40 and 7.39 we have

$$\|R(z, A)\| \leqslant \begin{cases} M_0(q)^{-1/q}(\operatorname{Re} z)^{-1/q} & \text{for } p > 1, \\ M_0 & \text{for } p = 1. \end{cases}$$

2.7.43. [418] Under the conditions of 7.40 and 7.39, but with $p < \infty$, we have

$$S(A) \leqslant -\delta = -1/(pM_0)^p.$$

For $z \in \mathbf{C}$ with $-\delta < \operatorname{Re} z \leq -\delta + M_0^{-p}$ we have

$$\|R(z, A)\| \leqslant (\operatorname{Re} z + \delta)^{-1}.$$

Definition 4. We define $H^\infty := \{f(z) : f(z)$ is bounded and analytic when $\operatorname{Re} z > 0$ and has values in $E\}$.

2.7.44. [418] Let $E = H$ be a Hilbert space and $R(z,A) \in H^\infty$. Then the OS exp(tA) is uniformly exponentially stable.

2.7.45. [418] Under the conditions of 7.33 we have $\omega(A) \leq -1 / (pM)^p$ with constant $M^p := \sup\left\{\int_0^\infty |(\exp(tA)x, y)|^p\,dt : \|x\|, \|y\| \leqslant 1\right\}$.

2.7.46. [126] Let the OS exp(tA) be positive, uniformly bounded, quasicompact, such that $S(A) = 0$. Then there exist a positive projector P of finite rank and constants $\delta > 0$ and $M \geq 0$ such that $\|\exp(tA) - P\| \leq M\exp(-\delta t)$ for all $t \geq 0$.

2.7.47. [126] Assume that the OS exp(tA) is positive, uniformly bounded, and continuous with respect to norm; also, assume that the space E is reflexive. Then the limit $\hat{P}x = \lim_{t\to\infty} \exp(tA)x$ exists for any $x \in E$, where $\hat{P}$ is the positive projector onto $\mathcal{N}(A)$.

2.7.48. [126] Assume that under the conditions of 7.46 we also assume that p is the order of the pole $\lambda_0 = S(A)$ of the resolvent $R(\lambda,A)$. Let $P(\lambda_0)$ be the projector onto $\mathcal{N}((\lambda_0 I - A)^p)$ along $\mathcal{R}((\lambda_0 I - A)^p)$. Then,

(i) for any $\eta \in (0,\varepsilon)$ there exists a constant $M(\eta)$ such that $\|\exp(tA)(I - P(\lambda_0))\| \leq M(\eta)\exp((\lambda_0 - \eta)t)$ for $t \in \tilde{\mathbf{R}}_+$;

(ii) if $p = 1$, then for any $\eta \in (0,\varepsilon)$ we have

$$\|\exp(-\lambda_0 t)\exp(tA) - P(\lambda_0)\| \leqslant M(\eta)\exp(-\eta t), \quad t \in \bar{\mathbf{R}}_+;$$

(iii) if the OS exp(tA) is irreducible and $Ax_0 = \lambda_0 x_0$, $A^* x_0^* = \lambda_0 x_0^*$, $\langle x_0^*, x_0 \rangle = 1$, then for any $\eta \in (0,\varepsilon)$ and $x \in E$

$$\| \exp(-\lambda_0 t) \exp(tA) x - \langle x, x_0^* \rangle x_0 \| \leqslant \mathbf{M}(\eta) \cdot \exp(-\eta t), \quad t \in \bar{\mathbf{R}}_+.$$

Definition 5. An OS exp(tA) is said to be ergodic if $t^{-1} \int_0^t \exp(sA) x ds$ strongly converges as $t \to \infty$ for any $x \in E$.

2.7.49. [278] An OS exp(tA) is ergodic if and only if $E = \{x \in E : \exp(tA) x = x, \ t \in \bar{\mathbf{R}}_+\} \oplus \left\{x \in E : t^{-1} \int_0^t \exp(sA) \times x ds \to 0 \text{ as } t \to \infty\right\}$.

2.7.50. [278] If an OS exp(tA) is ergodic, the following conditions are equivalent:

(i) $y \in \mathcal{R}(A)$;

(ii) the limit $\lim_{\lambda \to 0+} R(\lambda, A) y$ exists;

(iii) the limit $\lim_{\alpha \to \infty} \alpha^{-1} \int_0^{\alpha} \int_0^{\beta} \exp(sA) y ds d\beta$ exists;

(iv) for some sequence $\{\alpha_n\}$ that becomes infinite, (iii) holds in the sense of a w-lim.

2.7.51. [278] Assume that an OS exp(tA) is such that $t^{-1} \| \exp(tA) \| \to 0$ as $t \to \infty$ and $\sup_{t \geqslant 0} t^{-1} \left\| \int_0^t \exp(sA) \times x ds \right\| \leqslant \mathbf{M} \| x \|$ for all $x \in E$ and some $\mathbf{M} < \infty$. Then each of conditions (i)-(iv) of 7.50 is equivalent to the condition $y \in \mathcal{R}(A)_{\overline{\mathcal{R}(A)} \cap \mathcal{D}(A)}$).

2.7.52. [278] Let $A \in \mathcal{G}(\mathbf{M}, 0)$. Then the condition $y^* \in \mathcal{R}(A^*)$ is equivalent to condition $\sup_{t>0} \left\| \int_0^t \exp(tA)^* y^* ds \right\| < \infty$.

2.8. Positive Semigroups of Operators.

Definition 1. A subset E_+ of a Banach space E is said to be a cone if

(i) E_+ is convex;

(ii) together with any $x \in E_+$ the cone E_+ contains the ray $\{tx: t \geq 0\}$;

(iii) E_+ contains no opposing elements;

(iv) E_+ is closed.

Definition 2. For two elements $x, y \in E$, the relation $x \leqslant y$ is equivalent to the condition $y - x \in E_+$.

Definition 3. A cone E_+ is said to be a reproducing cone if $E = E_+ - E_+$. In this case there exists a number $C > 0$ such that for every $x \in E$ we have $x = x_1 - x_2$, where $\| x_j \| \leq C \| x \|$ $(x_j \in E_+, j = 1,2)$.

A special case of a reproducing cone is provided by a solid cone, i.e., a cone containing interior points.

Definition 4. A cone is said to be normal if for some number C the relations $0 \leqslant x \leqslant y$ imply that $\| x \| \leq C \| y \|$. If $C = 1$, the cone is sharp.

Definition 5. By the supremum of a set $S \subseteq E$ (written sup S) we mean the least of all the upper bounds of S (if it exists).

Definition 6. A Banach structure (lattice) is a partially ordered Banach space (i.e., a space with a reproducing cone E_+) for which

(i) the cone E_+ is sharp;

(ii) for each pair of elements $x, y \in E$ their sup is defined.

Definition 7. Let E be a Banach structure, $x \in E$. By the modulus of an element x we mean $|x| := x^+ + x^-$, where $x^+ = \sup(x,O)$ is the positive part of the element x, and $x^- = \sup(-x,O)$ is the negative part of x.

Definition 8. A set $S \subset E$ is said to be order bounded if there exists an x such that $y \leqslant x$ for all $y \in S$.

Definition 9. A partially ordered space is said to be order complete if any order bounded set in it has a sup. It is said to be σ-order complete if any countable order bounded set has a sup.

Definition 10. Two elements x and y of a Banach structure E are said to be disjunctive if $\inf\{|x|,|y|\} = 0$.

Definition 11. Let $A, B \in B(E)$. We will say that $A \leqslant B$, if $Ax \leqslant Bx$ for all $x \in E_+$.

Definition 12. If $0 \leq A$ we will say that A is positive.

Definition 13. An OS exp(tA) defined in a partially ordered space E is said to be positive if

$$\exp(tA)E_+ \subset E_+ \quad \text{for any} \quad t \geqslant 0.$$

2.8.1. [126] Assume that an OS exp(tA) is positive and $x \leqslant y$. Then

$$\exp(tA)x \leqslant \exp(tA)y \quad \text{for any} \quad t \geqslant 0.$$

2.8.2. [178] Positivity of an OS exp(tA) is equivalent to satisfaction of any of the following conditions:

(i) the resolvent is positive for all $\lambda > \omega(A)$, i.e., $R(\lambda,A)E_+ \subset E_+$;

(ii) the relation $x \leqslant y$ implies $R(\lambda,A)x \leq R(\lambda,A)y$ for $\mathrm{Re}\,\lambda > \omega(A)$.

Definition 14. A functional $y^* \in E^*$ is said to be positive if $\langle x,y^*\rangle \geq 0$ for all $x \in E_+$. We denote the set of positive functionals on E by E_+^*. If the cone E_+ is normal, then E_+^* is a cone in E^*.

2.8.3. [125, 146] If an OS exp(tA) is positive, then $D(A)_+ := D(A) \cap E_+$ is dense in E_+ and $D(A^*)_+^{*w^*}$ is dense in E_+^*.

Definition 15. The spectral bound of an operator A is defined to be

$$S(A) := \sup\{\mathrm{Re}\,\lambda : \lambda \in \sigma(A)\}, \quad \text{where} \quad S(A) = -\infty, \quad \text{if} \quad \sigma(A) = \varnothing.$$

2.8.4. [244] If an OS exp(tA) is positive, the cone E_+ is normal and reproducing. Then, for $S(A) < \mathrm{Re}\,\mu \leq \omega(A)$ Eq. (1.2) holds, i.e.,

$$R(\mu, A)x = \int_0^\infty e^{-\mu t}\exp(tA)x\,dt,$$

where the integral exists as a nonsingular Riemann integral.

2.8.5. [178] Given an OS exp(tA) that is positive and a Banach structure E, then

$$R(\mu, A)x \leqslant R(\mathrm{Re}\,\mu, A)|x| \quad \text{for all} \quad x \in E \quad \text{and} \quad \mathrm{Re}\,\mu > S(A).$$

Definition 16. A linear subset $J \subset E$ is an ideal if it follows from $x \in E$, $y \in J$, and $|x| \leqslant |y|$ that $x \in J$.

Definition 17. A positive semigroup exp(tA) is said to be irreducible if {O} and E are the only closed ideals that are invariant under the family exp(tA) with $t \geq 0$.

Definition 18. If $x \geqslant O$, the smallest ideal containing x is called the principal ideal J_x corresponding to x.

Definition 19. If J_x is dense in E, then x is said to be quasi-interior.

Definition 20. An OS exp(tA) is said to be stricly irreducible if for any $x > O$, $t > 0$, all elements exp(tA)x are quasi-interior for the cone E_+.

2.8.6. [244] If E is a Banach structure, the following statements are equivalent:

(i) exp(tA) is irreducible;

(ii) R(λ,A) is strictly irreducible for all $\lambda > S(A)$;

(iii) R(λ,A) is irreducible for all $\lambda > S(A)$.

2.8.7. [178] An irreducible positive analytic OS is strictly irreducible.

2.8.8. [146, 178]. Assume that an OS exp(tA) is positive and analytic in a Banach structure E. If $O \leqslant x \in E$ and $O \leqslant x^* \in E^*$ are such that $\langle \exp(t_0 A)x, x^* \rangle = 0$ for some $t_0 > 0$, then

$$\langle \exp(tA)x, x^* \rangle = 0 \quad \text{for any} \quad t \in \bar{\mathbf{R}}_+.$$

Definition 21. We say that a functional p(x) defined on a vector space E is sublinear if

$$\text{(i)} \quad p(x+y) = p(x) + p(y) \quad \text{for } x, y \in E;$$
$$\text{(ii)} \quad p(\lambda x) = \lambda p(x) \quad \text{for } x \in E, \ \lambda \geqslant 0.$$

Definition 22. For a partially ordered Banach space E we define the canonical functional with respect to the given partial ordering to be the functional

$$p_+(x) := \inf\{\|x+y\| : \ y \in E_+\}. \tag{8.1}$$

2.8.9. [178] Let E be a partially ordered Banach space. Then the canonical functional is sublinear and $p_+(x) + p_+(-x) > 0$. If E is a Banach structure, then $p_+(x) = \|x^+\|$.

Definition 23. Assume that in a Banach space E the functional p(x) is sublinear and continuous. An operator $A \in L(E)$ is said to be p-dissipative if $p(x) \leq p(x - tAx)$ for all $t \geq 0$ and $x \in D(A)$.

Definition 24. Let p be a sublinear functional. A mapping $T \in L(E)$ is said to be a p-contraction if $p(Tx) \leq p(x)$ for all $x \in E$.

2.8.10. [178] In a partially ordered Banach space any p_+-contraction is a positive operator, and any positive contraction is a p_+-contraction.

2.8.11. [178] In a partially ordered Banach space any OS exp(tA) is a p_+-contraction for all $t \geq 0$ if and only if its generator is p_+-dissipative.

2.8.12. [178] Let E_+ be normal. In order for a linear operator A to be the generator of a P_+-contraction OS it is necessary and sufficient for the following conditions to be satisfied:

(i) A is densely defined;

(ii) A is P_+-dissipative;

(iii) $\mathcal{R}(\lambda I - A) = E$ for some $\lambda > 0$.

Definition 25. An operator norm in B(E) is said to be positively attained if

$$\|B\| = \sup\{\|Bx\| : \ O \quad x \in E; \ \ \|x\| \leqslant 1\}$$

for every positive operator $B \in B(E)$.

2.8.13. [178] Assume that a cone E_+ is sharp and the norm in B(E) is positively attained. In order for A to be a generator of a positive contraction semigroup it is necessary and sufficient that

(i) A be densely defined;

(ii) A be P_+-dissipative;

(iii) $\mathcal{R}(\lambda I - A) = E$ for some $\lambda > 0$.

2.18.14. [178] In a Banach structure E the following statements are equivalent:

(i) A is the generator of a positive contraction semigroup;

(ii) A is densely defined, $\mathcal{R}(\lambda I - A) = E$ for some $\lambda > 0$, and A is p_+-dissipative.

2.8.15. [146, 178] Assume that a cone E_+ is sharp, and the norm in B(E) is positively attained. Then an operator $A \in \mathcal{G}(\mathbf{M}, \omega)$ and the OS exp(tA) are positive if and only if $A \in \mathcal{G}(\mathbf{E})$, $(\omega, \infty) \in \rho(A)$, and

$$p_+[R(\lambda, A)^n x] \leqslant \frac{\mathbf{M}}{(\lambda - \omega)^n} p_-(x) \tag{8.2}$$

for all $x \in E$, $n \in N$.

2.8.16. [146, 178] For an arbitrary Banach structure Proposition 6.15 remains valid when (8.2) is replaced by

$$\|[R(\lambda, A)^n x]^+\| \leqslant \frac{M}{(\lambda-\omega)^n}\|x^+\| \tag{8.3}$$

for all $x \in E$, $n \in \mathbf{N}$.

Definition 26. An operator $A \in \mathcal{G}(M, \omega)$ that is defined on a Banach structure E satisfies the positive minimum principle if, for all $x \in D(A)_+$ and $\varphi \in E_+^{\sim} := E_+^* \cap E^{\odot}$ the equation $\langle x, \varphi\rangle = 0$ implies the inequality $\langle Ax, \varphi\rangle \geq 0$.

2.8.17. [178, 206] The generator of a positive OS satisfies the positive minimum principle.

2.8.18. [162] Let $A \in \mathcal{G}(M, \omega)$ and let there exist a core $\mathcal{D}_0$ of operator A such that $x \in \mathcal{D}_0$ implies $|x| \in \mathcal{D}_0$. The satisfaction of the positive minimum principle for A implies that the OS exp(tA) is positive.

2.8.19. [121] Let Δ be the Laplace operator. Then for any $f \in L_1^{loc}(\mathbf{R}^d)$ such that $\Delta f \in L_1^{loc}(\mathbf{R}^d)$ we have

$$\mathrm{Re}[\mathrm{sign}\, f \cdot \Delta f] \leqslant \Delta |f|. \tag{8.4}$$

Definition 27. For each $x \in E$ we define the linear operator sign_x by the properties:

(i) $\mathrm{sign}_x x = |x|$

(ii) $|\mathrm{sign}_x y| \leqslant |y|$ for all $y \in E$;

(iii) $\mathrm{sign}_x y = 0$ if $\inf\{|x|, |y|\} = 0$.

2.8.20. [126, 178] In a σ-order complete structure, relative to any $x \in E$ the operator sign_x is uniquely defined.

2.8.21. [121, 363] Let $A \in \mathcal{G}(M, 0)$. Then the following conditions are equivalent:

(i) the OS exp(tA) is positive;

(ii) for all $x \in D(A)$ and all $0 \leqslant \varphi \in D(A^{\odot})$

$$\langle \varphi, \mathrm{sign}_{x^{**}}(Ax)^{**} \rangle \leqslant \langle |x|, A^{\odot}\varphi \rangle$$

for each $\lambda > 0$ there exists a set $K_\lambda \subset D(A^{\odot})_+$ such that $(\lambda I - A^{\odot})K_\lambda$ separates $E_+\backslash\{0\}$ from 0, i.e., for each $0 \leqslant x \in E$ there exists a $0 \leqslant \varphi \in D(A^{\odot})$ such that

$$\langle x, (\lambda I - A^{\odot})\varphi \rangle > 0.$$

2.8.22. [363] Assume that under the conditions of 8.21 E is also Dedekind complete (i.e., any majorizable subset $S \subset E$ has $(\sup S) \in E$). Then the following conditions are equivalent:

(i) the OS exp(tA) is positive;

(ii) for each $x \in D(A)$ and $0 \leq \varphi \in D(A)^*$ we have

$$\langle \mathrm{sign}_x(Ax), \varphi\rangle \leqslant \langle |x|, A^*\varphi\rangle,$$

and for each $\lambda > 0$ there exists a set $K_\lambda \in D(A^*)_+$ such that $(\lambda I - A^*)K\lambda$ separates $E_+\backslash\{0\}$ from 0.

Definition 28. A linear operator A in a σ-order complete Banach structure satisfies Kato's inequality if

$$\langle \mathrm{sign}_x Ax, \varphi\rangle \leqslant \langle |x|, A^*\varphi\rangle \tag{8.5}$$

for all $x \in D(A)$ and $0 \leq \varphi \in D(A^*)$.

2.8.23. [120, 121] The generator of a positive OS in a σ-order complete structure satisfies the Kato inequality (8.5).

In order to state sufficient conditions we introduce the following additional constructions.

Definition 29. A set $S \subset E_+^{\odot}$ is said to be strictly positive if for any $x \in E_+$ it follows from $\langle x, \varphi\rangle = 0$ $(\varphi \in S)$ that $x = 0$.

Definition 30. A function φ is said to be strictly positive if the set $\{\alpha\varphi : \alpha \in \tilde{\mathbf{R}}_+\}$ is strictly positive.

Definition 31. An element $x \in E$ is said to be a positive subvector of an operator $A \in L(E)$ if $0 \leqslant x \in D(A)$ and $Ax \leqslant \lambda x$ for some $\lambda \in \mathbf{R}$.

2.8.24. [178] If exp(tA) is a positive OS in a Banach structure E, there exists a strictly positive subset $K \subset A$ consisting of positive subvectors of A.

2.8.25. [178] If there is a strictly positive functional on E, then A^* has a positive subvector.

2.8.26. [178] Let exp(tA) be an OS in a σ-order complete Banach structure. Then the following conditions are equivalent:

(i) the OS exp(tA) is positive;

(ii) the operator A satisfies the Kato inequality and A^* has a strictly positive set of positive subvectors.

2.8.27. [178] If there is a strictly positive functional defined on E under the assumptions of 8.26, then positiveness of the OS exp(tA) is equivalent to the existence of a subvector φ of A^* for which inequality (8.5) is satisfied.

Definition 32. A linear operator has positive outer diagonal property if $\langle x,\varphi\rangle = 0$ for $0 \leqslant x \in D(A)$, $0 \leqslant \varphi \in E^*$, implies that $\langle Ax,\varphi\rangle \geq 0$.

2.8.28. [125, 146] Assume that the cone E_+ is solid. Then an OS exp(tA) is positive if and only if A has the positive outer diagonal property and $R(\lambda, A) \geqslant O$ for sufficiently large real λ.

2.8.29. [125, 146] For a space with a normal solid cone and a densely defined operator $A \in \mathcal{G}(\mathbf{M}, \omega)$ the following conditions are equivalent:

(i) A is the generator of a positive OS;

(ii) A has the positive outer diagonal property and $\mathcal{R}(\lambda I - A) = E$ for sufficiently large $\lambda \in \mathbf{R}$;

(iii) $R(\lambda, A) \geqslant O$ for sufficiently large $\lambda \in \mathbf{R}$.

2.8.30. [125, 146] Conditions (i)-(iii) of 8.29 imply $S(A) = \omega(A) = \inf\{\lambda \in \mathbf{R} : Ax \geq \lambda x$ for some $x \in D(A) \cap \operatorname{int} E_+\}$, where S(A) is the spectral bound of an OS.

2.8.31. [178] If an OS exp(tA) is positive in a Banach structure $L^1(\Omega,\mu)$ or $L^2(\Omega,\mu)$ then $s(A) = \omega(A)$.

2.8.32. [243, 244] Assume that exp(tA) is a positive OS in a Banach structure E. Then

(i) if $\sigma(A) \neq \emptyset$, then S(A) is a point of the spectrum;

(ii) If S(A) is a pole of order m of the resolvent $R(\lambda,A)$ and $\tilde{\lambda}$ is another pole with $\operatorname{Re} \tilde{\lambda} = S(A)$, then its order may not exceed m;

(iii) all of the numbers $S(A) + ik \cdot \operatorname{Im} \tilde{\lambda}$ ($k \in \mathbf{Z}$) appear in $\sigma(A)$;

(iv) if exp(tA) become compact, there exists an $\varepsilon > 0$ such that in the strip $S(A) - \varepsilon < \operatorname{Re} \lambda < S(A)$ there are no points of the spectrum of A;

(v) if exp(tA) is irreducible and S(A) is a pole of the resolvent, then S(A) is a characteristic number of algebraic multiplicity one and the corresponding characteristic vector is a quasi-interior element;

(vi) the spectral bound $B\sigma(A)$ (i.e., the part of $\sigma(A)$ on the line $S(A) + i\xi$, where $\xi \in \mathbf{R}$) under the assumptions of (v) consists of characteristic numbers of algebraic multiplicity one of the form $S(A) + i\nu k$ with fixed $\nu \geq 0$ and $k \in \mathbf{Z}$.

2.8.33. [178] If $S(A) < 0$ under the assumptions of 8.32, then there exists an $\omega > 0$ such that for any $x_0 \in D(A)$ we have

$$\|\exp(tA)x_0\| \leqslant \mathbf{M}_0 \exp(-\omega t),$$

where $\mathbf{M}_0$ may depend on x_0.

2.8.34. If an OS exp(tA) is positive in a Banach structure E, then $S(A) = \omega_1(A)$ (see 2.7).

Definition 33. A positive OS exp(tA) majorizes (or dominates) an OS exp(tG) if $|\exp(tG)x| \leqslant \exp(tA)|x|$ for all $x \in E$, $t \geq 0$.

2.8.35. [126] An OS exp(tG) is majorized by an OS exp(tA) if and only if $|R(\lambda, G)x| \leqslant R(\lambda, A)|x|$ for all $x \in E$ and large real λ.

2.8.36. [126] A semigroup exp(tG) is majorized by exp(tA) if and only if the generalized Kato inequality is satisfied:

$$\operatorname{Re}\langle \operatorname{sign}_x Gx, \varphi\rangle \leqslant \langle |x|, A^{\odot}\varphi\rangle \quad \text{for all} \quad x \in D(G), \tag{8.6}$$

$\varphi \in D(A^{\odot})_+$ and for each $\lambda > 0$ there exists a set $K_\lambda \subset D(A^{\odot})_+$ such that $\langle x,(\lambda I - A^{\odot})\varphi\rangle \geq 0$ for all $\varphi \in K_\lambda$ and $x \geqslant 0$.

2.8.37. [126] In case the space of 8.36 is Dedekind complete, Proposition 8.36 changes the same way as 8.21 and 8.22.

2.8.38. [121] Assume that exp(tA) is a positive OS, G is a densely defined linear operator, and (8.6) is satisfied. Then G admits the closure $\tilde{G}$.

2.8.39. [121] Assume that the conditions of 8.38 are satisfied. Then, if $(\lambda I - G)D(G)$ is dense in E for some $\lambda > \max\{0, S(A)\}$, then G generates an OS exp(tG) that is dominated by the OS exp(tA).

2.8.40. [126] Given positive semigroups exp(tA) and exp(tG), then the conditions

(i) $\exp(tG) \leqslant \exp(tA)$ $(t \geqslant 0)$; and

(ii) $\langle Gx, \varphi \rangle \leqslant \langle x, A^{\odot}\varphi \rangle$ for all $x \in D(G)_+$, $\varphi \in D(A^{\odot})_+$ are equivalent and imply satisfaction of the relation

(iii) $Gx \leqslant Ax$ for $0 \leqslant x \in D(A) \cap D(G)$.

2.8.41. [126] If, under the conditions of 8.40, we have $D(A) \subset D(G)$ or $D(G) \subset D(A)$, then (i) follows from (iii).

2.8.42. [178] Let E be a Banach structure with order continuous norm, and let exp(tA) be a positive OS defined on E. Assume that we are given a semigroup $\tilde{U}(t)$ such that the function $\tilde{U}(t)x$ is strongly measurable for every $x \in E$ and $\exp(tA) \leq \tilde{U}(t)$, $t \geq 0$, where $\sup\limits_{0<t<1} \|\tilde{U}(t)\| < \infty$. Then $\tilde{U}(t)$ is a positive OS.

Definition 34. By the modulus of an OS exp(tA) (denoted by $|\exp(tA)|$) we mean the minimum positive OS that majorizes the OS exp(tA).

2.8.43. There are examples of semigroups for which there are no moduli.

2.8.44. [147] If A is a bounded operator in an order complete complex Banach structure that can be represented by the difference of two positive operators, then a modulus exists for the semigroup exp(tA).

2.8.45. [147] If for some OS exp(tA) in an order complete Banach structure there exists a majorizing OS, then the modulus $|\exp(tA)|$ also exists.

2.8.46. [147] Assume that exp(tA) is an OS in an order complete Banach structure, and $Q \in B(E)$ is such that $Q \leq \gamma I$ for some $\gamma > 0$. Then for the semigroups exp(tA) and exp(t(A+Q)) moduli either simultaneously exist or do not exist, and the generators A_H and $(A + Q)_H$ are related by the expression

$$(A+Q)_H = A_H + \operatorname{Re} Q.$$

Definition 35. A linear operator A defined in a partially ordered space E with normal reproducing cone E is said to be resolvent positive if

(i) $(\zeta_0, \infty) \subset \rho(A)$ for some $\zeta_0 \in \mathbf{R}$;

(ii) $R(\lambda, A) \geqslant 0$ for all $\lambda > \zeta_0$.

Definition 36. We denote the infimum of ζ_0 from Def. 35 by $S_+(A)$.

2.8.47. [122] For resolvent positive operators with $S_+(A) < \lambda < \mu$, we have $0 \leqslant R(\mu, A) \leqslant R(\lambda, A)$.

2.8.48. [122] There are examples of resolvent positive operators that are not generators of OS's.

2.8.49. [122] If exp(tA) is a positive OS, then $S_+(A) = S(A)$.

Definition 37. A set $Q \subset E_+$ is said to be cofinal in E_+ if for each $x \in E_+$ there exists a majorant $x \leq y \in Q$.

2.8.50. [122] If A is a densely defined resolvent positive operator, $D(A)_+$ is cofinal in E_+, or $D(A^{\odot})_+$ is cofinal in $E_+^{\odot}$, where $S(A) = \omega(A)$.

2.8.51. [122] Assume that A is densely defined and resolvent positive. Then, if $\|R(\lambda_0, A)x\| \geq c\|x\|$, $(x \in E_+)$ for some $\lambda_0 > S(A)$ and $c > 0$, then A generates a positive OS and $S(A) = \omega(A)$.

2.8.52. [122] Assume that the norm of E is additive on the cone E_+ and $\overline{D(A)} = E$. Then the following statements are equivalent:

(i) A generates a positive group of operators;

(ii) A and $-A$ are resolvent positive and there exist λ $\operatorname{Max}\{S(A), S(-A)\}$ and $c > 0$ such that $\|R(\lambda \pm A)x\| \geq c\|x\|$ for all $x \in E_+$.

2.9. Functions of the Generating Operator of an OS. We consider the class $\mathfrak{B}$ of functions $\varphi(\tau)$ that are continuous on $\bar{\mathbf{R}}_+$, positive when $\tau > 0$, and representable in the form

$$\varphi(\tau)=\int_0^\infty \frac{\tau d\sigma(s)}{\tau+s}, \tag{9.1}$$

where σ is a measure that is positive on $[0,\infty)$ and such that

$$\int_0^\infty \frac{d\sigma(s)}{1+s}<\infty. \tag{9.2}$$

Definition 1. Let A be the generator of a contraction OS. Then for $x \in D(A)$ we set [85, 252]

$$\varphi(A)x=\int_0^\infty A(A+\lambda I)^{-1}x\,d\sigma(\lambda). \tag{9.3}$$

We call $\varphi(A)$ the closure of operator (8.3).

2.9.1. [22] The operator $-\varphi(A)$ is the generating operator for a contraction OS $\mathcal{U}_\varphi(t)$, for which

$$\mathcal{U}_\varphi(t)=\int_0^\infty \exp(\lambda A)\,d\mu_t(\lambda), \quad t>0,$$

where μ_t is a family of probability measures on $[0,\infty)$ ($\mu_0=\delta_0$, $\mu_{t+s}=\mu_t * \mu_s$, $\mu_t \to \delta_0$ weakly as $t \to 0$, $\mu(\bar{\mathbf{R}}_+)=1$). The integral converges in the strong sense.

2.9.2. [22] The measure μ_t can be found from the scalar representation

$$e^{-t\varphi}=\int_0^\infty e^{-\tau\lambda}d\mu_t(\lambda)$$

obtained from Bernshtein's theorem on completely monotonic functions.

2.9.3. [22] If $\mathcal{U}(t)$ is a bounded analytic semigroup, then so is $\mathcal{U}_\varphi(t)$.

Examples of the function φ. We represent the function $\varphi(\tau)=\tau^\alpha$ $(0<\alpha<1)$ in the form

$$\tau^\alpha=\frac{\sin(\pi\alpha)}{\pi}\int_0^\infty \frac{\tau}{\tau+s}s^{\alpha-1}ds. \tag{9.4}$$

Then from (9.4) we can obtain a definition of fractional powers of a generating operator:

$$(-A)^\alpha=\frac{\sin(\pi\alpha)}{\pi}\int_0^\infty A(A+\lambda I)^{-1}\lambda^{\alpha-1}d\lambda.$$

For the function $\varphi(\tau)=\ln(1+\tau)$ we have $\ln(1+\tau)=\int_1^\infty \frac{\tau}{\tau+s}\,ds/s$ and $\ln(1+A)=\int_1^\infty A(A+\lambda I)^{-1}d\lambda/\lambda$.

2.10. Boundary Values of Operator Subgroups. Let $\mathcal{U}(t)$ $(t>0)$ be a strongly continuous semigroup of bounded operators in a Banach space E, and let $-A$ be its generator. In general, strong continuity of the semigroup at zero and denseness of D(A) in E are not assumed. Also, assume that the following conditions are satisfied:

a) $\mathcal{N}(\mathcal{U}(t))=\{0\}$;

b) the semigroup is a contraction semigroup: $\|\mathcal{U}(t)\| \leqslant 1$;

c) for any $x \in E$ the function $\mathcal{U}(t)x$ is differentiable when $x \in E$.

In virtue of these conditions, the function $v(t) := \mathcal{U}(t)x$ with $x \in E$ satisfies the equation

$$v'(t) = -Av(t), \quad t > 0. \tag{10.1}$$

It turns out that this does not exhaust the functions satisfying (10.1).

We introduce the norm $\|x\|_{-t} := \|\mathcal{U}(t)x\| \leqslant \|x\|$ into the space E, and denote the completion relative to this norm by E_{-t}.

2.10.1. [22]. The operator $\mathcal{U}(t)$ can be extended by continuity to a contraction operator $\mathcal{U}(t)$ in E_{-t}.

2.10.2. [22] When $0 < s < t$, we have the dense imbedding $E_{-s} \subseteq E_{-t}$. The space $E_- := \operatorname{pr}\lim_{t \to 0} E_{-t}$ is a locally convex linear topological space.

Definition 1. A semigroup H(t) ($t > 0$) defined in a locally convex linear topological space is said to be equicontinuous if, for any continuous seminorm p, there exists a continuous seminorm q such that for all $t > 0$ and $x \in E$

$$p(H(t)x) \leqslant q(x)$$

2.10.3. [22] The semigroup $\tilde{\mathcal{U}}(t)$ is equicontinuous in E_-.

2.10.4. [22] For any $x \in E$ the function $\tilde{\mathcal{U}}(t)x$ satisfies Eq. (10.1) and, conversely, for any function v(t) that satisfies (10.1) there exists an $x \in E$ such that $v(t) = \tilde{\mathcal{U}}(t)x$.

2.10.5. [22] Let E be adjoint to some Banach space F and for a particular $t_0 > 0$ assume the operator $\mathcal{U}(t_0)$ is adjoint to some operator that is bounded in F.

Then $x \in E$ belongs to E if and only if $\sup_{0 < t < 1} \|\tilde{\mathcal{U}}(t)x\| < \infty$.

2.10.6. [22]. If E is reflexive, then $x \in E_-$ belongs to E if and only if the function $\|\tilde{\mathcal{U}}(t)x\|$ is bounded close to zero.

For the case of analytic semigroups it has been possible to establish a connection between the behavior of the function $\tilde{\mathcal{U}}(t)x$ and membership of bounded values of x in certain spaces intermediate between E and E_-. They are constructed like the space E_-, but the role of the operator A is played by a function $\varphi(A)$, where φ satisfies conditions (9.1) and (9.2). We denote the corresponding spaces by $E_-(\varphi)$.

2.10.7. [22] Assume that a semigroup satisfies the conditions of 10.5 and, moreover, is bounded analytic. In order for an $x \in E_-$ to belong to $E_-(\varphi)$ it is necessary and sufficient that there exist constants $\mu > 0$ and $C > 0$ such that

$$\|\tilde{\mathcal{U}}(t)x\| \leqslant Ce^{\max_{\tau \geqslant 0}(\mu\varphi(\tau) - t\tau)}.$$

A similar proposition holds for reflexive spaces E.

For other propositions of this type, see [22].

2.11. Supplementary Comments. The theory of C-semigroups and integrated semigroups has been the subject of a number of publications (see [191, 317, 318, 334, 335, 391-393]). Applications are also discussed in these papers.

Analytic OS's whose domains are not dense in E were considered in [113, 114, 187, 375].

Lipshits operator semigroups are discussed in [416, 417].

Special cases ($E = C_0(X)$, etc.) of mappings of the spectra of operator semigroups are considered in [126, 127].

Concerning groups of operators, see Propositions 2.6.14, 2.7.30, 3.1.32, 4.1.6, 5.1.3, 7.1.4. The analytic property of OS's is considered in Propositions 2.6.7, 2.6.13, 2.9.3, 2.10.7, 3.1.7, 8.1.4, 8.1.25, 8.1.46, 9.1.8. Concerning positive OS's, see also Propositions 2.7.10, 2.7.16, 2.7.17, 2.7.38, 2.7.46-48, 4.1.16, 4.1.17, 8.1.34, 8.1.35. Concerning differentiable OS's, see Propositions 2.5.2, 2.6.7, and 2.6.13. Semi-Fredholm operator semigroups also figure in Propositions 4.1.11, 8.1.28, and 8.1.30.

3. COSINE AND SINE OPERATOR FUNCTIONS (COF'S AND SOF'S)

Assume that in a Banach space E we are given a COF C(t,A) and the associated SOF $S(t, A) := \int_0^t C(s, A)ds$ (for the definitions, see 1.4). Recall that the generating operator (more accurately, the second generating operator) of a COF C(t,A) is defined as the limit

$$Ax := \lim_{h\to 0} \frac{2}{h^2}(C(h, A)x - x)$$

on all $x \in E$ for which it exists.

3.0.1. [137] we define

$$A_1 x := \lim_{h\to 0} \frac{C(2h, A) - 2C(h, A) + I}{h^2} x.$$

Then for $x \in D(A_1) \cap D(A)$ we have $Ax = A_1x$.

For a COF C(t,A) we can define the first generating operator

$$\mathring{\mathscr{C}}x := \lim_{h\to 0+} \frac{C(h, A)x - x}{h}$$

on those x for which the limit exists.

3.0.2. [381] For a COF C(t,A) we have $D(A) \subseteq D(\mathring{\mathscr{C}})$ and $\mathring{\mathscr{C}}x = 0$ for all $x \in D(A)$.

3.1. Fundamental Properties of COF's and SOF's.

3.1.1. [381] The operators C(t,A), C(s,A), S(t,A), and S(s,A) commute for all $t, s \in \mathbf{R}$.

3.1.2. [318] An SOF S(t,A) is continuous in the uniform operator topology.

3.1.3. [381] For all $t, s \in \mathbf{R}$ we have

$$\begin{gathered}
\text{(i)}\ C(t, A) = C(-t, A),\ S(-t, A) = -S(t, A),\ S(0, A) = O,\\
\text{(ii)}\ S(t+s, A) + S(t-s, A) = 2S(t, A)C(s, A),\\
\text{(iii)}\ S(t+s, A) = S(t, A)C(s, A) + S(s, A)C(t, A),\\
\text{(iv)}\ C(t+s, A) - C(t-s, A) = 2AS(t, A)S(s, A);\\
\text{(v)}\ C(2t, A) = 2C(t, A) - I,\ C(t, A)^2 - AS(t, A)^2 = I;\\
\text{(vi)}\ C((n+1)t, A) = b_0 I + b_1 C(t, A) + \ldots + b_{n+1} C^{n+1}(t, A),
\end{gathered}$$

where $b_0 + b_1 x + \ldots + B_{n+1}x^{n+1}$ is a Chebyshev polynomial of the first kind of order n + 1.

3.1.4. [381] For any COF C(t,A) there exist constants $M \geq 1$ and $\omega \geq 0$ such that for all $t \in \mathbf{R}$

$$\|C(t, A)\| \leqslant M \operatorname{ch}(\omega t),\ t \in \mathbf{R}. \tag{1.1}$$

Definition 1. The lower bound of the numbers ω of (1.1) is called the type of the COF and is denoted by $\omega^c(A)$.

3.1.5. [307] There may be no smallest ω satisfying (1.1) with an appropriate constant M_ω (i.e., generally speaking, the lower bound $\omega^c(A)$ is not attained).

3.1.6. [209, 226, 381] Let $\|C(t,A)\| \leq M \cosh(\omega t)$. Then $A \in \mathscr{G}(M, \omega^2)$, the OS exp(tA) can be analytically continued to the right half-plane, and

$$\exp(tA) = \frac{1}{\sqrt{\pi t}} \int_0^\infty e^{-\frac{s^2}{4t}} C(s, A)ds,\quad t \in \bar{\mathbf{R}}_+.$$

However, there are examples of analytic OS's whose generating operators do not generate COF's (see [329]).

3.1.7. [169] An operator A generates a uniformly bounded COF if and only if $A \in \mathcal{H}(\pi/2,0)$ and $\overline{EXP(A)} = E$.

3.1.8. [398] For any $x \in E$ and $t, s \in \mathbf{R}$ we have

$$\text{(i) } y := \int_s^t S(\tau, A)x\,d\tau \in D(A) \text{ and } Ay = C(t, A)x - C(s, A)x;$$

$$\text{(ii) } z := \int_0^t \int_0^s C(\tau, A)C(\zeta, A)x\,d\tau\,d\zeta \in D(A) \text{ and } Az = (C(t+s, A) - C(t-s, A))x/2;$$

$$\text{(iii) } S(t, A)x \in E^1.$$

3.1.9. [398] If an element x runs through all of E, the set of elements y of the form $y = \int_s^t S(\tau, A)x\,d\tau$ is dense in E.

3.1.10. [398] For any $x \in E$ we have $\lim_{t\to 0} t^{-1}S(t, A)x = x$ and $\lim_{t\to 0} 2t^{-2}\int_0^t S(\tau, A)x\,d\tau = x$.

3.1.11. [398] If $x \in E^1$, the for any $t \in \mathbf{R}$

$$\text{(i) } C(t, A)x \in E^1,\ S(t, A)x \in D(A) \text{ and } C'(t, A)x = AS(t, A)x;$$

$$\text{(ii) } \lim_{\tau\to 0} AS(\tau, A)x = 0 \text{ and } S''(t, A) = AS(t, A)x.$$

3.1.12. [398] Let $x \in D(A)$. Then for all $t \in \mathbf{R}$

$$\text{(i) } C(t, A)x \in D(A) \text{ and } C''(t, A)x = AC(t, A)x = C(t, A)Ax;$$

$$\text{(ii) } S(t, A)x \in D(A) \text{ and } S''(t, A)x = AS(t, A)x = S(t, A)Ax.$$

3.1.13. [396] For all $t, s \in \mathbf{R}$ we have

$$\text{(i) } C(2t, A) = C(t, A)^2 + C'(t, A)\cdot S(t, A);$$

$$\text{(ii) } C'(t, A)S(s, A) = C'(s, A)S(t, A);$$

$$\text{(iii) } C(t+s, A) - C(t-s, A) = 2C'(t, A)S(s, A).$$

$$\text{(iv) } (C(t, A) - I)\int_0^h S(r, A)\,dr = (C(h, A) - I)\int_0^t S(r, A)\,dr$$

$$\text{(v) } (A - \lambda^2)\int_0^t \operatorname{sh}(\lambda(t-s))\,C(s, A)\,ds = \lambda(C(t, A) - \operatorname{ch}(\lambda t))$$

3.1.14. [211] The domain of the generating operator of a COF C(t,A) coincides with E^2, and for each $x \in D(A)$

$$Ax = \lim_{\tau\to 0} C''(\tau, A)x. \tag{1.2}$$

Sometimes the generating operator of a COF is actually defined by (1.2).

We will denote the set of COF generators with bound (1.1) by $\mathcal{C}(\mathbf{M}, \omega)$.

3.1.15. [381] Let $A, G \in \mathcal{C}(\mathbf{M}, \omega)$. Then, if $D(A) \subset D(G)$ and $Ax = Gx$ for $x \in D(A)$, then $C(t,A) = C(t,G)$ for all $t \in \mathbf{R}$.

THEOREM 1 ([186, 209, 381, 387, 406]). An operator $A \in \mathscr{C}(E)$ generates a COF if and only if for some constants M and ω the resolvent $R(\lambda,A)$ exists when $\operatorname{Re} \lambda^2 > \omega$ and

$$\left\| \frac{d^n}{d\lambda^n}(\lambda R(\lambda^2, A)) \right\| \leqslant \frac{Mn!}{(\lambda-\omega)^{n+1}}, \quad n \in \mathbf{N}. \tag{1.3}$$

3.1.16. [381] Sometimes (1.3) is written in the form

$$\left\| \frac{d^n}{d\lambda^n}(\lambda R(\lambda^2, A)) \right\| \leqslant \frac{Mn!}{2}\left(\frac{1}{(\operatorname{Re}\lambda-\omega)^{n+1}} + \frac{1}{(\operatorname{Re}\lambda+\omega)^{n+1}} \right) \tag{1.4}$$

for all $\operatorname{Re} \lambda > \omega$, $n \in \mathbf{N}$.

In practice, it is difficult to verify (1.3) and (1.4), so it is of value to find other conditions that imply generation of COF's.

THEOREM 2 ([57]). An operator $A \in \mathscr{C}(E)$ generates a COF if and only if $\overline{D(A)} = E$ and there exist constants M, δ, and ω such that

$$\| R(\lambda^2, A) \| \leqslant \frac{M}{|\lambda|(\operatorname{Re}\lambda-\omega)} \quad \text{for all} \quad \operatorname{Re}\lambda > \omega, \tag{1.5}$$

and the following bound is uniformly satisfied with respect to $\tau \in (0,\delta)$:

$$\left\| \int_{\omega-i\infty}^{\omega+i\infty} e^{\lambda^2\tau} \operatorname{ch}(\lambda t) \lambda R(\lambda^2, A) x d\lambda \right\| \leqslant \xi(t) \| x \|, \tag{1.6}$$

where $\xi(t) \in C(\mathbf{R})$.

3.1.17. [77] In connection with bound (1.5), note that the condition

$$\| R(\lambda^2, A) \| \leqslant \frac{M}{|\lambda|^{1+\varepsilon}}, \quad \operatorname{Re}\lambda > \omega,$$

for some $\varepsilon > 0$ implies that the spectrum $\sigma(A)$ is bounded.

3.1.18. [310] When A is a normal operator in a Hilbert space, then it generates a COF if the following condition is met by its spectrum:

$\{z^2 : \operatorname{Re} z > \} \subset \rho(A)$ for some ω.

3.1.19. [396] For $\operatorname{Re}\lambda\ \omega^c(A)$, we have $\lambda^2 \in \rho(A)$ and

$$\text{(i)} \quad \lambda R(\lambda^2, A)x = \int_0^\infty e^{-\lambda t} C(t, A)x dt, \quad x \in E;$$

$$\text{(ii)} \quad R(\lambda^2, A)x = \int_0^\infty e^{-\lambda t} S(t, A)x dt, \quad x \in E.$$

3.1.20. [307] For any $A \in \mathscr{G}(M, \omega)$ and $x \in E$ we have $\lim_{\lambda\to\infty} \| \lambda^2 R(\lambda^2, A)x - x \| = 0$.

3.1.21. [211] If $x \in D(A^3)$, $y \in D(A)$, and $\omega > \omega^c(A)$, then

$$\text{(i)} \quad C(t, A)x = x + \frac{t^2}{2!}Ax + \frac{t^4}{4!}A^2x + \frac{1}{2\pi i}\int_{\omega-i\infty}^{\omega+i\infty} e^{\lambda t}\lambda^{-3} R(\lambda^2, A)A^3 x d\lambda;$$

$$\text{(ii) } C(t, A)y=\frac{1}{2\pi i}\int_{\omega-i\infty}^{\omega+i\infty} e^{\lambda t}\lambda R(\lambda^2, A)\,y\,d\lambda.$$

By writing the inverse Laplace transform in other forms we can obtain other analogous representations of C(t,A) and S(t,A).

3.1.22. [328] Let $x \in D(A^k)$ for some $k \in \mathbf{N}$. Then for $t \in \mathbf{R}$ we have Taylor's formula (for the analog for OS's see 2.1):

$$C(t, A)x=x+\frac{t^2}{2!}Ax+\ \dots\ +\frac{t^{2n-2}}{(2n-2)!}A^n x+$$

$$+\int_0^t \frac{(t-s)^{2n-1}}{(2n-1)!}C(s, A)A^n x\,ds.$$

3.1.23. [331] For any $x \in \tilde{E}_0$ (see (2.1.16), $A \in \mathscr{C}(\mathbf{M}, \omega)$, and $t \in \mathbf{R}$, we have

$$C(t, A)x=\sum_{k=0}^{\infty} t^{2k}A^k x/(2k)!,$$

and for each $\tilde{x} \in \tilde{E}_0$ the function $t \to C(t,A)\tilde{x}$ can be continued to a function that is analytic in the entire complex plane.

3.1.24. [209] If $x \in E$,

$$C(t, A)x=\lim_{k\to\infty}\frac{(-1)^k}{k!}\left(\frac{k}{t}\right)^{k+1}\frac{d^k}{d\lambda^k}(\lambda R(\lambda^2, A)x)\big|_{\lambda=\frac{k}{t}}, \tag{1.7}$$

where convergence is uniform with respect to t from any compact set in $\mathbf{R}\backslash\{0\}$.

3.1.25. [416] The expression $\mathfrak{M}(\lambda, k):=\frac{d^k}{d\lambda^k}(\lambda R(\lambda^2, A))$ in (1.7) can be written in the form

$$\text{(i) } \mathfrak{M}(\lambda, k)=k!\,(\lambda^{k+1}+c_{k+1}^2\lambda^{k-1}A+\ \dots\ +c_{k+1}^k\lambda A^{k/2})\times$$
$$\times(\lambda^2 I-A)^{-(k+1)},\quad n \text{ is even;}$$

$$\text{(ii) } \mathfrak{M}(\lambda, k)=-k!\left(\lambda^{k+1}+c_{k+1}^2\lambda^{k-1}A+\ \dots\ +A^{\frac{k+1}{2}}\right)(\lambda^2 I-A)^{-(k+1)},$$
$$n \text{ is odd;}$$

$$\text{(iii) } \mathfrak{M}(\lambda, k)=\sum_{j=k/2}^{k}(-1)^j\frac{(k+1)!\,j!\,\lambda\,(2\lambda)^{2j-k}}{(k-j)!\,(2!-k+1)!}(\lambda^2 I-A)^{-(j+1)},$$
$$n \text{ is even;}$$

$$\text{(iv) } \mathfrak{M}(\lambda, k)=\sum_{j=\frac{k-1}{2}+1}^{k}(-1)^j\frac{(k+1)^j j!\lambda\,(2\lambda)^{2j-k}}{(k-j)!\,(2j-k+1)!}(\lambda^2 I-A)^{-(j+1)},$$
$$n \text{ is odd.}$$

3.1.26. [142, 373] For a COF C(t,A) an SOF S(t,A), and any $x \in E$ and $t \in \mathbf{R}$ we have

$$\text{(i) } C(t, A)x=\lim_{k\to\infty}\sum_{l=0}^{k}\sum_{j=0}^{l}C_{2k}^{2l}C_l^j(-1)^{l-j}[I-(t/2k)^2A]^{-(2k-l+j)}x;$$

$$\text{(ii) } C(t, A)x = \lim_{n\to\infty} \sum_{k=0}^{n} \sum_{j=0}^{k} C_{2n+1}^{2k} C_k^j (-1)^{k-j} \times$$
$$\times \left[I - \left(\frac{t}{2n+1}\right)^2 A \right]^{-[2n+1-k+j]} x;$$

$$\text{(iii) } C(t, A)x = \lim_{n\to\infty} \exp(-nt) \sum_{m=0}^{\infty} \sum_{k=0}^{m} \sum_{j=0}^{k} \frac{(nt)^{2m}}{(2m)!} C_{2m}^{2k} C_k^j (-1)^{k-j} \times$$
$$\times \left[1 + \frac{nt}{2m-2k+1} (I - n^{-2}A)^{-1} \right] (I - n^{-2}A)^{-(2m-k+j)} x;$$

$$\text{(iv) } S(t, A)x = \lim_{n\to\infty} \frac{t}{n} \sum_{m=0}^{n-1} \sum_{k=0}^{m} \sum_{j=0}^{k} C_{2m+1}^{2k} C_k^j (-1)^{k-j} \times$$
$$\times \left(I - \left(\frac{t}{2n}\right)^2 A \right)^{-(n+m+1-k+j)} x;$$

where convergence in all cases is uniform with respect to $t \in \mathcal{K} \subset \mathbf{R}$.

3.1.27. [373] Under the conditions of 1.26 we have

$$C(t, A)x = \lim_{n\to\infty} \sum_{m=0}^{n} \sum_{k=0}^{m} \sum_{j=0}^{k} C_{2n}^{2m} C_{2m}^{2k} C_k^j (-1)^{k-j} t^{2m} (1-t)^{2n-2m-1} \times$$
$$\times \left[(1-t) + \frac{2n-2m}{2m-2k+1} t (I - (2n)^{-2}A)^{-1} \right] (I - (2n)^{-2}A)^{-(2m-k+j)} x;$$

uniformly with respect to $t \in [0,1]$. Other relationships of this type are given in [373].

Notation:

$$St(A) := \left\{ x \in E : \sum_{k=1}^{\infty} \| A^k x \|^{-\frac{1}{2k}} < \infty \right\} \text{ are Stieltjes vectors,}$$

$$\mathfrak{A}_p(A) := \left\{ x \in E : \sum_{k=0}^{\infty} \| A^k x \| \cdot t^{2k}/(2k)! < \infty \text{ for some } t > 0 \right\}$$

are semi-analytic vectors,

$$\mathfrak{A}_{pp}(A) := \left\{ x \in E : \sum_{k=0}^{\infty} \frac{t^k}{(2k)!} \| A^k x \| < \infty \text{ for all } t > 0 \right\}.$$

3.1.28. [168] Let $A \in \mathcal{C}(\mathbf{M}, \omega)$ and $\tilde{E}_0$ be constructed for a given OS exp(t,A). Then $\tilde{E}_0 \subseteq \mathfrak{A}_p(A)$.

3.1.29. [168] $\mathfrak{A}(A) \subseteq \mathfrak{A}_p(A) \subseteq St(A)$.

3.1.30. [168] Let $\overline{\mathfrak{A}_{pp}(A)} = E$. Then the set of vectors x of $D(A^\infty)$ with the property that $\| A^k x \|^{1/k} = o(k)$ is dense in E.

3.1.31. [168] Let $A \in \mathcal{C}(\mathbf{M}, \omega)$. . Then $\overline{\mathfrak{A}(A) \cap \mathfrak{A}_{pp}(A)} = E$.

3.1.32. [168] Let $\overline{\mathfrak{A}_{pp}(A)} = E$ and assume there exists an operator $G \in \mathcal{G}(E)$ such that

(i) $G^{-1} \in B(E)$;
(ii) $G^2 = A$, where the operators $\pm G$ are dissipative.

Then $A \in \mathcal{G}(\mathbf{M}, \omega)$ and C(t,A) = (exp(tG) + exp(—tG) + exp(—tG))/2, where $G \in \mathcal{GR}(1, 0)$.

3.1.33. [168]. Assume that A_1 is closed, that $St(A_1)$ is total in E, $A_2 \in \mathcal{G}(\mathbf{M}, \omega)$, and $A_1 \subset A_2$. Then $A_1 = A_2$.

3.1.34. [168] Let A be a closed symmetric and semibounded operator in a Hilbert space H. Then A is self-adjoint if and only if the set St(A) is total in H.

We define the following sets:

$$NBI_0 = \{ t > 0 : \text{the operator } C(t,A) \text{ does not have a bounded inverse}\},$$
$$NBI_1 = \{t > 0 : \text{the operator } S(t,A) \text{ does not have a bounded inverse}\}.$$

3.1.35. [250] Assume that the operator $C(t,A) - I$ is compact for all $t \in \mathbf{R}$. The sets NBI_0 and NBI_1 are simultaneously either empty or have cardinality of a continuum and there exist constants $\alpha_1, \alpha_2 > 0$ such that $NBI_j \subset (\alpha_j, \infty)$, $j = 0,1$.

3.2. Reduction of the Cauchy Problem for Second-order Equations to the Cauchy Problem for First-order Equations. In a Banach space E we consider uniformly well-posed Cauchy problem (1.1.)-(1.2) with $n = 2$ ($t \in \mathbf{R}$):

$$u''(t) = Au(t),\ u(0) = u^0,\ u'(0) = u^1. \tag{2.1}$$

We define the operator $\mathscr{A} := \begin{pmatrix} O & I \\ A & O \end{pmatrix} : E \times E \to E \times E$, to act on the element (x,y) according to the formula

$\mathscr{A}(x, y) = (y, Ax)$ with domain $D(\mathscr{A}) = D(A) \times E^1$.

3.2.1. [271] It is clear that $D(A) \subset E^1$.

3.2.2. [271] The set E^1 is dense in E.

THEOREM 1. [271] The space E^1 with norm

$$\| x \|_{E^1} = \| x \| + \sup_{0 \leqslant t \leqslant 1} \| C'(t, A)x \| \tag{2.2}$$

is a Banach space. The operator $\mathscr{A}$ generates the group of operators

$$\exp(t\mathscr{A})(x, y) := \begin{pmatrix} C(t, A) & S(t, A) \\ AS(t, A) & C(t, A) \end{pmatrix}(x, y) =$$
$$= (C(t, A)x + S(t, A)y,\ AS(t, A)x + C(t, A)y)$$

in the Banach space $E^1 \times E$.

3.2.3. [413, 414]. Assume we are given a COF C(t,A). Then E^1 coincides with the closure of D(A) in the norm

$$\| x \|_* := \| x \| + \sup_{\substack{z > \omega \\ n \in \mathbf{N}}} \frac{1}{n!} (z - \omega)^{n+1} \left\| \frac{d^n}{dz^n} A(z^2 - A)^{-1} x \right\|.$$

3.2.4. [271] The resolvent of the operator $\mathscr{A}$ is of the form

$$(\lambda I - \mathscr{A})^{-1} = \begin{pmatrix} \lambda(\lambda^2 I - A)^{-1} & (\lambda^2 I - A)^{-1} \\ A(\lambda^2 I - A)^{-1} & \lambda(\lambda^2 I - A)^{-1} \end{pmatrix} \text{ for } \lambda^2 \in \rho(A). \tag{2.3}$$

3.2.5. [271] If u(t) is a solution to problem (2.1) and $v(t) := u'(t)$, then the vector $(u,v)^T$ is the solution of the uniformly well-posed Cauchy problem

$$\begin{pmatrix} u \\ v \end{pmatrix}'(t) = \mathscr{A} \begin{pmatrix} u \\ v \end{pmatrix};\quad \begin{pmatrix} u \\ v \end{pmatrix}(0) = \begin{pmatrix} u^0 \\ u^1 \end{pmatrix}, \tag{2.4}$$

in the Banach space $E^1 \times E$. Assume that the Banach space $\tilde{E}^1$ is continuously and densely imbedded in the Banach space E, where $D(\tilde{A}) \subseteq \tilde{E}^1$ for some $\tilde{A} \in L(E)$. Then the Cauchy problem ($t \in \mathbf{R}$)

$$\begin{pmatrix} u \\ v \end{pmatrix}'(t) = \begin{pmatrix} O & I \\ A & O \end{pmatrix}\begin{pmatrix} u \\ v \end{pmatrix}(t) \equiv \tilde{\mathscr{A}}\begin{pmatrix} u \\ v \end{pmatrix}(t), \quad \begin{pmatrix} u \\ v \end{pmatrix}(0) = \begin{pmatrix} u^0 \\ u^1 \end{pmatrix} \tag{2.5}$$

is uniformly well-posed in the space $\tilde{E}^1 \times E$ and $\rho(\tilde{\mathscr{A}}) \neq \varnothing$.

3.2.6. [271] The OS corresponding to problem (2.5) can be represented in the form

$$\exp(t\tilde{\mathscr{A}}) := \begin{pmatrix} G_{11}(t) & G_{12}(t) \\ G_{21}(t) & G_{22}(t) \end{pmatrix}, \quad t \geqslant 0, \tag{2.6}$$

where the family $G_{22}(t)$ is the COF C(t,A) and coincides with $G_{11}(t)$ on E^1.

3.2.7. [271] For $x \in E$ and $y \in \tilde{E}^1$ we have

$$G_{12}(t)x = S(t,A)x \quad \text{and} \quad G_{21}(t)y = C'(t,A)y.$$

3.2.8. [271] The spaces $\tilde{E}^1$ and E^1 of Proposition 2.3 coincide up to equivalence of norms.

We should note, however, that investigation of problem (2.1) by reduction to a system is extremely awkward, since the space E^1 is defined either in terms of a COF C(t,A) or as a power of the resolvent. As a rule, we have information only about the operator A. Thus, other methods of reducing problem (2.1) to a system are of interest.

3.2.9. [398] Assume we are given a Hilbert space E and a self-adjoint negative definite operator A. Then $A \in \mathscr{G}(\mathbf{M}, \omega)$ and the corresponding space E^1 coincides with $D(A^{1/2})$.

Assume that a uniformly well-posed problem (2.1) is of the form

$$u'' = G^2 u; \quad u(0) = u^0, \quad u'(0) = u^1, \tag{2.8}$$

where $G \in \mathscr{G}(E)$.

Definition 1. A solution u(t) of problem (2.8) is said to satisfy condition (K) if $u'(t) \in C([0, T], \mathscr{D}(G))$.

3.2.10. [59] Problem (2.8) has a unique solution satisfying condition (K) if and only if the following Cauchy problem is uniformly well-posed:

$$\begin{pmatrix} v \\ w \end{pmatrix}'(t) = \begin{pmatrix} G & O \\ O & -G \end{pmatrix}\begin{pmatrix} v \\ w \end{pmatrix}(t), \quad \begin{pmatrix} v \\ w \end{pmatrix}(0) = \begin{pmatrix} v_0 \\ w_0 \end{pmatrix}. \tag{2.9}$$

The analog of condition (K) that allows us to simplify investigation of problem (2.1) by means of OS's is condition (F):

Definition 2. A COF C(t,A) satisfies condition (F) if the following conditions are satisfied:

(i) there exists an operator $G \in \mathscr{G}(E)$ such that $G^2 = A$ and G commutes with any operator of B(E) that commutes with A;

(ii) the SOF S(t,A) maps E into D(G) for all $t \in \mathbf{R}$;

(iii) the function GS(t,A) is continuous in $t \in \mathbf{R}$ for any fixed $x \in E$.

3.2.11. [211] When condition (F) is satisfied for every $t \in \mathbf{R}$ we have $GS(t,A) \in B(E)$ and $D(kG) \subseteq E^1$.

3.2.12. [211] There exist a Banach space E and a uniformly bounded COF C(t,A) such that condition (F) is not satisfied.

3.2.13. [211] We can always use $A_b := A - b^2 \cdot I$ for $b > \omega^c(A)$ to construct A_b and G_b such that $G_b^2 = A_b$ and G_b commutes with any operator of B(E) that commutes with A_b.

3.2.14. [210] The operator G_b of 2.13 can be constructed, for example, thus:

$$G_b x := \frac{-i}{\pi} \int_0^\infty \lambda^{-1/2} (\lambda I - A_b)^{-1} (-A_b x) d\lambda.$$

THEOREM 2. [396] Let A and G be operators satisfying condition (i) of Definition 2, and assume that $0 \in \rho(G)$. The following conditions are equivalent:

(i) the COF C(t,A) satisfies condition (F);

(ii) the operator G generates the OS exp(tG) on E;

(iii) the operator $\begin{pmatrix} O & G \\ G & O \end{pmatrix}$ with domain D(A) × D(G) generates a C_0-group on E × E;

(iv) the operator $\mathscr{A} := \begin{pmatrix} O & I \\ A & O \end{pmatrix}$ with domain D(A) × D(G) generates the C_0-group $\exp(t\mathscr{A})$ on $\mathscr{D}(G) \times E$,

where $\mathscr{D}(G)$ is the Banach space D(G) with the graphic norm;

(v) we have $D(G) \supseteq E^1$;

(vi) $D(G) = E^1$.

3.2.15. [211] Let $A \in \mathscr{C}(M, 0)$ and assume that E = H is a Hilbert space. Then condition (F) is satisfied.

3.2.16. [414] The following condition is equivalent to conditions (i)-(vi) of Theorem 2:

D(G) is dense in E and there exist constants $M > 0$ and $\omega \geq 0$ such that $\lambda^2 \in \rho(A)$ for all $\lambda > \omega$, the operator functions $\lambda(\lambda^2 I - A)^{-1}$ and $G(\lambda^2 I - A)^{-1}$ are strongly differentiable an infinite number of times when $\lambda > \omega$, and for $n \in N \cup \{0\}$

$$\left\| \frac{(\lambda - \omega)^{n+1}}{n!} \left(\frac{d}{d\lambda} \right)^n (\lambda(\lambda^2 I - A)^{-1}) \right\| \leqslant M,$$
$$\left\| \frac{(\lambda - \omega)^{n+1}}{n!} \left(\frac{d}{d\lambda} \right)^n (G(\lambda^2 I - A)^{-1}) \right\| \leqslant M.$$

3.2.17. [396] Under the conditions of Theorem 2 we have

$$\text{(i)} \quad \exp(tG) = C(t, A) + GS(t, A),$$
$$C(t, A) = (\exp(tG) + \exp(-tG))/2;$$
$$\text{(ii)} \quad \exp(t\mathscr{A}) = \begin{pmatrix} G^{-1} & O \\ O & I \end{pmatrix} \exp\left(t \begin{pmatrix} O & G \\ O & G \end{pmatrix} \right) \begin{pmatrix} G & O \\ O & I \end{pmatrix}, \quad t \in \mathbf{R}.$$
$$\text{(iii)} \quad \exp(t\mathscr{A})(x, y) = (C(t, A)x + S(t, A)y,$$
$$AS(t, A)x + C(t, A)y), \quad (x, y) \in D(G) \times E, \quad t \in \mathbf{R}.$$

3.3. Spectral Properties in the Theory of Cosine Operator Functions. As in the case of operator semigroups, necessary and sufficient conditions for A to generate a COF can be stated in terms of conditions on the spectrum and bounds for the resolvent — see 3.1.

3.3.1. [327] Assume that we are given a COF C(t,A). Then

$$\begin{aligned} &\text{(i)} \quad \operatorname{ch}(t, \overline{\sigma(A)}) \subseteq \sigma(C(t, A)); \\ &\text{(ii)} \quad \operatorname{ch}(t, \overline{P\sigma(A)}) = P\sigma(C(t, A)); \\ &\text{(iii)} \quad \operatorname{ch}(t, \overline{R\sigma(A)}) \subseteq R\sigma(C(t, A)). \end{aligned} \tag{3.1}$$

3.3.2. [328, 4] If $\mu \in R\sigma(C(t,A))$ and $\{\lambda_n\}_{n\in\mathbf{N}}$ is the set of roots of the equation $\mu = \exp(\lambda_n \cdot t)$, then $\lambda_{\tilde{n}} \in R\sigma(A)$ for some $\tilde{n}$ and $\lambda_n^2 \notin P\sigma(A)$ for any $n \in N$ and $\mu \in P\sigma(C(t,A)^*)$.

3.3.3. [327, 4] If $\mu \in C\sigma(C(t,A))$ and λ_n are as in Proposition 3.2, then $\lambda_n^2 \in C\sigma(A) \cup \rho(A)$. It is possible that $\lambda_n^2 \in (A)$ for all $n \in \mathbf{Z}$ in some cases.

3.3.4. [215, 280] If E = H is a Hilbert space and $A \in \mathscr{C}(\mathbf{M}, 0)$ or C(t,A) is a family of normal operators, then

$$\sigma(C(t, A)) = \operatorname{ch}(t] \overline{\sigma(A)}), \quad t \in \mathbf{R}.$$

3.3.5. [167] Let a COF C(t,A) satisfy condition (F) and assume that E = H is a Hilbert space. Then $\mu \in \rho(C(t,A))$ if and only if $\{z^2 : \cosh(zt) = \mu\} \subseteq \rho(A)$ and $\sup\{\| zR(z,A) \| : \cosh(zt) = \mu\} < \infty$.

3.3.6. [306] Let $A \in \mathscr{C}(\mathbf{M}, 0)$. Then

(i) $\sigma(A) \subset \mathbf{R}_-$;

(ii) if $E \neq \{0\}$, then $\sigma(A) \neq \varnothing$;

(iii) the spectrum $\sigma(A)$ is bounded if and only if $A \in B(E)$.

3.3.7. [13] For $A \in \mathscr{C}(\mathbf{M}, \omega)$ and the corresponding matrix operator $\mathscr{A} = \begin{pmatrix} O & I \\ A & O \end{pmatrix}$, that appears in the reduction of Cauchy problem (3.2.1) and is given in 3.2 we have

$$\{\lambda^2 : \lambda \in \sigma(\mathscr{A})\} = \sigma(A).$$

3.3.8. [4] There exist a COF C(t,A) and a Banach space E such that the sets $r_1 := \{t : 0 \in \rho(C(t, A))\}$, $r_2 := \{t : 0 \in P\sigma(C(t, A))\}$, and $r_3 := \{t : 0 \in C\sigma(C(t, A))\}$ are dense in R, where $R = r_1 \cup r_2 \cup r_3$.

3.3.9. [133] For a COF $C(t, A) = \sum_{k=0}^{\infty} \frac{t^{2k} A^k}{(2k)!}$, $A \in \mathscr{B}$, defined in a Banach algebra with identity, we have

(i) $0 \leqslant \omega^c(A) < \infty$;

(ii) $\lambda R(\lambda^2, A) = \int_0^\infty \exp(-\lambda t) C(t, A)\, dt$ for $\operatorname{Re}\lambda > \omega^c(A)$;

(iii) $R(\lambda^2, A) = \int_0^\infty \exp(-\lambda t) S(t, A)\, dt$ for $\operatorname{Re}\lambda > \omega^c(A)$;

(iv) $C(t, A) = \frac{1}{2\pi i} \int_\gamma \exp(\lambda t) \lambda R(\lambda^2, A)\, d\lambda$, $t \in \mathbf{R}$;

(v) $S(t, A) = \frac{1}{2\pi i} \int_\gamma \exp(\lambda t) R(\lambda^2, A)\, d\lambda$, $t \in \mathbf{R}$,

where γ is some curve that includes the spectrum of $A \in \mathscr{B}$.

3.3.10. [133] Under the assumptions of Proposition 3.9 we have

$$\omega^c(A)^2 = \sup_{\lambda \in \sigma(A)} (|\lambda| + \operatorname{Re}\lambda)/2.$$

3.4. Supplementary Comments. The foundations of the theory of COF's were laid in [280-286, 208-216, 381, 396-399].

For more on reduction of the problem $u'' + Bu' + Au = 0$, $u(0) = u^0$, $u'(0) = u^1$, to a system of equations, see [164, 254-256].

The case of $\rho(A) = \varnothing$ for problem (2.1) was considered in [389]. The construction of solution families is analogous to COF's, but the existence of the resolvent $(\lambda^2 \cdot I - A)^{-1}$ is assumed on some $Y \subset E$ with a norm stronger than the original. A different approach to definition of the sine and cosine operators was taken in [56].

An attempt to define a COF in terms of easily verified properties of the operator A has led (see [238]) to an unbounded COF.

It was shown in [349] that $D(A_b^{1/2+\varepsilon}) \subseteq E^1 \subseteq D(A_b^{1/2-\varepsilon})$, where $A_b := A - bI$ with appropriate b and $\varepsilon > 0$.

The properties of $\mathfrak{A}_p$ are used in automatic control theory — see [400]. A saturation theorem for COF's was proved in [201].

Linear affine COF's were discussed in [234-235].

4. OPERATOR SEMIGROUPS AND COSINE OPERATOR FUNCTIONS THAT ARE CONTINUOUS IN THE UNIFORM OPERATOR TOPOLOGY

4.1. Operator Semigroups that are Continuous in the Uniform Operator Topology.

4.1.1. For any bounded operator $B \in B(E)$ the series $\sum_{k=0}^{\infty} (tB)^k/k!$ converges uniformly and the operator function

$$\exp(tB) = \sum_{k=0}^{\infty} (tB)^k/k!, \quad t \geqslant 0, \tag{1.1}$$

is an OS with generator B.

Definition 1. An OS exp(tA) is said to be continuous in the uniform operator topology (continuous with respect to norm) beginning at t_0 (sometimes we will say "becomes continuous with respect to norm") if there exists a $t_0 > 0$ such that the function $t \to \exp(tA)$ from $[t_0, \infty)$ into B(E) is continuous in the uniform operator topology. We say that an OS is continuous with respect to norm if $t_0 = 0$.

4.1.2. If $B \in B(E)$, then the OS exp(tB) is continuous with respect to norm.

THEOREM 1. [35] An OS exp(tA) is continuous with respect to norm if and only if $A \in B(E)$.

4.1.3. [41] If $A \in \mathcal{G}(\mathbf{M}, \omega)$ and D(A) is a set of the second category in E, then $\lim_{\tau \to 0-} \| \exp(\tau A) - I \| = 0$ and $A \in B(E)$.

4.1.4. [101] If $\tilde{E}_{\alpha_0} \subseteq D(A)$ for some α_0, then $A \exp(tA) \in B(E)$ for $t \geq \alpha_0$ and the OS exp(tA) is continuous with respect to norm when $t > \alpha_0$.

4.1.5. [341] If $\tilde{E}_\alpha \subseteq D(A)$ for all $\alpha > 0$, then the OS exp(tA) is n-times continuously differentiable in the uniform operator topology with $t > 0$ no matter what the $n \in \mathbb{N}$.

4.1.6. [126] If exp(tA) is a group of operators and $\overline{\lim}_{\tau \to 0-} \| \exp(\tau A) - I \| < 2$, then the generating operator $A \in B(E)$.

4.1.7. [341] If $\tilde{E}_\alpha \subseteq D(A)$ for all $\alpha > 0$ and $\overline{\lim}_{t \to 0+} t \| A \exp(tA) \| \leqslant 1/e$, then $A \in B(E)$.

4.1.8. [10] For a C_0-semigroup exp(tA) the following conditions are equivalent (the same holds for w-continuous semigroups):

(i) there exist $\varepsilon > 0$ and $\delta > 0$ such that

$$\| \exp(tA) - I \| \leqslant 1 - \varepsilon \text{ for } 0 \leqslant t < \delta;$$

(ii) $\lim_{t \to 0+} \| \exp(tA) - I \| = 0$.

4.1.9. [231] If $A \in \mathcal{H}(\theta, \omega)$ and $-A \in \mathcal{G}(\mathbf{M}, \omega)$, then $A \in B(E)$.

4.1.10. [101] If there exists an unbounded open connected set Ω that contains zero and numbers $0 \le \alpha < \beta$ such that $\sigma(\exp(tA)) \cap \Omega = \varnothing$ for $t \in [\alpha,\beta]$, then $A \in B(E)$.

4.1.11. [219] If an OS $\exp(tA)$ is continuous with respect to norm beginning with some t_0 and semi-Fredholm, then $A \in B(E)$.

4.1.12. [101] Assume that the semigroup $\exp(tA)$ is strongly continuous for $t > 0$ and $\exp(t_0A) \in B_0(E)$ for some $t_0 > 0$. Then the operator $\exp(tA)$ is compact for all $t > t_0$ and the semigroup $\exp(tA)$ is continuous with respect to norm for $t > t_0$. Note that here we have not assumed that the semigroup $\exp(tA)$ is a C_0-semigroup (OS).

Definition 2. A Banach space E is said to be a Grothendieck space if any sequence $\{x_n^*\}$, $x_n^* \in E^*$, that converges in the *-weak topology to zero converges to zero in a weak topology.

Definition 3. We will say that a space E has the Dunford—Pettis property if for any other Banach space X each weakly compact operator $T:E \to X$ maps weakly convergent sequences into sequences that converge with respect to norm.

4.1.13. [300] An example of a Grothendieck space with the Dunford-Pettis property is L^∞.

4.1.14. [300] An OS $\exp(tA)$ defined on a Grothendieck space with the Dunford—Pettis property is continuous with respect to norm, i.e., $A \in B(E)$.

4.1.15. [300] Let an OS $\exp(tA)$ be given in a space E with the Dunford—Pettis property. If the family $\exp(tA)^{**}$ is an OS on E^{**}, the OS $\exp(tA)$ is continuous with respect to norm.

4.1.16. [363] Let $A \in B(E)$, where E is a Banach structure. The following conditions are equivalent:

(i) the OS $\exp(tA)$ is positive;

(ii) $\langle \varphi, \operatorname{sign}_{x^{**}}(Ax)^{**} \rangle \leqslant \langle A|x|, \varphi \rangle$ for all $x \in E$ and $o \leqslant \varphi \in E^*$;

(iii) if $0 \leqslant x \in E$, $0 \leqslant \varphi \in E^*$, are such that $\langle x,\varphi \rangle = 0$, the $\langle Ax,\varphi \rangle \ge 0$;

(iv) the operator $A + \|A\| \cdot I \ge 0$.

4.1.17. [363] If $A, B \in B(E)$, where E is a Banach structure, the the following conditions are equivalent:

(i) $|\exp(tB)x| \leqslant \exp(tA)|x|$ for any $x \in E$;

(ii) $\langle A|x|, \varphi \rangle \geqslant \langle \varphi, \operatorname{sign}_{x^{**}}(Bx)^{**} \rangle$ for all $x \in E$ and $0 \leqslant \varphi \in E^*$;

(iii) $A \geqslant B$ and $A+B \geqslant -\|A+B\| \cdot I$.

4.1.18. [363] If A, B $\in$ B(E) and the operators C and D are invertible in B(H), where $\sup\limits_{t \in R} \|I - C\exp(-tA) \times D\exp(tB)\| \leqslant 1$, then the operators A and B are similar.

4.2. Cosine Operator Functions that are Continuous in the Uniform Operator Topology.

Definition 1. A COF C(t,A) is continuous in the uniform operator topology (continuous with respect to norm) if the function $C(t,A) : \mathbf{R} \to B(E)$ is continuous in the operator norm.

4.2.1. Let A COF C(t,A) be continuous in the uniform operator topology. Then $A \in B(E)$ and

$$C(t, A) = \sum_{k=0}^{\infty} t^{2k}A^k/(2k)!, \quad t \in \mathbf{R}, \tag{2.1}$$

where the series converges uniformly with respect to t on each finite segment [0,T]. Sometimes, by analogy with the scalar case, series (2.1) is written as $\cosh(tA^{1/2})$.

4.2.2. [396] Let $A \in B(E)$. Then series (2.1) is a COF with generating operator A.

THEOREM 2. [134] Each of the following conditions is equivalent to continuity with respect to norm for C(t,A):

(i) $\lim\limits_{t \to 0} \|C(t, A) - I\| = 0$;

(ii) $\lim\limits_{t \to 0} \|t^{-1}S(t, A) - I\| = 0$;

(iii) the generating operator A is bounded;

(iv) $\mathcal{R}(C(t, A)) \subset E^1$ for all $t \in (\alpha,\beta)$ for some $\alpha < \beta$;

(v) the inclusion $\mathcal{R}(S(t, A)) \subset D(A)$ and strong continuity of the function $r \to AS(t,A)$ occur for all $t \in (\alpha,\beta)$ for some $\alpha < \beta$.

4.2.3. [7] The generating operator A of a COF with a nonquasianalytic weight χ is bounded if and only if one of the following conditions is satsified:

(i) for some $\varepsilon > 0$ we have $\sup_{0<t<\varepsilon} \|C(t,A) - I\| < 1$;

(ii) the COF C(t,A) is the restriction to $\mathbf{R}$ of an entire function $\tilde{C}: \mathbf{C} \to B(E)$ of exponential type (equal to r^*A)).

4.2.4. [397] Assume that the COF C(t,A) is strongly twice differentiable at zero. Then $A \in B(E)$.

4.2.5. [306] Let $A \in \mathscr{C}(\mathbf{M}, 0)$. Then the operator $A \in B(E)$ if and only if the spectrum $\sigma(A)$ is bounded.

4.2.6. Let the COF C(t,A) be continuous with respect to norm. Then

(i) $\lim\limits_{t \to 0} \left\| (2/t^2) \int\limits_0^t S(\tau, A)\, d\tau - I \right\| = 0$;

(ii) for sufficiently small h the operator $\int\limits_0^h S(\tau, A)\, d\tau$ has a bounded inverse;

(iii) for sufficiently small h we have

$$A = (C(h, A) - I)\left(\int_0^h S(\tau, A)\, d\tau \right)^{-1}.$$

4.2.7. [371] Any COF C(t,A) defined on a Grothendieck space with the Dunford—Pettis property is continuous with respect to norm, i.e., $A \in B(E)$.

4.2.8. [7] If $B \in B(E)$ and the COF C(t,B) is uniformly bounded with respect to $t \in \mathbf{R}$, we have the Bernshtein inequality

$$\| B \| \leqslant r(B) \cdot \sup_{t \in \mathbf{R}} \| C(t, B) \|,$$

where r(B) is the spectral radius of the operator B.

4.2.9. [34, 342] Let $B \in B(E)$. Then the functions $C(t, -B^2)x$ and $S(t, -B^2)y$ are uniformly bounded with respect to $t \in \mathbf{R}$ for all $x, y \in E$ if and only if on E there exists an equivalent norm $\| \cdot \|_*$ such that $\| \exp(itB) \|_* \leq 1$ for $t \in \mathbf{R}$ (such operators are said to be Hermitian equivalent on E).

4.2.10. [34, 342] An operator B is Hermitian equivalent on E if and only if there exists a constant $C > 0$ such that

$$\| \sin(tB) \| \leqslant C, \quad t \in \mathbf{R}.$$

4.2.11. [133] Assume were are given, in a Banach algebra $\mathscr{B}$ with identity, a COF $C(t, A)$, $A \in \mathscr{B}$. Then, for $\nu^2 > \sup_{\lambda \in \sigma(A)} (|\lambda| + \operatorname{Re} \lambda)/2$ we have

$$C(t, A) = \frac{1}{2\pi i} \int_{\nu - i\infty}^{\nu + i\infty} e^{\lambda t} \lambda R(\lambda^2, A)\, d\lambda, \quad t \geqslant 0,$$

$$S(t, A) = \frac{1}{2\pi i} \int_{\nu - i\infty}^{\nu + i\infty} e^{\lambda t} R(\lambda^2, A)\, d\lambda, \quad t \geqslant 0.$$

4.3. Supplementary Comments. In [299] there is a discussion of the behavior of the spectrum $\sigma(T(t))$ of an operator function T(t) that is a continuous function of t in the uniform operator topology.

Conditions under with an OS $\exp(tA)$ is representable in the form $\exp(tA) = P \exp(tB)$, where $P, B \in B(E)$, $PB = BP$ and the operator $I - P$ is of finite rank, were considered in [300]. See also Propositions 2.6.13, 2.7.25, 2.7.47, and 8.1.4.

5. ALMOST PERIODIC AND PERIODIC OPERATOR SEMIGROUPS AND COSINE OPERATOR FUNCTIONS

Assume we are given a function $f(t) : \mathbf{R} \to E$. A number $\tau \in \mathbf{R}$ is said to be an ε-period of the function f(t) if

$$\|f(t+\tau)-f(t)\| \leqslant \varepsilon, \ t \in \mathbf{R}.$$

We denote the set of all ε-periods of the function f by $J(f,\varepsilon)$.

Definition 1. We say that a function f(t) is almost periodic (and write a.p.) if for any $\varepsilon > 0$ the set $J(f,\varepsilon)$ is relatively dense in **R**. This means that for any $\varepsilon > 0$ there exists a number $l_\varepsilon > 0$ with the following property: any subinterval of length l in **R** intersects $J(f,\varepsilon)$, i.e.,

$$[a, a+l_\varepsilon] \cap J(f, \varepsilon) \neq \varnothing \quad \text{for all } a \in \mathbf{R}.$$

5.0.1. For any a.p. function f(t) and any $\varepsilon > 0$ the set of ε-periods $\{\tau_\varepsilon\}$ is closed.

5.0.2. Every periodic function is almost periodic.

5.0.3. [118] Assume that an almost periodic function $f(t) : \mathbf{R} \to E$ is continuous. The function f(t) is uniformly continuous on **R**.

5.0.4. [118] If f(t) is an a.p. function, then the set of values $R_f := \{x \in E : x = f(t), t \in \mathbf{R}\}$ is relatively compact in E.

5.0.5. [118] If $f : \mathbf{R} \to E$ is an a.p. function that has a uniformly continuous derivative f'(t) on **R**, then f' is an a.p. function.

THEOREM 1. ([107, 318]) Let $f \in C(\mathbf{R};E)$. Then f is an a.p. function if and only if the family of function $\{f(t + h)\}_{-\infty<h<\infty}$ is compact in $C(\mathbf{R};E)$.

5.0.6. [61] The sum f(t) + g(t) of two a.p. functions is an a.p. function. The product of an a.p. function f and a numerical a.p. function $\varphi(t)$ is an a.p. function, $\varphi(t)f(t) : \mathbf{R} \to E$.

5.0.7. [118] For every a.p. function f the following limit exists:

$$a(\lambda, f) := \lim_{T\to\infty} (2T)^{-1} \int_{-T}^{T} f(t) e^{-i\lambda t} dt \quad \text{for any} \quad \lambda \in \mathbf{R}.$$

5.0.8. [118] Under the assumptions of 0.7 the following limit exists:

$$a(0, f) := \lim_{T\to\infty} (2T)^{-1} \int_{c-T}^{c+T} f(t) dt \ \text{uniformly with respect to } c \in \mathbf{R}.$$

Definition 2. The function $a(\lambda,f)$ is called the Bohr transform of f. We call the numbers λ for which $a(\lambda,f) \neq 0$ the Fourier spectrum of F and denote them by SP(f).

5.0.9. [118] The Fourier spectrum of any a.p. function is no larger than countable.

5.0.10. [118] If, under the assumptions of 0.7, SP(f) does not contain a sequence converging to zero, the function $\int_0^t f(s) ds$ is almost periodic.

Definition 3. A continuous function $f \in C(\mathbf{R};E)$ is said to be recurrent if for any numbers ε and T the set

$$L(\varepsilon, T; f) := \{\tau \in \mathbf{R} : \sup_{|t|<T} \| f(t+\tau) - f(t) \| \leqslant \varepsilon\}$$

is relatively dense in **R**.

Definition 4. A subset $\mathscr{E} \subset \mathbf{R}$ is said to be harmonious if for any $\varepsilon > 0$ the set $\bigcap_{\lambda \in \mathscr{E}} \{\tau : |e^{i\lambda\tau} - 1| \leqslant \varepsilon\}$ is relatively dense in R.

Definition 5. A family of functions $\mathscr{F} = \{f\}$ that maps **R** into E is said to be uniformly almost periodic if the set $\bigcap_{f \in \mathscr{F}} J(f, \varepsilon)$ is relatively dense in **R** for each $\varepsilon > 0$.

5.0.11. [144] If a set $\mathscr{E}$ is harmonious, there exists a number $\delta > 0$ such that the distance between arbitrary points of $\mathscr{E}$ is greater than δ.

Definition 6. A function $f : \mathbf{R} \to E$ is said to be weakly a.p. if $\langle f(t), y^* \rangle$ is a.p. for all $y^* \in E^*$. Weak periodicity is defined similarly.

5.1. Almost Periodic and Periodic Operator Semigroups.

Definition 1. An OS exp(tA) is said to be an almost periodic semigroup (and we write a.p. OS) if for each $x \in E$ the function $\exp(tA)x : \mathbf{R} \to E$ is an a.p. function.

Definition 2. An OS exp(tA) is said to be uniformly almost periodic if the family of functions $\{\exp(tA)x : x \in E, \|x\| = 1\}$ is uniformly almost periodic.

Weak almost periodicity for an OS is defined analogously by way of a.p. functions of the form $\langle \exp(tA)x, x^* \rangle$ for all $x \in E$, $x^* \in E^*$.

5.1.1. [15] For an a.p. OS exp(tA) the space E can be factored into a direct sum $E = E_{st} \oplus E_{ap}$ (see 2.7 for definitions).

Definition 3. The subspaces E_{st} and E_{ap} are called the interior and the boundary, respectively.

5.1.2. [15] The subspaces E_{st} E_{ap} are invariant with respect to the OS exp(tA).

5.1.3. [144] An a.p. OS exp(tA) can always be imbedded in an a.p. group $\exp(t\tilde{A})$ as follows:

$$\exp(t\tilde{A}) = \begin{cases} \exp(tA), & t \geqslant 0, \\ \exp(-tA)^{-1}, & t < 0. \end{cases}$$

THEOREM 1. [144] A groups of operators exp(tA) is a.p. if and only if the following three conditions are satisfied:

i) the group exp(tA) is uniformly bounded;

ii) the spectrum $\sigma(A) \subseteq i\mathbf{R}$;

(iii) the set of linear combinations of the characteristic vectors of the generating operator A is dense in the space E.

5.1.4. [144] Let E be weakly sequentially complete (for example, in particular, E might be reflexive). In this case a group exp(tA) is an a.p. group if and only if it is weakly a.p.

5.1.5. [144] Let exp(tA) be a uniformly bounded group in a reflexive space E. If the function exp(tA)x is almost periodic for all $x \in \mathscr{R}(A)$, then exp(tA) is an a.p. group.

5.1.6. [144] Assume that the group exp(tA) is almost periodic. Let $i\eta$ ($\eta \in \mathbf{R}$) be an isolated point of the spectrum $\sigma(A)$. Then $i\eta$ is a simple pole of $R(\lambda, A)$, $i\eta \in P\sigma(A)$, and the corresponding Riesz projector is given by the formula

$$P(i\eta)x = \lim_{t \to \infty} t^{-1} \int_0^t e^{-i\eta s} \exp(sA)x\, ds. \tag{1.1}$$

5.1.7. [15] Let $A \in \mathscr{G}(\mathbf{M}, 0)$ and assume that $\sigma(A) \cap i\mathbf{R}$ is no larger than countable; also, for each characteristic value $i\mu \in \sigma(A)$ ($\mu \in \mathbf{R}$) of the operator A^* and each functional $f \in \mathscr{N}(i\mu I^* - A^*)$, $f \neq 0$, assume there exists an element $x \in \mathscr{N}(i\mu I - A)$, such that $\langle x, f \rangle \neq 0$. Then the OS exp(tA) is a.p.

5.1.8. [15] Let $A \in \mathscr{G}(\mathbf{M}, 0)$ and let the space E be reflexive. If $\sigma(A) \cap i\mathbf{R}$ is no larger than countable, the OS exp(tA) is a.p.

5.1.9. [15] If an OS exp(tA) is a.p. then

$$\operatorname{nul}(i\mu I - A) = \operatorname{nul}(i\mu I^* - A^*)$$

for all $\mu \in \mathbf{R}$. In particular, $\mathcal{N}(i\mu I - A) = \{0\}$ if and only if $\mathcal{N}(i\mu I^* - A^*) = \{0\}$.

5.1.10. [144] If a group exp(tA) is almost periodic and $P\sigma(A)$ is bounded, then the generating operator $A \in B(E)$, i.e., is bounded.

5.1.11. [144] An a.p. OS (see 1.3) can be uniformly imbedded in an a.p. group.

5.1.12. [144] A group exp(tA) is uniformly almost periodic if and only if the following three conditions are satisfied:

(i) the group exp(tA) is uniformly bounded;

(ii) the set $(1/i)\sigma(A)$ is a harmonious subset of $\mathbf{R}$;

(iii) the set of linear combinations of the characteristic vectors of the generating operator A is dense in the space E.

5.1.13. [144] If exp(tA) is uniformly almost periodic, then $\sigma(A)$ consists of simple poles of the resolvents $R(\lambda,A)$ and $\sigma(A) = P\sigma(A)$.

5.1.14. [144] Let $A \in B(E)$. In this case the group exp(tA) is a uniformly a.p. group if and only if it is uniformly bounded and E is representable as the direct sum of a finite number of characteristic subspaces of the operator A.

5.1.15. [143] A uniformly bounded OS exp(tA) is periodic with period T if and only if the following two conditions are satisfied:

(i) the set $(T/2\pi i)\sigma(A) \subseteq \mathbf{Z}$, the integers;

(ii) the set of linear combinations of characteristic vectors of the generating operator A is dense in E.

5.1.16. [143] Assume that the OS exp(tA) is periodic with period T. Then

(i) the resolvent $R(\lambda,A)$ is a meromorphic function with only first-order poles;

(ii)

$$R(\lambda, A) = (1 - e^{-\lambda T})^{-1} \int_0^T e^{-\lambda s} \exp(sA)\, ds;$$

(iii) the orthogonal spectral projectors corresponding to the poles λ_k are of the form

$$P(\lambda_k) = \frac{1}{T} \int_0^T e^{-\lambda_k s} \exp(sA)\, ds;$$

(iv) for any $x \in E$

$$AP(\lambda_k)x = \lambda_k P(\lambda_k)x \quad \text{and} \quad \exp(tA)P(\lambda_k)x = e^{\lambda_k t} p(\lambda_k)x.$$

5.12.17. [143] The following statements are equivalent:

(i) the OS exp(tA) is periodic;

(ii) the OS exp(tA) is weakly periodic;

(iii) the OS exp(tA) is periodic as an operator function.

5.1.18. [143] Assume that the function $\exp(tA)x : \mathbf{R} \to E$ is periodic for every $x \in D(A)$. The OS exp(tA) is periodic.

5.1.19. [143] Assume that E = H and the OS exp(tA) is periodic. Then the following conditions are equivalent:

(i) the OS exp(tA) is a contraction OS;

(ii) the OS exp(tA) is unitary (see def in 7.1);

(iii) the OS exp(tA) is normal.

If the OS exp(tA) is also self-adjoint, then $\exp(tA) \equiv I$ for all $t \geq 0$.

5.1.20. [143] Let $A \in \mathcal{G}(\mathbf{M}, \omega)$, and let P be a projector. Then $\exp(2\pi A) = P$ if and only if the following conditions are satisfied:

(i) $\sigma(A) \subseteq i\mathbf{Z}$;

(ii) the spectrum $\sigma(A)$ consists of simple poles of the resolvent;

(iii) there exists a constant $M \geq 0$ such that

$$\|(I-P)R(\lambda, A)\| \leqslant M \max\{1, e^{-2\pi \operatorname{Re}\lambda}\}, \quad \operatorname{Re}\lambda \neq 0;$$

(iv) the set of linear combinations of the characteristic vectors of the generating operator is dense in E.

Definition 4. A function $f(t) : \mathbf{R} \to E$ is said to be almost automorphic if for any sequence $\{\alpha_n'\}_{n \in \mathbf{N}}$ there exists a subsequence $\{\alpha_n\} \subseteq \{\alpha_n'\}$ such that

$$\lim_{n \to \infty} f(t+\alpha_n) = g(t) \quad \text{for any} \quad t \in \mathbf{R};$$
$$\lim_{n \to \infty} g(t-\alpha_n) = f(t) \quad \text{for any} \quad t \in \mathbf{R}.$$

5.1.21. [427] Let $B \in B(E)$ and let u(t) be an almost automorphic solution of the equation $u'(t) = B(u(t))$, $t \in \mathbf{R}$. Then either $\inf_{t \in \mathbf{R}} \|u(t)\| > 0$, or $u(t) \equiv 0$ for $t \in \mathbf{R}$.

5.1.22. [428] Proposition 1.21 remains true if $A \in \mathcal{G}(\mathbf{M}, \omega)$, and u(t) is an almost periodic solution.

5.1.23. [426] Proposition 1.22 remains true if the condition of almost periodicity is replaced by the requirement that the solution $u(t) : \mathbf{R} \to D(A)$ be weakly almost periodic.

5.2. Almost Periodic and Periodic Cosine Operator Functions.

Definition 1. A COF or SOF is said to be a.p. (uniformly a.p.) if for all $x \in E$ the corresponding function C(t,A)x or S(t,A)x is a.p. (uniformly a.p.).

5.2.1. [165] If E is weakly sequentially complete, then a weakly a.p. COF is almost periodic.

THEOREM 1. [287] A COF C(t,A) is almost periodic if and only if the following three conditions are satisfied:

(i) the COF C(t,A) is uniformly bounded;

(ii) $\sigma(A) \subset \mathbf{R}_-$;

(iii) the set of linear combinations of the characteristic vectors of the generating operator A is dense in the space E.

If, under these conditions, $\mu \in \sigma(A)$ is an isolated spectral point, then μ is a simple pole of the resolvent $R(\lambda,A)$ and $E = \overline{\mathcal{R}(\mu I - A)} \oplus \mathcal{N}(\mu I - A)$.

5.2.2. [169] A COF C(t,A) is a.p. if and only if $A \in \mathcal{H}(\pi/2, 0)$ and the system of characteristic vectors of the generating operator A is complete in the space E.

THEOREM 2. [82] Cauchy problem (3.2.1) has an a.p. generalized solution for all $u^0, u^1 \in E$ if and only if conditions (i)-(iii) of Theorem 1 are satisfied and $0 \in \rho(A)$.

THEOREM 3. [82] A COF C(t,A) or an SOF S(t,A) is uniformly a.p. if and only if the following three conditions are satisfied:

(i) the COF C(t,A) or SOF S(t,A) are uniformly bounded with respect to $t \in \mathbf{R}$;

(ii) the set $(1/i)\sigma(A)$ is a harmonious subset in $\mathbf{R}$ and $0 \in \rho(A)$.

(iii) the set of linear combinations of characteristic vectors of the operator A is dense in E.

5.2.3. [82] If the COF C(t,A) is uniformly a.p., then $\sigma(A)$ consists of simple poles of the resolvent $R(\lambda,A)$. In this case $\sigma(A) = P\sigma(A)$.

5.2.4. [253] The following conditions are equivalent:

(i) the COF C(t,A) is periodic as an operator function;

(ii) the COF C(t,A) is strongly periodic;

(iii) the COF C(t,A) is weakly periodic.

THEOREM 4. [82, 218, 305]. The COF C(t,A) is periodic with period 2π if and only if the following three conditions are satisfied:

(i) $\sigma(A) \subseteq \{l\ ;\ l = -k^2, k \in \mathbf{Z}\}$;

(ii) the spectrum $\sigma(A)$ consists of simple poles of the resolvent;

(iii) the set of linear combinations of characteristic vectors of the operator A is dense in the space E.

When conditions (i)-(iii) are satisfied the Riesz projectors are given by the formulas

$$P(-k^2)x=\begin{cases}\frac{1}{\pi}\int_0^{2\pi}\cos(ks)C(s,A)x\,ds & \text{for } k\neq 0,\\ \frac{1}{2\pi}\int_0^{2\pi}C(s,A)\,ds & \text{for } k=0,\end{cases}$$

where, for $x \in D(A)$,

$$C(t,A)x=\sum_{k=0}^{\infty}\cos(kt)P(-k^2)x, \tag{2.1}$$

where the series converges uniformly with respect to $t \in \mathbf{R}$.

5.2.5. [218] When E = H and C(t,A) is 2π periodic, Eq. (2.1) holds for all $x \in E$, and convergence of the series is uniform with respect to $t \in \mathbf{R}$.

THEOREM 5. [83] The function $C(t,A)u^0 + S(t,A)u^1$ is 2π periodic for all $u^0, u^1 \in E$ if and only if conditions (i)-(iii) of Theorem 4 are satisfied and $0 \in \rho(A)$.

5.2.6. [218] A COF C(t,A) is periodic with period T if and only if the function $F(z) := (1 - e^{-Tz})zR(z^2,A)$ can be analytically continued to an entire function $\hat{F}(z)$ such that for $|z| > r$ we have

$$\|\hat{F}(z)\| \leqslant M \exp(q|z|^{2-\varepsilon}), \text{ where } q, M, r, \varepsilon > 0. \tag{2.2}$$

5.2.7. [218] A uniformly well-posed Cauchy problem (2.1) for n = 2 has only periodic solutions of period T if and only if $A \in \mathscr{C}(M,\omega)$ and the function F(z)/z (see Prop. 2.6) can be analytically continued to an entire function Q(z) such that for $|z| > r$ bound (2.2) is valid for the function Q(z).

5.2.8. [215] Assume that a COF C(t,A) is given in a Hilbert space H and C(t,A) is weakly a.p. Then $C(t,A) = Q^{-1}C(t,V)Q$, where V is a self-adjoint operator, $V := \Sigma_{\lambda \geq 0}\lambda P(\lambda)$, and $P(\lambda)$ is a family of orthogonal projectors.

5.2.9. [305] If a COF C(t,A) is periodic for every $x \in D(A)$, then C(t,A) is periodic.

5.2.10. [7] Let $(\sigma(-A))^{1/2} \cap \mathbf{R}_+$ be no larger than countable. Then all of the solutions of problem (3.2.1) will be almost periodic if and only if the following conditions are satisfied:

(i) the COF C(t,A) is uniformly bounded for $t \in \mathbf{R}$;

(ii) $0 \in \rho(A)$;

(iii) for every limit point λ_0 of the set $(\sigma(A))^{1/2}$ there exists a sequence $\varepsilon_n \in \mathbf{R}$ that converges to zero and is such that $\lim_{n\to\infty}\varepsilon_n(\varepsilon_n + i\lambda_0)[(\varepsilon_n + i\lambda_0)\cdot I - A]^{-1}x = 0$ for every $x \in E$.

5.2.11. [7] A COF C(t,A) is a.p. in the uniform operator topology if and only if it is uniformly bounded on $\mathbf{R}$ and $(\sigma(A))^{1/2}$ is a harmonious subset of $\mathbf{R}$.

5.2.12. [7] Let $A \in \mathscr{C}(M,\omega)$ and assume that $\sigma(-A)$ has no limit points in $\mathbf{R}_+$. Then

(i) the linear hull of characteristic and root vectors of A is dense in E if there exists a function $\chi(t)$ such that

$$\|C(t,A)\| \leqslant \chi(t) \text{ and } \chi(t) \leqslant k(1+|t|)^{\gamma} \text{ for } t \in \mathbf{R},\ \gamma \geqslant 0;$$

(ii) when condition $\lim \chi(t)/t = 0$ is satisfied, the COF C(t,A) is periodic with period 1 if and only if $\sigma(A) \subseteq \{-(2\pi k)^2, k \in \mathbf{N}\}$.

5.2. Supplementary Comments. Proposition 1.21 is also true when $A(t) : \mathbf{R} \to B(E)$ with separable E (see [428]).

6. COMPACTNESS IN THE THEORY OF OPERATOR SEMIGROUPS AND COSINE OPERATOR FUNCTIONS

We will denote the set of compact operators defined on E by $B_0(E)$ (or $B_0(E,F)$ if we are dealing with different spaces).

6.1. Compactness in the Theory of Operator Subgroups.

Definition 1. An OS exp(tA) is said to be compact beginning (or to become compact) at t_0 if, for $t > t_0$, the operator exp(tA) is compact. If $t_0 = 0$ then the OS exp(tA) is said to be compact.

6.1.1. [126] Assume that an OS exp(tA) becomes compact. Then for any number $\alpha > 0$ there exists a factorization of the space $E = E_f \oplus E_d$ such that E_f and E_d are invariant under the OS exp(tA), the subspace E_f is finite dimensional, and the restriction of the OS $\exp(tA)|_{E_d}$ to E_d satisfies the condition

$$\lim_{t\to\infty} e^{\alpha t}\exp(tA|_{E_d}) = O.$$

6.1.2. [126] If the generating operator of an OS exp(tA) has a compact resolvent, and the OS itself is continuous with respect to norm at the point t_0, then the OS exp(tA) is compact beginning at t_0.

THEOREM 1. [124] An OS exp(tA) is compact if and only if it is continuous with respect to norm for any $t > 0$ and the generating operator has a compact resolvent.

6.1.3. [124] If an OS exp(tA) is compact, then intersection $\sigma(A) \cap \{z \in \mathbf{C} : a < \operatorname{Re} z < b\}$ for any fixed $a, b \in \mathbf{R}$ contains a finite number of characteristic values of the operator A.

6.1.4. [126] If an OS exp(tA) is compact, then the spectrum $\sigma(A) = P\sigma(A)$.

We define com(A) := {t >0 : exp(tA) — I is a compact operator}.

6.1.5. [237] if $\operatorname{Com}(A) \neq \varnothing$, then the OS exp(tA) is invertible ($t \geq 0$).

THEOREM 2. [237] For an OS exp(tA) the following conditions are equivalent:

(i) $\operatorname{Com}(A) = \bar{\mathbf{R}}_+$;

(ii) the generating operator A is compact;

(iii) the operator $\lambda R(\kappa, A) - I$ is compact for all $\lambda > \omega(A)$.

6.1.6. [237] If Com(A) is a dense subset of $\mathbf{R}_+$ without interior points, then A is unbounded.

6.1.7. [237] If a generating operator A is unbounded and $\operatorname{Com}(A) \neq \varnothing$, then $\overline{\lim}_{\tau\to 0+} \| \exp(\tau A) - I \| \geqslant 2$.

Definition 2. An OS exp(tA) is said to be quasicompact if

$$\lim_{t\to\infty} \inf \{\| \exp(tA) - K \| : K \in B_0(E)\} = 0.$$

6.1.8. [126] For an OS exp(tA) the following conditions are equivalent:

(i) the OS exp(tA) is quasicompact;

(ii) $E\omega(\exp(tA)) < 0$ for any $t \geq 0$;

(iii) there exist $t_0 > 0$ and $K \in B_0(E)$ such that $\| \exp(t_0 A) - K \| < 1$.

6.1.9. [126] Let exp(tA) be a quasicompact OS. Then the set $\{\lambda \in \sigma(A) : \operatorname{Re} \lambda \geq 0\}$ is finite (possibly empty) and contains only poles of finite multiplicity. If we denote the poles by $\lambda_1, \ldots, \lambda_m$ and their multiplicities by K(1),...,k(m), we obtain

$$\exp(tA) = T_1(t) + \ldots + T_m(t) + R_m(t), \quad t \geqslant 0, \tag{1.1}$$

where $T_l(t) = \exp(\lambda_l t) \sum_{j=0}^{k(l)-1} t^j (A - \lambda_l I)^j P(\lambda_l)/j!$, $t \geqslant 0$ and $\| R_m(t) \| \leq c e^{-\varepsilon t}$ for some constants c and $\varepsilon > 0$.

6.1.10. [126] Assume that the OS exp(tA) becomes compact. Then the spectrum $\sigma(A)$ is a countable set $\{\lambda_1, \lambda_2, \ldots\}$ (it may be finite or empty) and contains poles of only finite multiplicity. Moreover, the set $\{\mu \in \sigma(A) : \operatorname{Re} \mu \geq r\}$ is finite for only $r \in \mathbf{R}$. If we denote the multiplicity of the poles by $k(l)$ and the corresponding projectors by $P(\lambda_k)$, we have, for each m, expansion (1.1), where $\| R_m(t) \| \leq c \exp((\varepsilon + \operatorname{Re} \lambda_m)t)$, $(\operatorname{Re} \lambda_{p+1} \leq \operatorname{Re} \lambda_p)$ for some constants c and ε.

6.2. Compactness in the Theory of Cosine Operator Functions.

Definition 1. A COF C(t,A) is said to be compact if the operator C(t,A) is compact for any $t \in \mathbf{R}$. An SOF S(t,A) is said to be compact if the operator S(t,A) is compact for any $t \in \mathbf{R}$.

6.2.1. [397] If an operator C(t,A) is compact for every $t \in (\alpha,\beta)$ for some $\alpha < \beta$, then the COF C(t,A) is compact.

6.2.2. [397] If an operator S(t,A) is compact for every $t \in (\alpha,\beta)$ for some $\alpha < \beta$, then the SOF S(t,A) is compact.

6.2.3. [307] If $\dim E = \infty$ and A is unbounded, then the operators $C(t_0,A)$ and $C(2t_0,A)$ cannot simultaneously be compact for any $t_0 > 0$.

6.2.4. [397] Under the assumptions of 6.2.1, it is necessary that $\dim E < \infty$.

THEOREM 1. [397] The following conditions are equivalent:

i) the SOF S(t,A) is compact;

(ii) the resolvent $R(\lambda^2,A)$ is compact for any $\lambda > \omega^c(A)$.

6.2.5. [397] Assume that an SOF S(t,A) is compact and the generating operator $B \in B(E)$. Then the operator $A+B \in \mathscr{C}(\mathbf{M},\omega)$ and the associated COF is compact.

THEOREM 2. [81, 250] The following conditions are equivalent:

(i) the generating operator A is compact;

(ii) the operator $\lambda^2 R(\lambda^2,A) - I$ is compact for every $\lambda > \omega^c(A)$;

(iii) the operator $S(t,A) - tI$ is compact for any $t \in \mathbf{R}$;

(iv) the operator $C(t,A) - I$ is compact for any $t \in \mathbf{R}$.

6.2.6. [250] Assume we are given an operator $C(t,A) - I$ that is compact for any $t \in \mathbf{R}$. Then operator $\lambda R(\lambda,A) - \mu R(\mu,A)$ is compact for all $\lambda, \mu \in \rho(A)$ such that $\operatorname{Re}\lambda, \operatorname{Re}\mu > \omega^c(A)$.

6.2.7. [250] Let $C(t,A) - I$ be compact for every $t \in (\alpha,\beta)$ for some $\alpha < \beta$. Then the operator $C(t,A) - I$ is compact for any $t \in \mathbf{R}$.

6.2.8. [250] If $S(t) - tI$ is a compact operator for every $t \in (\alpha,\beta)$ for some $\alpha < \beta$, then the operator $S(t) - tI$ is compact for any $t \in \mathbf{R}$.

6.2.9. [810] If S(t,A) is an SOF and the operator $C(t,A) - I$ is compact, then the space E must be finite dimensional.

6.3. Supplementary Remarks. Theorem 6.1.1 can be transferred to the case of nonlinear OS's [264]. The property $\operatorname{Com}(A) \neq \varnothing$ is essential for the study of Markov semigroups [237].

In [357] there are sufficient conditions for the operator

$$A = \sum_{|j|,|k| \leq 1} (-1)^j D^j a_{jk} D^k + q$$

with $q(x) \to \infty$ as $x \to \infty$ to have a compact resolvent. The case of compact $R(\lambda,A) - R(\lambda_0,A)$ is also considered, but here it is the behavior of the natural spectra with perturbations that is discussed.

See also Propositions 2.2.5, 2.5.7, 2.6.7, 2.6.13, 2.7.46, 2.8.32, 3.1.35, 4.1.12, 8.1.4., and 8.1.21.

7. UNIFORMLY BOUNDED OPERATOR SEMIGROUPS AND COSINE OPERATOR FUNCTIONS

For every element $y \in E$ we define the adjoint set

$$\theta(y) := \{y^* \in E^* : \|y^*\|^2 = \|y\|^2 = \langle y, y^* \rangle\}.$$

7.0.1. [301] The set $\theta(y)$ is nonempty for all $y \in E$, convex, closed, and such that $\theta(0) = \{0\}$.

Consider a mapping $\varphi : E \to E^*$ such that $\varphi(y) \in \theta(y)$ for all $y \in E$. Such a mapping φ is called the dual map. We fix some φ and set $[x,y] := \langle x, \varphi(y) \rangle$ for all $x,y \in E$. The function $[x,y]$, which maps $E \times E$ into $\mathbf{R}$ or $\mathbf{C}$, is called the semiscalar product.

7.0.2. [301] The semiscalar product has the following properties:

$$[x+y, z] = [x, z] + [y, z], \quad [\lambda x, y] = \lambda [x, y],$$
$$[x, x] = \|x\|^2, \quad |[x, y]| \leqslant \|x\| \cdot \|y\| \quad \text{for any} \quad x, y, z \in E.$$

Definition 1. An operator $A \in L(E)$ is said to be dissipative (with respect to the semiscalar product $[x,y]$) if

$$\mathrm{Re}[Ax, x] \leqslant 0 \quad \text{for all} \quad x \in D(A). \tag{0.1}$$

If the $\leq$ sign is replaced by $\geq$ in (0.1) we say that A is accretive.

Definition 2. By the numerical domain of an operator A we mean the set

$$W(A) := \{[Ax, x] : x \in D(A),\ \|x\| = 1\}.$$

We also set $\Theta(A) := \sup\{\mathrm{Re}\,\lambda : \lambda \in W(A)\}$.

7.0.3. [44] If an operator $B \in B(E)$ is dissipative, then the resolvent $R(\lambda, B)$ exists for any λ such that $\mathrm{Re}\,\lambda > 0$.

Definition 3. A dissipative operator A is said to be m-dissipative if $\rho(A) \cap \mathbf{R}_+ \neq \emptyset$. The operator $-A$ is said to be m-accretive if A is m-dissipative.

7.1. Uniformly Bounded Operator Semigroups.

THEOREM 1. [301] An operator A generates a contraction OS if and only if $\overline{D(A)} = E$ and A is m-dissipative.

7.1.1. [301] An operator A is dissipative if and only if for any $\alpha > 0$ the operator $I - \alpha A$ is injective and $\|(I - \alpha A)^{-1}\| \leq 1$.

7.1.2. [126] Let A be a dissipative operator. Then:

(i) if A is closed, then the closure $\bar{A}$ is also dissipative;

(ii) if $\overline{D(A)} \supseteq \mathcal{R}(A)$, in particular, if A is densely defined, then the operator A is closed;

(iii) A is m-dissipative if and only if $\mathcal{R}(I - hA) = E$ for all $h > 0$.

Definition 1. An operator is said to be T-accretive if it satisfies the condition

$$\|((I - \lambda A)^{-1}u - (I - \lambda A)^{-1}v)^+\| \leqslant \|(u - v)^+\|$$

for $\lambda > 0$ and $u, v \in E$.

7.1.3. [160] Any T-accretive operator in a Banach structure E is accretive if and only if dim E is less than three or the norm is p-additive for $p \geq 0$, i.e., $\|x + y\|^p = \|x\|^p + \|y\|^p$ $(p < \infty)$ or $\|x + y\| = \max(\|x\|, \|y\|)$ $(p = \infty)$.

7.1.4. [213] Assume that $A \in \mathcal{GR}(1, 0)$. Then

$$\mathrm{Re}\langle Ax, y\rangle = 0 \quad \text{for all} \quad x \in D(A) \ \text{ and } \ y \in \theta(x), \tag{1.1}$$

and the equation

$$(\lambda \cdot I - A) D(A) = E \tag{1.2}$$

is satisifed for all $\lambda \in \mathbf{R}$ $(\lambda \neq 0)$.

7.1.5. [213] If A is densely defined and both (1.1.) and (1.2) are satisifed for $\lambda_0 > 0$ and $\lambda_1 < 0$, then $A \in \mathcal{GR}(1, 0)$.

7.1.6. [126] The contraction group $\|\exp(tA)\| \leq 1$ is an isometry, i.e., $\|\exp(tA)x\| = \|x\|$ for all $x \in E$, $t \in \mathbf{R}$.

Definition 2. An operator A is equivalent to an m-accretive operator if there exists an equivalent norm in which it is m-accretive.

7.1.7. [296] Let A and A^2 both be equivalent to m-accretive operators. Then A generates a uniformly bounded analytic OS with angle $\pi/4$. In this case the operator A^2 generates an analytic OS with an angle θ if and only if A generates a uniformly bounded OS with angle $\pi/4 + \theta/2$ $(0 < \theta < \pi/2)$.

7.1.8. [296] Let A^{2^k} be equivalent to an m-accretive operator for $k = \overline{0, n}$. Then A generates a uniformly bounded analytic OS with angle $\pi(1 - 2^{-n})/2$.

Definition 3. An operator A is said to be totally accretive if for every natural $n \in \mathbf{N}$ the operator A^n is accretive. Total m-accretiveness is analogously defined.

7.1.9. [296] The following conditions are equivalent:

(i) A is equivalent to a totally m-accretive operator;

(ii) A generates an analytic OS with angle $\pi/2$;

(iii) for all $n \in N$ the operator A^n generatives an analytic OS with angle $\pi/2$.

7.1.10. [296] In a Hilbert space H the following conditions are equivalent:

(i) A is m-accretive and totally accretive;

(ii) A is totally m-accretive;

(iii) A is self-adjoint and positive.

7.1.11. [296] If A is m-accretive in a Hilbert space H, there exists only one m-accretive B such that $B^2 = A$. If A has spectrum $\sigma(A)$ and W(A) belongs to $\Sigma(\theta)$, then $\sigma(B)$ and W(B) are contained in $\Sigma(\theta/2)$.

7.1.12. [88] Assume that the operator semigroups $\exp(tA), \exp(tB), \exp(t(\overline{A+B})) \in \mathcal{G}(1, 0)$. Then

$$\exp(t(A+B))x = \lim_{k\to\infty} \{\exp\left(\frac{t}{k}A\right)\exp\left(\frac{t}{k}B\right)\}^k x$$

for each $x \in E$ and uniformly with respect to t in any compact set of **R**.

7.1.13. [229] Let $A \in \mathcal{G}(\mathbf{M}, 0)$, $x_1 \in \mathcal{N}(A)$, $x_2 \in \mathcal{R}(A)$. Then

$$\frac{1}{t}\int_0^t \exp(sA)(x_1+x_2)\,ds = x_1 + O(1/t) \quad \text{for } t\to\infty.$$

7.1.13. [229] Let $A \in \mathcal{G}(\mathbf{M}, 0)$ and assume that E is reflexive. Then $E = \mathcal{N}(A) \oplus \overline{\mathcal{R}(A)}$, $\mathcal{N}(A) \cap \mathcal{R}(A) = \{0\}$ and the projector $P(0) : E \to \mathcal{N}(A)$ along $\mathcal{R}(A)$ has norm $\|P(0)\| \le M$. Moreover, any $x \in E$ can be represented in the form

$$x = P(0)x + (I - P(0))x \in \mathcal{N}(A) \oplus \overline{\mathcal{R}(A)}$$

and

$$\lim_{t\to\infty} t^{-1}\int_0^t \exp(sA)x\,ds = P(0)x.$$

7.1.14. [229] For $A \in \mathcal{G}(\mathbf{M}, 0)$ the following statements are equivalent:

(i) $E = \mathcal{N}(A) \oplus \overline{\mathcal{R}(A)}$;

(ii) $\lim_{t\to\infty} t^{-1}\int_0^t \exp(sA)x\,ds$ exists for every $x \in E$;

(iii) $\lim_{\lambda\to 0+} R(\lambda, A)x$ exists for every $x \in E$;

(iv) $\lim_{\lambda\to 0+} R(\lambda, A)x$ exists for every $x \in \mathcal{R}(A)$.

7.1.15. [229] Assume that family of contractions (onto E) $\{\mathcal{V}(t)\}$, $t \ge 0$, has properties $\mathcal{V}(0) = I$, $\mathcal{V}'(0)x$ exists for all $x \in \mathfrak{B} \subset E$ and the closure A of the operator $\mathcal{V}'(0)|_{\mathfrak{B}}$ generates a contraction OS. Then for each $x \in E$ we have

$$\exp(tA)x = \lim_{k\to\infty} \mathcal{V}(t/k)^k x$$

uniformly with respect to t in any compact set of $\bar{\mathbf{R}}_+$.

7.1.16. [360] Assume there exists a $\delta > 0$ such that $\|\exp(tA) - I\| < 2$ for all $t \in (0,\delta)$. Then for every $t \in (0,\delta)$ and $x \in E$ we have

$$\lim_{t\to\infty} t^{-1}\int_0^t \exp(sA)x\,ds = \lim_{k\to\infty} k^{-1}\sum_{j=0}^{k} \exp(jtA)x$$

if one of the limits exists.

7.1.17. [44] Assume A_1, A_2, $\overline{A_1+A_2}\in\mathcal{G}(1, 0)$. Then

$$s\text{-}\lim_{k\to\infty}\left[\exp\left(\frac{t}{k}A_1\right)\exp\left(\frac{t}{k}A_2\right)\right]^k=\exp(t(A_1+A_2)),\ t\geqslant 0.$$

7.1.18. [341] Let $A\in\mathcal{G}(\mathbf{M}, 0)$. Then

$$\|Ax\|^2\leqslant 4\mathbf{M}^2\|x\|\cdot\|A^2x\|,\quad x\in D(A^2).$$

7.1.19. [229] Assume that under the assumptions of 1.18 the space E = H is a Hilbert space. Then for each $x\in D(A^2)$ we have

$$\|Ax\|\leqslant 2\cdot\|x\|\cdot\|A^2x\|, \tag{1.3}$$

where equality occurs if and only $A^2x + \lambda Ax + \lambda^2x = 0$ and $\mathrm{Re}(A^2x,x) = 0$ for some $\lambda\in\mathbf{R}_+$. If A generates an isometry group, the constant 2 in (1.3) can be replaced by 1. This inequality (along with Prop. 1.18) belongs to a series of moment inequalities (see [59]).

Definition 4. An OS exp(tA) is said to be weakly (strongly, uniformly) (C,α)-ergodic at infinity if the operator

$CH(t,\alpha)x:=\alpha t^{-\alpha}\int_0^t(t-s)^{\alpha-1}\exp(sA)x\,ds$ exists for all $t>0$; $\int_0^\infty e^{-\lambda t}\|CH(t,\alpha)x\|\,dt<\infty$ for all $x\in E$ and $\lambda >$ $\max(0,\omega(A))$ and if the limit $(C,\alpha)\text{-}\lim\exp(tA):=\lim_{t\to\infty}CH(t,\alpha)$ exists in the weak (strong, uniform) operator topology.

This is the so-called Cesaro limit.

Definition 5. An OS exp(t,A) is said to be weakly (strong, uniformly) Abel ergodic at infinity

$$A\text{-}\lim_{t\to\infty}\exp(tA):=\lim_{\lambda\to 0+}\lambda\int_0^\infty\exp(tA)\,dt\equiv\lim_{\lambda\to 0+}\lambda R(\lambda, A)$$

exists in the corresponding operator topology.

By setting $t\to 0+$ instead of $t\to\infty$ or $\lambda\to\infty$ instead of $\lambda\to 0+$, we obtain the definition of ergodicity at zero.

7.1.20. [101] An OS exp(tA) is weakly Abel ergodic at zero (infinity) if and only if it is strongly Abel ergodic at zero (infinity).

7.1.21. [101] If an OS exp(tA) with $\alpha\geq 0$ is weakly (C,α)-ergodic at zero, it is strongly (C,α)-erogodic at zero.

7.1.22. [101] Let exp(tA) be an OS that is strongly Abel ergodic at infinity and $A\text{-}\lim_{t\to\infty}\exp(tA)x:=Px$ for any $x\in E$. Then

(i) the operator P is a projector, i.e., $P = P^2$;

(ii) $P\exp(tA) = \exp(tA)P$ for all $t > 0$;

(iii) $PAx = 0$ for all $x\in D(A)$ and $APx = 0$ for all $x\in E$;

(iv) $\mathcal{R}(P)=\mathcal{N}(A)$;

(v) $E=\mathcal{R}(P)\oplus\mathcal{N}(P)$, $\mathcal{R}(P)\cap\mathcal{N}(P)=\{0\}$, $E=\overline{\mathcal{R}(A)}\ominus\mathcal{N}(A)$.

7.1.23. [101] The following conditions are equivalent for an OS exp(tA) with $\omega(A)\leq 0$:

(i) exp(tA) is strongly Abel ergodic at infinity;

(ii) for each $x\in E$ the set $\{\lambda R(\lambda,A)x\}_{0<\lambda<\omega}$ is relatively sequentially weakly compact;

(iii) $\|\lambda R(\lambda, A)\|\leqslant\mathbf{M}$ for $0<\lambda<\omega$ and $E=\mathcal{N}(A)\oplus\overline{\mathcal{R}(A)}$.

For convenience we will denote the strong, weak, -weak, and uniform limits by s-lim, w-lim, w*-lim, and u-lim, respectively.

7.1.24. [101] The following conditions are equivalent for an OS exp(tA) with $\omega\leq 0$:

(i) the OS exp(tA) is uniformly Abel ergodic at infinity and $u\text{-}(A)\text{-}\lim_{t\to\infty}\exp(tA)=P$, where $P^2 = P\in B(E)$;

(ii) the point $\lambda = 0$ is a simple pole of the resolvent $R(\lambda,A)$ and the residue of $R(\lambda,A)$ at zero is P;

(iii) $s\text{-}\lim\limits_{\lambda\to 0-} \lambda^2 R(\lambda, A) = O$ and $E = \mathscr{R}(A) \dotplus \mathscr{N}(A)$, and the subspace is closed;

(iv) $s\text{-}\lim\limits_{\lambda\to 0+} \lambda^2 R(\lambda, A) = O$ and $\mathscr{R}(A^2)$ is closed.

7.1.25. [101] If $\omega(A) \leq 0$ and the OS $\exp(tA)$ is uniformly Abel ergodic at infinity, then $\mathscr{R}(A^k) = \mathscr{R}(A)$ for $k \in \mathbf{N}$.

7.1.26. [300] Let E be a Grothendieck space with the Dunford–Pettis property. If the space $E/\bar{E}_0$ is separable, there exist operators $B, P \in B(E)$, such that $I - P$ is a projector of finite rank, $PB = BP$, and $\exp(tA) = P\exp(tB)$, $t \geq 0$.

7.2. Uniformly Bounded Cosine Operator Functions.

7.2.1. [232] There exist operators $A \in \mathscr{C}(\mathbf{M}, \omega)$, such that for any $b \in \mathbf{R}$ the operator $A + bI$ does not generate a uniformly bounded COF.

For all $x \in E$ and $a \in \bar{\mathbf{R}}_+$ we define the operator

$$F_a x := \frac{2}{\pi} \int_0^\infty \left(\frac{\sin(at)}{t}\right)^2 C(2t, A)\, x\, dt,$$

which is bounded (see [348]) when $A \in \mathscr{C}(\mathbf{M}, 0)$.

7.2.2. [348] Let $0 \leq a \leq b$. Then

$$F_a F_b x = F_b F_a x = 2\int_0^a F_u x\, du + (b-a) F_a x, \quad x \in E.$$

7.2.3. [348] Assume that for some $0 \leq a \leq b$ the equation $2\int_a^b F_t x\, dt = (b-a)(F_a x + F_b x)$ is satisfied for all $x \in E$. Then the open interval $(-b^2, -a^2) \subseteq \rho(A)$.

7.2.4. [348] The following equations are satisfied for the operator F_a with $a \in \mathbf{R}_+$:

$$F_a^n = (n-1)\, n \int_0^a (a-t)^{n-2} F_t\, dt, \quad n = 2, 3, \ldots,$$

$$\exp(itF_a) = I + itF_a - t^2 \int_0^a \exp(it(a-s))\, F_s\, ds.$$

Assume that a COF $C(t,A)$ is such that the operator

$$E_a x := \lim_{\alpha\to 0+} E_{a,\alpha} x := \lim_{\alpha\to 0-} \int_{\alpha+i0}^{\alpha+ia} [\lambda R(\lambda^2, A) + \bar{\lambda} R(\bar{\lambda}^2, A)]\, x\, d\lambda \tag{2.1}$$

(where $\lambda = \alpha + i\tau$) is linear and continuous for all $a \geq 0$.

7.2.5. [348] There are examples of uniformly bounded COF's for which the family $\{E_\alpha\}$ of (2.1) does not exist.

7.2.6. [405] Assume that for the COF $C(t,A)$ the family (2.1) is defined and $E_a = E_b$ for some $0 < a < b$. then

$$(-b^2, -a^2) \cap (R\sigma(A) \cup P\sigma(A)) = \varnothing.$$

7.2.7. [405] Let $E_{a,\alpha} x$ be bounded for $a \in [0, \bar{a}]$ and $\alpha \in [0, \bar{\alpha}]$ with any $\bar{a}, \bar{\alpha} \geq 0$, $\bar{\alpha} \neq 0$. Then, for all $x \in E$ and $a \in [0, \bar{a}]$, E_a exists, the operator E_a is bounded, and for all $0 \leq a \leq b$ we have $E_a E_b = E_b E_a = \pi i E_a$, where for almost all

$a \geq 0$ we have $E_a x = \frac{2}{\pi}\int_0^\infty \frac{\sin(at)}{t} C(t, A)x\,dt$ if the integral converges. Now, we define, for $0 \leq a \leq b$, $\Delta := (-b^2, -a^2)$ and $E_\Delta := E_b - E_a$.

7.2.8. [405] Under the assumptions of 2.7 with any two intervals Δ_1 and Δ_2 we have

$$E_{\Delta_1} E_{\Delta_2} = E_{\Delta_2} E_{\Delta_1} = E_{\Delta_1 \cap \Delta_2}.$$

7.2.9. [405] Let $\| E_{\psi,\varphi} x \| \leq \text{const}$ for any $\psi \geq 0$ and $\varphi > 0$, where $E_a = E_b$ for some $0 \leq a \leq b$. Then $(-b^2, -a^2) \subseteq \rho(A)$.

7.2.10. [405] Assume that under the assumptions of 2.7 $E_a x$ for any $x \in E$ is continuous at $a_0 \geq 0$. The $-a_0^2 \notin R\sigma(A)$.

7.2.11. [348] Let E be reflexive and strongly convex with Gateaux differentiable norm, $A \in \mathscr{C}(\mathbf{M}, 0)$, and let the operator C(t,A) for any $t \in \mathbf{R}$ have a real spectrum. Then $R\sigma(A) = \varnothing$.

7.2.12. [12] Let $A \in \mathscr{C}(1, 0)$ and let $\mathscr{A}$ be the operator of 3.2.1. Then, for $t \in [\ln 2, \infty)$,

$$\| C(t, A) \|_{B(E,E)} \leqslant 1, \quad \| C'(t, A) \|_{B(E^1,E)} \leqslant t \cdot \ln 2,$$
$$\| S(t, A) \|_{B(E,E^1)} \leqslant t + 1,$$

and the resolvent admits the bound $(\lambda \cdot I - \mathscr{A})^{-1}$

$$\| R(\lambda, \mathscr{A}) \| \leqslant (\operatorname{Re}\lambda - \ln 2)^{-1} \quad \text{for} \quad \operatorname{Re}\lambda > \ln 2.$$

7.2.13. [7] Let $0 \notin \sigma(A)$ and $A \in \mathscr{C}(M, 0)$. Then

$$\sup_{t \in \mathbf{R}} \| S(t, A) \| \leqslant \frac{\pi}{2} \operatorname{dist}(0, \overline{\sigma(A)}) \sup_{t \in \mathbf{R}} \| C(t, A) \|.$$

7.2.14. [7] In order for the set $\mathfrak{R} := \{x \in E: \sup \| S(t, A)x \| < \infty\}$ to be dense in E, it is necessary and sufficient that one of the following conditions be satisfied:

(i) $\lim_{n\to\infty} \varepsilon_n R(\varepsilon_n, A)x = 0$ for any $x \in E$ and some sequence $\varepsilon_n \in \mathbf{R}_+$ such that $\lim_{n\to\infty} \varepsilon_n = 0$;

(ii) the set $\mathscr{R}(A)$ is dense in E;

(iii) $\mathscr{N}(A^*) = \{0\}$.

7.2.15. [169] An operator $A \in \mathscr{C}(\mathbf{M}, 0)$ satisfies condition (F) if and only if the following condition is satisfied on every bounded interval $[a,b]$ (see Definition 3.2.2):

$$\sup \{ \| \exp(tG|_{D(G,\mu)}) \|_{D(G,\mu)} : \mu \in \mathbf{N}, \quad t \in [a, b] \} < \infty.$$

7.2.16. [215] Let E = H. Then there exist a self-adjoint operator and constant $M > 0$ such that $(e^{1/2}(2M + 1))^{-1} \cdot I \leq Q \leq MI$ and the operator $QC(t,A)Q^{-1}$ is self-adjoint for every $t \in \mathbf{R}$. In this case $C(t,A) = Q^{-1}\cos(tG)Q$ and $G^* = B \geq O(G := QAQ^{-1})$.

7.2.17. [236] Let $A \in \mathscr{C}(\mathbf{M}, 0)$. For any $x = y + z \in \mathscr{R}(A) \oplus \mathscr{N}(A)$

$$\left\| \frac{1}{T}\int_0^T C(t, A)x\,dt - z \right\| = O(|T|^{-1}),$$

$$\left\| \frac{1}{T}\int_0^T S(t, A)x\,dt - \frac{T}{2}z \right\| = O(|T|^{-1}) \quad \text{as} \quad T \to \infty.$$

7.2.18. [236] When $A\in\mathscr{G}(\mathbf{M},0)$ and E is reflexive we have $E=\overline{\mathscr{R}(A)}\oplus\mathscr{N}(A)$, where, for each $x\in E$, we have strong convergence of $T^{-1}\int_0^T C(t,A)x\,dt$ as $T\to\infty$.

7.2.19. [236] When $A\in\mathscr{G}(\mathbf{M},0)$ and $x\in D(A)$ we have the inequality

$$\sup_{t\geq 0}\|S(t,A)Ax\|^2\leq 4\sup_{t\geq 0}\|C(t,A)Ax\|\sup_{t\geq 0}\|C(t,A)x\|.$$

For $x\in E$ we set $x\in E$ $P_c x:=\lim_{t\to\infty}t^{-1}S(t,A)x$,

$$P_a x:=\lim_{\lambda\to 0+}\lambda\int_0^\infty e^{-\lambda t}C(t,A)x\,dt \text{ and } P_t x:=\lim_{n\to\infty}\frac{1}{n}\sum_{k=0}^{n-1}C(kt,A)x.$$

7.2.20. [370] The two operator P_c and P_a coincide and are projectors. We have

$$\mathscr{R}(P_c)=\mathscr{N}(A)=\bigcap_{s>0}\mathscr{N}(C(s,A)-I),$$
$$\mathscr{N}(P_c)=\overline{\mathscr{R}(A)}=\overline{\bigcup_{s>0}\mathscr{R}(C(s,A)-I)},$$
$$D(P_c)=\bigcap_{s>0}\mathscr{N}(C(s,A)-I)\oplus\bigcup_{s>0}\mathscr{R}(C(s,A)-I)=$$
$$=\{x\in E:\exists\{t_n\},\ t_n\to\infty,\ w\text{-}\lim_{n\to\infty}(S(t_n,A)x)/t_n \text{ exists}\}.$$

7.2.21. [370] For every $t>0$ the operator P_t is a projector and

$$\mathscr{R}(P_t)=\mathscr{N}(C(t,A)-I),\quad \mathscr{N}(P_t)=\overline{\mathscr{R}(C(t,A)-I)}$$
$$D(P_t)=\mathscr{N}(C(t,A)-I)\oplus\overline{\mathscr{R}(C(t,A)-I)}=$$
$$=\left\{x\in E:\exists\{n_k\},\ k\to\infty,\ w\text{-}\lim_{k\to\infty}n_k^{-1}\sum_{l=0}^{n_k-1}C(lt,A)x \quad \text{exists}\right\}.$$

7.2.22. [370] Assume that there exists a $\delta>0$ such that the operator $C(t,A)+I$ is invertible for $t\in(0,\delta)$ (in particular, this occurs when $\|C(t,A)-I\|<2$). The $P_t=P_c$ for all $t\in(0,2\delta)$.

7.2.23. [306] Let $A\in\mathscr{G}(1,0)$, and

$$C_m(t):=\left[\sum_{j=0}^{m}C_{2m}^{2j}\left(\frac{t}{2m}\right)^{2j}A^j\right]\left[\left(\frac{2m}{t}\right)^2 R\left(\left(\frac{2m}{t}\right)^2,A\right)\right]^{2m}.$$

Then $\lim_{m\to\infty}C_m(t)x=C(t,A)x$ for all $x\in E$ and $\|[C_m(t)-C(t,A)]x\|\leq t^2\|Ax\|/m^{1/2}$ for all $x\in D(A)$, $m\geq 2$.

7.2.24. [327] The functions C(t,A) and S(t,A) are uniformly bounded for all $x\in E$ if and only if there exists a constant $M\geq 1$ with the property

$$\left\|\frac{d^k}{dz^k}R(z^2,A)\right\|,\quad \left\|z\frac{d^k}{\partial z^k}R(z^2,A)+k\frac{d^{k-1}}{dz^{k-1}}R(z^2,A)\right\|\frac{Mk!}{k+1}$$
$$(k=0,1,2,\ldots).$$

7.3. Supplementary Comments. See also Propositions 2.2.12-13, 2.2.16, 2.7.22, 2.7.29, 2.7.46-48, 2.7.52, 4.2.5, 5.1.7, 5.1.8, 5.1.14, 7.1.4, 8.1.14. Formula (1.20) was generalized in [257].

In [59, 418] there are different forms of the moment inequalities for generating operators of OS's and COF's (see Proposition 7.1.18), and, for certain cases, the coefficients appearing in the inequalities are presented in explicit form.

8. THEORY OF PERTURBATIONS FOR OPERATOR SEMIGROUPS AND COSINE OPERATOR FUNCTIONS

If an operator $A \in \mathscr{C}(E)$, then by a perturbation of this operator we mean a linear operator G for which $D(G) \supseteq D(A)$. The operator A + G is said to be perturbed. If $B \in B(E)$, then it is clear that $A+B \in \mathscr{C}(E)$.

8.0.1. If a perturbation G is a closed operator, then there exist constants a and b such that

$$\|Gx\| \leqslant a\|x\| + b\|Ax\| \quad \text{for any} \quad x \in D(A).$$

Definition 1. If the preceding inequality is satisfied for some perturbation G, we say that the operator G is bounded with respect to A (or simply A-bounded, or subordinate to A). The smallest value b_0 of all possible constants in the preceding inequality is called the A-bound of the operator G (or the norm of G relative to A).

8.0.2. [388] A-boundedness of an operator G is equivalent to $D(G) \supseteq D(A)$ and $GR(\lambda, A) \in B(E)$ for some $\lambda \in \rho(A)$.

8.0.3. [44] Let $A, G \in L(E)$, where G is A-bounded with an A-bound less than 1. The operator A + G can be closed if and only if the operator A can be closed. In this case the closures of the operators A and A + G have the same domains. In particular, A + G is closed if and only if A is closed.

Let $\mathfrak{S}(t)$ denote either an OS or a COF.

Definition 2. A class of operators $\mathfrak{M}(\mathfrak{S})$ is said to be a class of Miyadera perturbations for the family $\mathfrak{S}(t)$ with generator A if for any operator $G \in \mathfrak{M}(\mathfrak{S})$ we have $D(G) \supseteq D(A)$ and there exists a constant $K > 0$ such that

$$\int_0^1 \| G\mathfrak{S}(t)x \| dt \leqslant K \| x \| \quad \text{for any} \quad x \in D(A).$$

We use K_∞ to denote the limit of $K_\lambda := \sup \left\{ \int_0^1 e^{-\lambda t} \| G\mathfrak{S}(t)x \| dt : \| x \| = 1, x \in D(A) \right\}$, as $\lambda \to \infty$; this limit exists because K_λ monotonically decreases with λ and is positive.

Definition 3. An operator $G \in \mathscr{C}(E)$ is said to be subordinate to the operator $A \in \mathscr{C}(E)$ with order α ($\alpha \in [0,1]$) if $D(G) \supseteq D(A)$ and $\| Gx \| \leq C_\alpha(G) \| x \|^{1-\alpha} \| Ax \|^\alpha$.

Definition 4. (see [59]) An operator $G \in \mathscr{C}(E)$ is said to be totally subordinate to an operator $A \in \mathscr{C}(E)$, if it is subordinate with order 1 and for any sufficiently small $\eta > 0$ we have

$$\| Gx \| \leqslant \Phi_\eta(x) + \eta \| Ax \| \quad \text{for any} \quad x \in D(A),$$

where Φ_η is a continuous convex functional on E.

8.1. Theory of Perturbations for Operator Semigroups.

THEOREM 1. [44]. Let $A \in \mathscr{G}(\mathbf{M}, \omega)$ and $B \in B(E)$. Then $A+B \in \mathscr{G}(\mathbf{M}, \omega + \mathbf{M}\|B\|)$. Also, the OS $\exp(t(A + B))$ with fixed $t \geq 0$ is a holomorphic function of B. In particular, $\exp(t(A + zB))$ is an entire function of the complex variable z.

8.1.1. [213] Theorem 1 is also valid for groups of operators.

8.1.2. [193] The perturbed OS in Theorem 1 satisfies the equation

$$\exp(t(A+B)) = \exp(tA) + \int_0^t \exp((t-s)A) B \exp(s(A+B)) ds, \quad t \geqslant 0$$

and can be represented as a series that converges absolutely and uniformly with respect to $t \in [0,T]$,

$$\exp(t(A+B)) = \sum_{n=0}^{\infty} S_n(t), \text{ где } S_0(t) = \exp(tA),$$
$$S_n(t) = \int_0^t \exp((t-s)A) B S_{n-1}(s)\, ds, \quad n \in \mathbf{N}, \tag{1.1}$$

or, in a different form (see [193]),

$$\exp(t(A+B)) = \exp(tA) + \sum_{k=0}^{\infty} \int_0^t \int_0^{t_1} \dots \int_0^{t_k} \exp(t_{k+1}A) B \exp((t_k - t_{k+1})A) B \dots \dots \exp((t-t_1)A)\, dt_{k+1} \dots dt_1. \tag{1.2}$$

8.1.3. [341] Let $A \in \mathcal{G}(\mathbf{M}, \omega)$ and $B \in B(E)$. Then

$$\|\exp(t(A+B)) - \exp(tA)\| \leqslant \mathbf{M} \exp(\omega t)(\exp(\mathbf{M} \cdot \|B\| \cdot t) - 1), \quad t \geqslant 0.$$

8.1.4. [34, 126] Let exp(tA) be an OS that is either analytic, or continuous with respect to norm, or compact, and $B \in B$. Then the OS exp(t(A + B)) is, correspondingly, either analytic, or continuous with respect to norm, or compact.

8.1.5. [126] Proposition 1.4 remains valid if $B \in B(\mathcal{D}(A))$. If in this case, we also have $B \in B_0(\mathcal{D}(A))$ and the OS exp(tA) is continuous with respect to norm, then the OS exp(t(A + B)) is also continuous with respect to norm.

8.1.6. [44] Let $A \in \mathcal{G}(\mathbf{M}, \omega)$, and assume that the operator G is bounded relative to A. Then, generally speaking, the operator A + G does not necessarily generate an OS.

8.1.7. [44] Let $A, G \in \mathcal{G}(1, 0)$ and assume that operator G is A-bounded with A-bound $b_0 \leq 1$. Then $A+G \in \mathcal{G}(1, 0)$.

8.1.8. [229] Assume that A is m-dissipative and densely defined, and that G is m-dissipative and A-bounded with A-bound $b_0 = 1$, while G^* is densely defined. The $\overline{A+G}$ is m-dissipative.

8.1.9. [229] In Proposition 1.8 the condition of denseness of $D(G^*)$ in E^* can be replaced by reflexivity of the space E.

8.1.10. [44] Assume that $A, G \in \mathcal{G}(1, 0)$, the set $D(A) \cap D(G)$ is dense in E, and the operator $A + G + \lambda \cdot I$ has a dense domain for sufficiently large λ. If we take the closure of A + G we have $\overline{A+G} \in \mathcal{G}(1, 0)$.

8.1.11. [35] Let $A \in \mathcal{G}(\mathbf{M}, \omega)$, $G \in \mathcal{C}(E)$, $D(A) \subseteq D(G)$, and $\int_0^t \|G \exp(tA)\|_*' dt < \infty$, where $\|G\|_*' :=$ $\sup\{\|Gx\| / \|x\| : \|x\| \leq 1, x \in D(A)\}$. Then the operator a + G has a closure and generates on OS that is representable in the form (1.1).

8.1.12. [316] If exp(tA) is a group of operators, and the operator G is such that D(D) = D(A), G is A-bounded and $\int_0^1 \|G \exp(tA)\|_*' dt < \infty$, then $\|G\|_*'$ is finite.

8.1.13. [193] Let $A \in \mathcal{G}(\mathbf{M}, \omega)$, $D(G) \supseteq \tilde{E}_0$, and $\int_0^1 \|G \exp(tA)\| dt < \infty$. Then $D(A) \subseteq D(G)$ and A + G generates an OS.

8.1.14. [229] Let $A \in \mathcal{G}(1, 0)$, assume that G is dissipative and totally subordinate to A. Then $A+G \in \mathcal{G}(1, 0)$.

8.1.15. [316] Let $A \in \mathcal{G}(\mathbf{M}, \omega)$ and $G \in \mathfrak{M}(\exp(tA))$, where $GR(\lambda, A) \in B(E)$ for some $\lambda > \omega(A)$. Then there exists a finite or infinite ε_0 such that for each $|\varepsilon| < \varepsilon_0$ the operator A + εG generates an OS. The proposition remains valid for groups of operators.

We now turn our attention to the fact that perturbations of the class $\mathfrak{M}$ may be unbounded operators (cf. Prop. 1.12).

8.1.16. [10] Assume that we are given two w-continuous OS's or C_0^*-semigroups such that

$$\|\exp(tA)-\exp(tG)\| = o(t) \quad \text{as} \quad t\to 0.$$

Then exp(tA) = exp(tG) for all $t \geq 0$.

8.1.17. [10] Given two w-continuous groups or C_0^*-groups exp(tAP) and exp(tG), the following conditions are equivalent:

(i) there exist $\varepsilon_1 > 0$ and $\delta_1 > 0$ such that

$$\|\exp(tA)\exp(-tG)-I\| \leqslant 1-\varepsilon_1 \quad \text{for} \quad 0 \leqslant t \leqslant \delta_1.$$

(ii) there exist $\varepsilon_2, \delta_2 > 0$ and $P, W \in B(E)$ such that $W^{-1} \in B(E)$, $A = W(G+P)W^{-1}$ and $\|\exp(tA)W^{-1}\exp(-tA)W - I\| \leq 1 - \varepsilon_2$ for all $t \in [0,\delta_2]$.

If these conditions are satisfied, then $W = \frac{1}{\delta_1}\int_0^{\delta_1} \exp(tA)\exp(-tG)\,dt$ and, consequently, $\|I - W\| \leq 1 - \varepsilon_1$.

Moreover,

$$\|\exp(tA)\,W^{-1}\exp(-tA)\,W-I\| = \|\exp(tA)\exp(-tG)-I\|+O(t)$$

and

$$\|\exp(tA)\,W\exp(-tG)-W\| = O(t) \quad \text{as} \quad t\to 0.$$

8.1.18. [423] If the operators $A, G \in \mathcal{G}(\mathbf{M}, \omega)$ and the element $\tilde{x} \in D(A)$ are such that $\exp(sA)\tilde{x} \in D(G)$ for $0 < s < t$ and $\int_0^1 \|G\exp(sA)x\|\,ds < \infty$, then

$$\exp(tG)\tilde{x}-\exp(tA)\tilde{x} = \int_0^t \exp((t-s)G)(G-A)\exp(sA)\tilde{x}\,ds.$$

8.1.19. [126] There are examples of differentiable compact OS's exp(tA) such that for some $B \in B(E)$, exp(t(A + B)) is not continuous with respect to norm.

8.1.20. [126] Let exp(tA) be an OS that is continuous with respect to norm beginning with some t_0 and $B \in B_0(E)$, i.e., is compact. Then the OS exp(t(A + B)) is continuous with respect to norm beginning with t_0.

Definition 1. An OS exp(tA) is said to be quasicompact if

$$\lim_{t\to\infty} \inf\{\|\exp(tA)-K\| : K \in B_0(E)\} = 0.$$

8.1.21. [126] Assume that exp(tA) is a quasicompact OS and $B \in B_0(E)$. Then exp(t(A + B)) is a quasicompact OS.

8.1.22. [126] Let $A \in \mathcal{G}(\mathbf{M}, \omega)$ and $B \in B(\mathcal{D}(A))$. Then the operator A + B generates and OS and, what is more, there exists a $B_1 \in B(E)$ such that A + B is subordinate to $A + B_1$. More precisely, for some $\lambda_0 \in \rho(A)$

$$A+B = U(A+B_1)U^{-1}, \text{ where } U = (I-BR(\lambda_0, A)).$$

THEOREM 2. [44] For any $A \in \mathscr{H}(\theta, \omega)$ and $\varepsilon \in (0,\theta)$ there exist positive constants γ and δ such that if the operator G is bounded relative to A with a, $b < \delta$, then $A + G \in \mathscr{H}(\theta-\varepsilon, \gamma)$. In particular, if $\omega = 0$ and $a = 0$, then $G \in \mathscr{H}(\theta-\varepsilon, 0)$.

8.1.23. [59] If $A \in \mathscr{H}(\theta, \omega)$, and the operator G is totally subordinate to A, then the operator A + G also generates an analytic OS.

8.1.24. [341] Under the assumptions of Theorem 2 we have

$$\|\exp(t(A+G))\| \leqslant \mathbf{M} \exp((\omega + v(b)t), \qquad t \geqslant 0,$$

where $\lim_{b \to 0} v(b) = 0$.

Definition 2. Let $A \in \mathscr{C}(E)$. An operator K with $D(K) \supseteq D(A)$ is compact relative to A if there exists a $\lambda_0 \in \rho(A)$ such that the operator $K \cdot R(\lambda_0, A)$ is compact in E.

8.1.25. [11] Let $A \in \mathscr{H}(\theta, \omega)$ and assume that the operator B is compact relative to A. If one of the sets $D(A^*)$ or $D(B^*)$ is dense in E^*, the A + B generates an analytic OS.

8.1.26. [11] Let $A \in \mathscr{H}(\theta, \omega)$, and assume that the operator B is such that $D(B) \supseteq D(A)$ and $R(\lambda_0, A)B \in B_0(E)$ for some λ_0. Then the operator A + B generates an analytic OS.

8.1.27. [101] Let $A \in \mathscr{H}(\theta, \omega)$, and assume the operator G is A-bounded and $\int_0^1 \|G \exp(tA)\| dt < \infty$. Then A + G generates an analytic OS and (1.1) holds.

8.1.28. [219] Let $A, G \in \mathscr{G}(\mathbf{M}, \omega)$ and $\lim_{t \to 0+} \|\exp(tA) - \exp(tG)\| = 0$. If exp(tA) is a semi-Fredholm OS, then:

(i) the OS exp(tG) is semi-Fredholm;

(ii) nul(exp(tA)) = nul(exp(tG)) = 0 if nul(exp(t_0A)) = 0 for some t_0,
nul(exp(tA)) = nul(exp(tG)) = ∞ if nul(exp(t_0A)) = ∞ for some t_0;

(iii) def(exp(tA)) = def(exp(tG)) = 0 if dep(exp(t_0A)) < ∞ for some t_0,
def(exp(tA)) = def(exp(tG)) = ∞ if dep(exp(t_0A)) = ∞ for some t_0.

8.1.29. [59] Let $A \in \mathscr{C}(E)$ and assume the resolvent R(λ,A) has bound $\|R(\lambda,A)\| \leq M(1 + |\lambda|^{-\beta})$ for some $0 < \beta < 1$ and all $\lambda \in \rho(A)$. If operator G is subordinate to A with order $\alpha \in (0,\beta)$, then $\|R(\lambda, A+G)\| \leqslant \frac{M}{1-\delta} (1+|\lambda|)^{-\beta}$,

where $\|GR(\lambda,A)\| \leq \delta$ for Re $\lambda \geq \omega$.

8.1.30. [219] Let exp(tA) be a semi-Fredholm OS and let $\mathscr{L}$ be A-subordinate and $\int_0^1 \|\mathscr{L} \exp(tA)\| dt < \infty$. Then $A + \mathscr{L}$ generates a semi-Fredholm OS and Proposition 1.28 holds with $G = A + \mathscr{L}$.

8.1.31. [219] Let exp(tA) be a semi-Fredholm OS, and let $G \in \mathfrak{M}(\exp(tA))$ and be A-bounded. Then there exists an $\varepsilon_0 > 0$ such that for $|\varepsilon| < \varepsilon_0$ the operator A + εG generates a semi-Fredholm OS, where

$$\text{nul}(\exp(tA)) = \text{nul}(\exp(t(A+\varepsilon G)))$$

and

$$\text{def}(\exp(tA)) = \text{def}(\exp(t(A+\varepsilon G))) \quad \text{for all} \quad t \geqslant 0.$$

8.1.32. [219] Let E be a Hilbert space, and assume $A \in \mathscr{G}(1, 0)$, and the operator G is dissipative and A-bounded with A-bound b < 1, while $\inf_{\|x\| \leqslant 1} \{\text{Re}((A+G)x, x)\} > -\infty$. Then the OS's exp(tA) and exp(t(A + B)) have bounded inverses and def(exp(tA)) = def(exp(t(A + G)) for t ≥ 0.

8.1.33. [219] There are examples of Fredholm OS's with an A-bounded operator G with A-bound less than 1 and such that the OS exp(t(A + G)) is not Fredholm.

8.1.34. [101] Let exp(tA) be a positive OS in a Banach structure E, and let G be an A-bounded operator with $\int_0^1 \|G\exp(tA)\|\,dt < \infty$. If, for some $\xi \in \mathbf{R}$, the operator G + ξI is positive, then the OS exp(t(A + B)) is also positive.

8.1.35. [178] Let exp(tA) be a positive OS, and let B ∈ B(E) be a positive operator. The OS exp(t(A + B)) is irreducible if and only if J = {0} and J = E are closed ideals with the properties

$$\text{(i)}\ \exp(tA)J \subseteq J,\ t \geqslant 0;$$
$$\text{(ii)}\ BJ \subseteq J.$$

8.1.36. [178] Assume that $A \in \mathscr{G}(\mathbf{M}, \omega)$ and the space E is ⊙-reflexive with respect to A (see §9). Let B ∈ B(E,E$^{\odot *}$). Then there exists an OS U(t) such that for all t ≥ 0 and every x ∈ E we have

$$U(t) = \exp(tA)x + \int_0^t \exp((t-s)A)^{\odot *} BU(s)x\,ds.$$

Definition 3. Let B(E,F) be a space of linear bounded operators mapping E into F. Also, let $\|U(t)\| \leq \mathbf{M}\exp(\tilde{\omega}t)$, where $\tilde{\omega} := \omega + \mathbf{M}\|B\|$.

8.1.37. [178] Under the assumptions of 1.36, and with the same notation, U(t) = exp(tA$^{\times}$). Then E is ⊙-reflexive with respect to A$^{\times}$ and

$$\text{(i)}\quad D(A^{\times}) = \{x \in D(A^{\odot *}) : A^{\odot *}x + Bx \in E\},\quad A^{\times}x = A^{\odot *}x + Bx,\ x \in D(A^{\times});$$
$$\text{(ii)}\ D(A^{\times *}) = D(A^*),\ A^{\times *}x^{\odot} = A^*x^{\odot} + B^*x^{\odot},\ x^{\odot} \in D(A^{\times *});$$
$$\text{(iii)}\ D(A^{\times \odot}) = \{x \in D(A^{\times *}) : A^*x^{\odot} + B^*x^{\odot} \in E\},\ A^{\times \odot}x^{\odot} = A^*x^{\odot} + B^*x^{\odot},\ x^{\odot} \in D(A^{\times \odot});$$
$$\text{(iv)}\ D((A^{\times})^{\odot *}) = D(A^{\odot *}),\ (A^{\times})^{\odot *}x = A^{\odot *}x + Bx,\ x \in D((A^{\times})^{\odot *}).$$

We write $\hat{U}(t) := \int_0^t \exp((t-s)A)^{\odot *} B\exp(s(A+B))\,ds$, $K(\lambda) := R(\lambda; A^{\odot *})B$.

8.1.38. [178] Assume that $\hat{U}(t)$ is an operator function that is continuous with respect to norm on $[t_0,\infty)$ and $K(\lambda)$ is a compact operator function for sufficiently large λ. Then the operator $\hat{U}(t)$ is compact for all $t \geq t_0$.

8.1.39. [178] In the notation of 1.38, if the operator function $\hat{U}(t)$ becomes compact, then $E\sigma(A + B) \leq \omega(A)$.

8.1.40. [178] Assume E is ⊙-reflexive, B ∈ B(E,E$^{\odot *}$), and for any λ with Re λ > S(A) the operator R(λ,A$^{\odot *}$)B is compact. Then the set {λ ∈ σ(A + B) : Re λ > S(A)} contains no more than a countable number of points {λ$_k$}, each of which lies outside Eσ(A + B) or, equivalently, $E\sigma(A + B) \subseteq \{\lambda \in \mathbf{C} : \operatorname{Re}\lambda \leq S(A)\}$.

8.1.41. [174] Assume that exp(tA)$^{\odot}$ and B ∈ B(E$^{\odot}$,E$^{\odot}$) are OS's on E$^{\odot}$. Then the equation

$$\exp(t\tilde{A})x^{\odot} = \exp(tA)^{\odot}x^{\odot} + \int_0^t \exp((t-s)A)^* B\exp(s\tilde{A})x^{\odot}\,ds,\ x^{\odot} \in E^{\odot},$$

uniquely determines the adjoint of the OS exp(tÃ) on E$^{\odot}$. Here $w^*\text{-}\lim_{t\to 0-} t^{-1}(\exp(t\tilde{A}) - I)x^{\odot} = A^{*\odot}x^{\odot} + Bx^{\odot}$ if and only if x$^{\odot}$ ∈ D(A$^{\odot}$).

8.1.42. [174] Under assumptions of 1.41, the operator Ã is part of operator A$^{\odot}$ + B in E$^{\odot}$ (i.e., $D(\tilde{A}),\ \mathscr{R}(\tilde{A}) \subseteq E^{\odot}$).

8.1.43. [574] Let exp(tA) and exp(tG) be two OS's on E^* that are w^*-continuous for each $t \geq 0$. If $t^{-1}\exp(tA) - \exp(tG) \| = O(1)$ for $t \to 0+$, then there exists an operator $B \in B(E^*)$ such that

$$\exp(tA) = \exp(tG) + \sum \int_0^t \int_0^{t_1} \dots \int_0^{t_k} \exp(t_{k+1}G) B \exp((t_k - t_{k+1})G) B \dots$$
$$\dots B \exp((t_1 - t_2)G) B \exp((t - t_1)G) dt_{k+1} \dots dt_1,$$

where the integrals are taken in the strong sense.

Definition 4. We define the **LΦ** perturbation of an operator A by means of the following conditions:

(i) let E and F be Banach spaces, and let D(A) be dense in E, where D(A), equipped with the norm $|\cdot|$ is complete (we will denote this space by $(D(A),|\cdot|)$;

(ii) $A \in B((D(KA),|\cdot|),E)$ and $L \in B((D(A),|\cdot|),F)$;

(iii) $\mathcal{R}(L) = F$;

(iv) the restriction of A to $\mathcal{N}(L)$ (we denote it by A_0) generates the OS $\exp(tA_0)$ on E;

(v) for a bounded operator $\Phi \in B(E,F)$ we define the operator A_Φ thus: $A_\Phi x := Ax$, $D(A_\Phi) := \{x \in D(A) : Lx = \Phi x\}$.

8.1.44. [242] Assume that there exist constants $\gamma > 0$ and $\lambda_0 \in \mathbf{R}$ such that

$$\|Lx\| \geqslant \lambda\gamma\|x\| \quad \text{for all} \quad x \in \mathcal{N}(\lambda \cdot I - A) \text{ and } \lambda > \lambda_0 \tag{1.3}$$

Then the **LΦ**-perturbation A_Φ generates an OS for any $\Phi \in B(E,F)$, $\Phi \in B(E,F)$.

8.1.45. [242] If inequality (1.3) in Prop. 1.44 is replaced by $\|Lx\| \geq |\lambda| \cdot \gamma \|x\|$ for all $x \in \mathcal{N}(\lambda \cdot I - A)$ and $\lambda \in \{z \in \mathbf{C} : |z| \geq r, |\arg z| < \pi/2\}$ for some r, then $A_0 \in \mathcal{H}(\theta, \omega)$ implies that A generates an analytic OS.

8.1.46. [242] Let $A_0 \in \mathcal{H}(\theta, \omega)$. If the operator Φ in the **LΦ** perturbation is compact, then A_Φ generates an analytic OS.

8.2. Theory of Perturbations for Cosine Operator Functions.

8.2.1. [328, 413] Let $A \in \mathcal{C}(\mathbf{M}, \omega)$ and $B \in B(E)$. The operator A + B generates a COF and $\|C(t,A+B) - C(t,A)\| \to 0$ as $\|B\| \to 0$ uniformly with respect to any compact subset of **R**.

8.2.2. [327] If, under the conditions of 2.1, $\tilde{\omega} > \omega + \frac{\mathbf{M}}{\mathfrak{G}} \|B\|$, there exists a number $\tilde{\mathbf{M}} = \tilde{\mathbf{M}}(\omega)$ such that

$$\|C(t, A+B)\| \leqslant \tilde{\mathbf{M}} \exp(|t| \tilde{\omega}), \quad t \in \mathbf{R}.$$

8.2.3. [208] Let $A \in \mathcal{C}(\mathbf{M}, \omega)$. Then, for $x \in E$,

$$C(t, \zeta^2 I + A)x = C(t, A)x + \zeta t \int_0^t \frac{I_1(\zeta\sqrt{t^2 - s^2})}{\sqrt{t^2 - s^2}} C(s, A)x\, ds,$$

where I_1 is a Bessel function and

$$\|C(t, A + \zeta^2 I)\| \leqslant \mathbf{M} \operatorname{ch}(|\sqrt{\zeta^2 + \omega^2}|\, t) \quad \text{for } \zeta \in \mathbf{C}.$$

Other more accurate estimates for the expression $C(t,A \pm \xi^2 I)$ are given in [208], 382-383] for Banach and Hilbert spaces.

8.2.4. [413] Let $A \in \mathcal{C}(\mathbf{M}, \omega)$ and $B \in B(E^1,E)$. Then $A + B \in \mathcal{C}(\mathbf{M}_1, \omega_1)$.

8.2.5. [365, 414] Let $A \in \mathcal{C}(\mathbf{M}, \omega)$ and $D(A) \subseteq D(G)$. If there exists $\omega' \geq 0$ and $\mathbf{M}' \geq 1$ such that $\rho(A)$ contains the set $\{z : z > \omega'\}$ and the function $GR(z^2,A)$ is infinitely differentiable, where

$$\frac{1}{n!} \left\| (z - \omega')^{n+1} \left(\frac{d}{dz}\right)^n (GR(z^2, A)x) \right\| \leqslant \mathbf{M}' \|x\|$$

for $x \in E$ and any $n \in \mathbf{N}$ and $z > \omega'$, then A + G generates a COF.

8.2.6. [229, 232] If $A,\ G\in\mathscr{C}(\mathbf{M},\ \omega)$, then the operator A + G (or its closure) generally does not generate a COF, even when C(t,A) and C(t,G) commute. However, we always have $\overline{A+B}\in\mathscr{H}(\pi/2,\ \omega)$.

8.2.7. [28] Let $A,\ G\in\mathscr{C}(\mathbf{M},\ \omega)$ and assume that $D_1 := D(A)\cap D(G)$ is dense in E. Then a "generalized" COF (in the sense that conditions (i)-(ii) of 1.4.1 are satisfied) is defined on D as follows:

$$\tilde{C}(t,\ A+G)x=C(t,\ A)x+\frac{t^2}{2}\int_0^1 j_1(t\sqrt{1-s^2},\ A)\,C(t\cdot s,\ G)xds,$$

where $j_1(t,\ A)x:=\frac{4}{\pi}\int_0^1\sqrt{1-s^2}C(ts,\ A)xds,\quad x\in D_1$. If A + G generates the COF C(t,A + G), then C(t,A + G) = $\tilde{C}$(t,A + G), $t\in\mathbf{R}$.

8.2.8. [399] Let $A\in\mathscr{C}(\mathbf{M},\ \omega)$ and the operator $G\in\mathscr{C}(E)$ be such that

(i) for the SOF, $R(S(t,A))\subseteq D(G)$ for all $t\in\mathbf{R}$;
(ii) the function GS(t,A) is strongly continuous with respect to $t\in\mathbf{R}$.

Then the operator A + G generates a COF. In this case

$$C(t,\ A+G)x=\sum_{k=0}^{\infty}\hat{C}_k(t)\ \text{и}\ S(t,\ A+G)x=\sum_{k=0}^{\infty}\hat{S}_k(t),\tag{2.1}$$

where $\hat{C}_0(t):=C(t,\ A),\ \hat{C}_k(t):=\int_0^t C(t-s,\ A)G\hat{S}_{k-1}(s)xds$, and $\hat{S}_0(t):=S(t,\ A),\ \hat{S}_k(t):=\int_0^t S(t-s,\ A)G\hat{S}_{k-1}\times(s)xds$, and series (2.1) converges absolutely in B(E).

8.2.9. [399] Under the conditions of 2.8,

$$R(\lambda^2,\ A+G)=R(\lambda^2,\ A)\sum_{k=0}^{\infty}(GR(\lambda^2,\ A))^k.$$

8.2.10. [397] If, under the assumptions of 2.8, the SOF S(t,A) is compact, then the SOF S(t,A + G) is also compact.

8.2.11. [387] Let $A_1,\ A_2\in\mathscr{C}(\mathbf{M},\ \omega)$ and $D(A_1)\subseteq D(A_2)$. Then

$$C(t,\ A_1)x-C(t,\ A_2)x=\int_0^t S(t-s,\ A_2)(A_1-A_2)S(s,\ A_1)xds$$

for all $x\in D(A_1)$.

THEOREM 1. [397, 399] Let $A\in\mathscr{C}(\mathbf{M},\ \omega)$ and $G\in\mathscr{C}(E)$ be such that
(i) $D(A)\subseteq D(G)$;
(ii) there exists a continuous function D(t) such that

$$\|GS(t,\ A)x\|\leqslant K(t)\|x\|\quad\text{for all}\quad x\in D(A).$$

Then the operator A + G generates a COF.

8.2.12. [399] Assume that the operator $A\in\mathscr{C}(\mathbf{M},\ \omega)$ satisfies condition (F) with the operator $G\in\mathscr{C}(E)$, and for some $Q\in\mathscr{C}(E)$ the condition $D(G)\subseteq D(Q)$ is satisfied. Then A + Q generates a COF.

8.2.13. [399] Assume that in Prop. 2.12 the condition of inclusion of domains of definition is replaced by the condition $D(A)\subseteq D(Q)$. If the operator Q is G-bounded, then the operator A + Q generates a COF.

8.2.14. [374, 388] Let $A \in \mathcal{C}(\mathbf{M}, \omega)$. $G \in \mathfrak{M}(C(t, A))$, $GR(\lambda, A) \in B(E)$, and $|k| < K_\infty^{-1}$. Then the operator A + G generates a COF, $\lim_{\varepsilon \to 0} \| C(t, A+\varepsilon G) - C(t, A) \| = 0$ uniformly on any compact set in $\mathbf{R}$, and

$$C(t, A+\varepsilon G) = \sum_{k=0}^{\infty} \varepsilon^k \bar{C}_k(t),$$

where $\bar{C}_0(t) := C(t, A)$; $C_k(t)x = \int_0^t \bar{C}_{k-1}(t-s) GS(s, A)x\, ds$, $x \in D(A)$, and $\bar{C}_k(t)$ is a continuous extension of $C_k(t)$ to all of E.

8.2.15. [308] Let $C(t,A_1)$ and $C(t,A_2)$ be COF's such that $\| C(t,A_1) \| \leq Me^{\omega t}$ and $\| C(t,A_2) \| \leq Ne^{\nu t}$ for $t \geq 0$ and $\| (A_1 - A_2)x \| \leq a \| x \| + b \| A_1 x \|$ for $x \in D(A_1)$. Then for $z > \omega$ we have

$$\| (C(t, A_1) - C(t, A_2))(z^2 I - A)^{-1} \| \leqslant$$
$$\leqslant \begin{cases} \frac{MN}{2\nu} Qt \operatorname{sh}(\omega t), & \omega = \nu, \\ \frac{MN}{\omega^2 - \nu^2} Q(\operatorname{ch}(\omega t) - \operatorname{ch}(\nu t)), & \omega \neq \nu, \end{cases}$$

where $Q := (1 + M)b + (a + b\omega^2)M/(z^2 - \omega^2)$.

8.2.16. [308] If $a, b \to 0$ under the assumptions of 2.15, then

$$\lim_{a,b \to 0} C(t, A_2)x = C(t, A_1)x \text{ for } x \in D(A_1).$$

8.2.17. [371] Let K(t,A) be a w^*-continuous COF (see §9) and let B be a w^*-w^*-continuous operator on E^*. Then A + B is the w^*-generating operator of a w^*-continuous COF K(t,A + B) and $\lim_{\|B\| \to 0} \| K(t, A+B) - K(t, A) \| = 0$ uniformly on any compact [0,T].

8.3. Supplementary Comments. Problems on perturbations of adjoint OS's were discussed in [174-177].

In was proved in [261] that when $\alpha \in (0,1)$ the condition $t^{-\alpha} \| \exp(tA) - \exp(tg) \| = O(1)$ as $t \to 0$ is not sufficient for validity of Prop. 1.43, even for unitary OS's or commuting OS's.

For information on limit values of perturbations in theorems on perturbations of Fredholm operators, see [362].

Perturbation of the well-posed Cauchy problem $\sum_{j=0}^{n} A_j u(t)^{(n-j)} = 0$, $t \geqslant 0$, $u^{(j)}(0) = u^j$, $j = \overline{0, n-1}$ was studied in [132] by replacement of the operators A_n by $A_n + B(t)$, where $B(t) : \bar{\mathbf{R}}_+ \to B(E)$. This paper also considers restrictions on B(T) under which the perturbed problem remains well-posed.

Concerning perturbations of the form A · B, see [242]

In [414] there are conditions, for the case $E = C([\hat{a}, \hat{b}])$, under which the assumptions of Prop. 2.5 are satisfied by operators $Au := a(x)u'' + \frac{1}{2}a'(x)u'$, $D(A) = \{u : u,\ u'' \in C([\hat{a}, \hat{b}]),\ u'(x) = 0$ with $x = \hat{a}$ and $x = \hat{b}\}$ and $Gu := \left(b(x) - \frac{1}{2}a'(x)\right)u' + c(x)u$, $D(G) = \{u : u,\ u' \in C([\hat{a}, \hat{b}]):\ u' \in C([\hat{a}, \hat{b}])\}$. See also Props. 2.1.34, 2.1.7-9, 2.1.17, 6.2.5, 7.1.12, and 9.2.7.

9. ADJOINT OPERATOR SEMIGROUPS AND COSINE OPERATOR FUNCTIONS

9.1. Adjoint OS's. Let E^* be the adjoint of a Banach space E. For each member $x \in E$ the formula $\langle x, y^* \rangle$ defines on the space E^* a functional $x^{**} \in E^{**}$ by means of the rule $\langle y^*, x^{**} \rangle := \langle x, y^* \rangle$. We call the mapping $\pi : x \to x^{**}$ the natural mapping of E into E^{**}.

9.1.1. The natural mapping π is isometric, i.e., $\| x \| = \| x^{**} \|$ for all $x \in E$.

Assume that the operators of $\exp(tA)^*$ are adjoint to the operators of the OS $\exp(tA)$. The family $\exp(tA)^*$ satisfies conditions (i)-(ii) of Def. 1.3.1 and the bound $\| \exp(tA)^* \| \leq Me^{\omega t}$, although it generally does not necessarily

satisfy condition (iii), i.e., it is not necessarily a C_0-semigroup in E^* (continuity is taken to be with respect to the strong topology in E^*).

9.1.2. In the case of a reflexive E, the assumption $A \in \mathcal{G}(\mathbf{M}, \omega)$ implies that $D(A^*)$ is dense in E^*. This is not so in the general case.

9.1.3. [101] $(R)\lambda, A^*) = R(\lambda, A)^*$.

Let $E^\odot$ denote the (closed) subspace of E^* on which the family $\exp(tA)^*$ is strongly continuous with respect to t, and define the OS $\exp(tA)^\odot$ to be $\exp(tA)^*|_{E^\odot}$.

9.1.4. [174, 424] for an OS exp(tA) we have

(i) the mapping $t \to \exp(tA)^* x^*$ is w^*-continuous with respect to $t \in \bar{\mathbf{R}}_+$ for all $x^* \in E^*$;

(ii) the mapping in (i) is w^*-differentiable from the right at zero if and only if the element $x^* \in D(A^*)$;

(iii) if $x^* \in D(A^*)$, then if $t > 0$ we have $\exp(tA)^* x^* \in D(A^*)$ and the function in (i) is w^*-differentiable, where

$$\frac{d}{dt} \langle x, \exp(tA)^* x^* \rangle = \langle x, A^* \exp(tA)^* x^* \rangle = \\ = \langle x, \exp(tA)^* A^* x^* \rangle$$

for all $x \in E$ and $t \geq 0$;

(iv) the domain $D(A^\odot)$ is w^*-dense in E^*, where $A^\odot$ is the generating operator of an OS $\exp(tA)^*$ defined on $E^\odot$;

(v) the family $\exp(tA)^*$ operates according to the formula

$$\exp(tA)^* = (\lambda \cdot I - A^*) \exp(tA)^\odot (\lambda \cdot I - A^*)^{-1}.$$

9.1.5. [178] The following statements are true:

(i) $E^\odot$ is the closure with respect to norm of $D(A^*)$;

(ii) $E(A^\odot) = \{\varphi \in D(A^*) : A^* \varphi \in E^\odot\}$;

(iii) $A^\odot$ and A^* coincide on $D(A^\odot)$.

THEOREM 1. [424] For $A \in \mathcal{G}(\mathbf{M}, \omega)$ we have:

(i) the operator $A^\odot$ is the largest restriction of A^* with domain and range in the subspace $E^\odot$;

(ii) the subspace $E^\odot = \overline{D(A^*)}$, where the closure is taken in the strong topology of E^*;

(iii) the subspace $E^\odot$ is invariant relative to the family $\exp(tA)^*$, and the above-defined family $\exp(tA)^\odot$ is a C_0-semigroup of operators on $E^\odot$;

(iv) if E is reflexive, then $E^\odot = E^*$ and $A^\odot = A^*$.

We now introduce a new norm in E:

$$\|x\|' : \sup \{|\langle x, x^\odot \rangle| : x^\odot \in E^\odot, \ \|x^\odot\| \leq 1\} \quad \text{for } x \in E.$$

9.1.6. [300] Let E be a Grothendieck space. Then the families $\exp(tA)^*$ and $\exp(tA)^{**}$, $t \in \bar{\mathbf{R}}_+$, are strongly continuous in E^* and E^{**}, respectively.

9.1.7. [178] Let $\tilde{C} := \lim_{\lambda \to \infty} \|\lambda R(\lambda, A)\|$. Then

$$\|x\|' \leq \|x\| \leq \tilde{C} \|x\|' \quad \text{for any } x \in E.$$

These norms coincide for contraction semigroups.

9.1.8. [158] Let $A \in \mathcal{H}(\theta, \omega)$. Then the operator function $\exp(tA)^*$ has a holomorphic continuation into the sector $\Sigma(\theta)$, and

(i) $\exp(\zeta_1 A)^* \exp(\zeta_2 A)^* = \exp((\zeta_1 + \zeta_2) A)^*$ for any $\zeta_1, \zeta_2 \in \Sigma(\theta)$

(ii) $w^*\text{-}\lim_{\zeta \to 0} \exp(\zeta A)^* x^* = x^*$ for all $x^* \in E^*$ and $|\arg \zeta| \leq \alpha < \theta$.

Definition 1. A Banach space E is said to be $\odot$-reflexive (solar reflexive) according to $A \in \mathcal{G}(\mathbf{M}, \omega)$, if $E = E^{\odot\odot}$.

9.1.9. [175] A space E is $\odot$-reflexive with respect to an operator $A \in \mathcal{G}(\mathbf{M}, \omega)$ if and only if one of the following conditions is satisfied:

(i) the resolvent $R(\lambda,A)$ is $\sigma(E,E^{\odot})$-weakly compact for all $\lambda \in \rho(A)$ (here $\sigma(X,Y)$ is the locally convex topology on X induced by the functionals of Y);

(ii) the space $E^{\odot}$ is $\odot$-reflexive with respect to the operator A.

Definition 2. We will call a subgroup in E^* that is continuous at zero in the w^*-topology a $C_0{}^*$-semigroup.

9.1.10. [10] For a $C_0{}^*$-semigroup exp(tA) there exist constants $M \geq 1$ and $\beta \geq \inf_{t>0}(t^{-1}\ln \| \exp(tA) \|)$ such that

$$\|\exp(tA)\| \leqslant \mathbf{M} \exp(\beta t), \quad t \geqslant 0.$$

9.1.11. [10] For a $C_0{}^*$-semigroup exp(tA), assume we are given a set $\mathcal{D} \subset D(A)$, that is w^*-dense in E^* and invariant under exp(tA). Then the $\mathcal{D}$-core of the operator A, i.e., the w^* — w^*-closure of the restriction $A|_{\mathcal{D}}$, coincides with A.

Definition 3. We say that $\bar{x} \in E$ is holomorphic for an operator A if $\bar{x} \in D(A^{\infty})$ and the function

$$z \to \sum_{k=0}^{\infty} z^k \| A^k \bar{x} \| / k!, \quad z \in \mathbf{C}$$

is analytic.

9.1.12. [10] An operator $A \in \mathcal{C}(E^*)$ is the generator of a $C_0{}^*$-group if and only if it is w^*-densely defined, w^* — w^* closed, and satisfies the following conditions:

(i) there exist $M \geq 1$ and $\beta \geq 0$ such that

$$\| (I - \alpha A)^k x \| \geqslant \mathbf{M}^{-1}(1 - \alpha\beta)^k \|x\| \quad \begin{array}{l} \text{for all } x \in D(A^k), \\ \text{all } \alpha \text{ with } |\alpha|\beta < 1 \text{ and } k \in \mathbf{N}; \end{array}$$

(ii) either $\mathcal{R}(I - \alpha A)$ for all α with $0 < \alpha\beta < 1$ coincides with the space, or A has a w^*-dense set of holomorphic elements and any member of the space can be approximated by a uniformly bounded net of holomorphic elements.

9.1.3. [10] For a $C_0{}^*$-semigroup exp(tA) the following conditions are equivalent:

(i) $x \in D(A)$;

(ii) $\sup\{ \| (\exp(tA) - I)x \| / t : t \in [0,1] \} < \infty$. The analog of this statement for OS's is generally false.

9.1.14. [10] For a C_0- or $C_0{}^*$-group exp(tA) the following conditions are equivalent:

(i) there exist $\varepsilon > 0$ and $\delta > 0$ such that

$$\|\exp(tA) - I\| \leqslant 1 - \varepsilon \quad \text{for all } t \in [0, \delta);$$

(ii) $\lim_{t \to 0} \| \exp(tA) - I \| = 0.$

9.1.15. [10] For two given $C_0{}^*$-semigroups exp(tA) and exp(tG) the following conditions are equivalent:

(i) $\| \exp(tA) - \exp(tG) \| = O(t)$ as $t \to 0$;

(ii) $D(A) = D(G)$ and the operator A is a G-bounded operator from D(G) into E^*.

9.1.16. [369] Let exp(tA) be an OS given on a Grothendieck space E and

a) $\overline{\lim_{t \to \infty}} \| CH(t, 1) \| / t < \infty$;

b) $s\text{-}\lim_{t \to 0} (u/t) \exp(tA) CH(u, 1)x = o$ for all $x \in E$, $u > 0$.

Then:

(i) the operator $P := s\text{-}(C, 1) \lim_{t \to \infty} \exp(tA)$ is a projector into E, where $\mathcal{R}(P) = \bigcap_{t>0} \mathcal{N}(\exp(tA) - I)$, $\mathcal{N}(P) = \overline{\operatorname{span}\left(\bigcup_{t>0} \mathcal{R}(\exp(tA) - I) \right)}$ and $D(P) = \{x \in E : \exists t_n \to \infty,\ w\text{-}\lim_{n \to \infty} t_n^{-1} CH(t_n, 1)x \text{ exists}\}$;

(ii) the operator $Qx^* = w^*\text{-}\lim_{t\to\infty} CH(t,1)^* x^*$ is a projector into E^*, where $\mathcal{R}(Q) = \bigcap_{t>0} \mathcal{N}(\exp(tA) - I)$,

$\mathcal{N}(Q) = \operatorname{span}\{\bigcup_{t>0} \mathcal{R}(\exp(tA)^* - I^*)\}$ and $D(Q) = \{x^* \in E^* : \exists t_n \to \infty,\ w^*\text{-}\lim CH(t_n, 1)^* x^* \text{ exists}\}$. Also,

$\mathcal{N}(A) = \mathcal{R}(P)$, $\mathcal{N}(A^*) = \mathcal{R}(Q)$, $\mathcal{R}(A) = \mathcal{N}(P)$, $\overline{\mathcal{R}(A^*)} = \mathcal{N}(Q)$.

9.1.17. [369] Let E be a Grothendieck space. An OS exp(tA) is strongly (C,1)-ergodic if and only if conditions a) and b) of 1.16 are satisfied and $w^*\text{-cl}(\mathcal{R}(A^*)) = \overline{\mathcal{R}(A^*)}$.

9.2. Adjoint Cosine Operator Functions. Following 9.1, we let $C(t,A)^\odot$ denote restriction $C(t,A)^*|_{E^\odot}$, $t \in \mathbf{R}$, where $E^\odot \subseteq E$ is a subspace on which the adjoint family $C(t,A)^*$ is strongly continuous at zero.

9.2.1. [328] For a COF C(t,A) defined on E we have

(i) if $x^* \in D(A^*)$, then for any $t \in \mathbf{R}$

$$C(t,A)^* x^* \in D(A^*) \text{ and } A^* C(t,A)^* x^* = C(t,A)^* A^* x^*$$

and for any $x \in E$ we have

$$\langle x, (C(t,A)^* - I^*)x^* \rangle = \int_0^t (t-s) \langle x, C(s,A)^* A^* x^* \rangle \, ds;$$

(ii) $x^* \in D(A^*)$ if and only if the limit $w^*\text{-}\lim_{s\to 0+} (2/s^2)(C(s,A)^* - I^*)x^* = y^*$, exists, where $A^* x^* = y^*$.

THEOREM 1. [328, 371] In the notation of Prop. 2.1 we have

(i) the subspace $E^\odot = \overline{D(A^*)}$, where closure is taken in the strong topology of the space E^*;

(ii) the subspace $E^\odot$ is invariant under $C(t,A)^*$, and $C(t,A)^\odot$, $t \in \mathbf{R}$, is a COF on $E^\odot$;

(iii) the generating operator $A^\odot$ of the COF $C(t,A)^\odot$ is a maximal restriction of the operator A^* to $E^\odot$ (i.e., $A^\odot$ is the part of the operator A^* on $E^\odot$);

(iv) if E is reflexive, then $E^\odot = E^*$ and $A^\odot = A^*$;

(v) for each $t > 0$ the operator $C(t,A)^*$ is the w^*-closure of the operator $C(t,A)^\odot$.

9.2.2. [328] For any $x^* \in E(A^*)$ we have

$$\|(C(t,A)^* - I^*)x^*\| \leqslant (t^2/2)\|A^* x^*\| \cdot \sup_{0 \leqslant s \leqslant t} \|C(s,A)\|.$$

Definition 1. An operator function K(t), $t \in \mathbf{R}$, defined on E^* and satisfying the conditions $K(0) = I^*$ and (1.4.1) is said to be a w^*-continuous COF if, for each $t > 0$, the operator K(t) is continuous in E^* in the w^*-topology (we write is $w^* - w^*$-continuous) and for any $x^* \in E^*$ the function $t \to K(t)x^*$ is w^*-continuous in E^* with respect to $t \in \mathbf{R}$.

9.2.3. An operator $Q \in L(E^*)$ is $w^* - w^*$-closed if and only if it is adjoint to a densely defined and closed operator $\mathcal{P} \in L(E)$. Moreover, $D(A) = E^*$ if and only if $D(\mathcal{P}) = E$, and, in this case, both $\mathcal{P}$ and Q are bounded, where $\|\mathcal{P}\| = \|Q\|$.

THEOREM 2. [371] In order for an operator function K(t) on E to be a w^*-continuous COF it is necessary and sufficient for $K(t) = C(t,A)^*$, where C(t,A) is some COF. If Q is the generating operator of K(t), $t \in \mathbf{R}$ (in the sense of the w^*-topology) and A is the generating operator of the COF C(t,A), then $Q = A^*$.

We use the notation K(t,Q) for the w^*-continuous COF with generating operator Q.

9.2.4. [37] In the notation of Theorem 2, the following conditions are equivalent:

$$\text{(i)}\ \ x^* \in D(Q);$$
$$\text{(ii)}\ \|K(t,Q)x^* - x^*\| = O(t^2) \text{ as } t \to 0;$$
$$\text{(iii)}\ \varlimsup_{t\to 0+} t^{-2}\|K(t,Q)x^* - x^*\| < \infty.$$

9.2.5. [371] Let $Q \in L(E^*)$. The operator Q generates a w^*-continuous COF if and only if it is w^*-densely defined, $w^* - w^*$ closed, and there exist constants $M > 0$ and $\omega \geq 0$ such that for $\lambda > \omega$ the point $\lambda^2 \in \rho(Q)$ and

$$\left\| \frac{d^n}{d\lambda^n} (\lambda (\lambda^2 I^* - Q)^{-1}) \right\| \leqslant M n! (\lambda - \omega)^{-n-1}, \quad n \in \mathbf{N}.$$

9.2.6. [371] If the set $\mathcal{D} \subseteq D(Q)$ is w^*-dense in D(A) and invariant under the w^*-continuous COF K(t,Q), then $\mathcal{D}$ is a w^*-core of the operator Q.

9.2.7. [371] For w^*-continuous COF's $K(t,Q_1)$ and $K(t,Q_2)$ the following conditions are equivalent:

$$\begin{gathered} \text{(i) } D(Q_1) \subseteq D(Q_2); \\ \text{(ii) } \| (K(t, Q_1) - K(t, Q_2)) x^* \| = O(t^2) \quad \text{for} \quad t \to 0 \quad \text{for any} \\ x^* \in D(Q_1). \end{gathered}$$

9.2.8. [371] The following conditions are equivalent:

(i) $\| K(t,Q_1) - K(t,Q_2) \| = O(t^2)$ as $t \to 0$;
(ii) $D(Q_1) = D(Q_2)$ and $Q_2 - Q_1$ is a bounded operator from $D(Q_1)$ into E^*;
(iii) $D(A_1) \subseteq D(Q_2)$ and $Q_2 - Q_1$ is a bounded operator from $D(Q_1)$ into E^*;
(iv) $D(Q_2) \subseteq D(Q_1)$ and $Q_2 - Q_1$ is a bounded operator from $D(Q_2)$ into E^*;

Moreover, in these cases,

$$\begin{gathered} \| Q_2 - Q_1 \| \leqslant \varlimsup_{t \to 0} 2t^{-2} \| K(t, Q_1) - K(t, Q_2) \| \leqslant \\ \leqslant \sup_{t > 0} 2t^{-2} \| K(t, Q_1) - K(t, Q_2) \| \end{gathered}$$

where equality is achieved, for example, in the case of contraction w^*-continuous COF's.

9.2.9. [371] If $K(t,Q_1) - K(t,Q_2) = o(t^2)$ as $t \to 0$, then $K(t,Q_1) = K(t,Q_2)$, $t \in \mathbf{R}$.

9.2.10. [371] For a w^*-continuous COF K(t,Q), $t \in \mathbf{R}$, we have $\mathcal{N}(Q) = \bigcap_{t>0} \mathcal{N}(K(t, Q) - I^*)$, and the w^*-closure of $\mathcal{R}(Q)$ is the w^*-closure $\bigcup_{t>0} \mathcal{R}(K(t, Q) - I^*)$. If E is a Grothendieck space, then

$$\mathcal{R}(Q) = w - \mathrm{cl} \left(\bigcup_{t>0} \mathcal{R}(K(t, Q) - I^*) \right) = \overline{\operatorname{span} \{ \mathcal{R}(K(t, Q) - I^*) : t > 0 \}},$$

where the closure is taken in the strong topology of E^*.

Notation:

$$\begin{gathered} T(t, A) := \int_0^t (t - s) C(s, A)\, ds; \quad Q_s^1 := s\text{-}\lim_{t \to \infty} t^{-1} S(t, A)^*; \\ Q_w^1 := w\text{-}\lim_{t \to \infty} t^{-1} S(t, A)^*; \quad Q_{w^*}^1 := w^*\text{-}\lim_{t \to \infty} t^{-1} S(t, A)^*. \end{gathered} \tag{2.1}$$

9.2.11. [371] Assume that for $Q = A^*$ the following conditions are satisfied:

a) $\| S(t,A) \| = O(t)$ as $t \to \infty$;

b) $w^* - \lim_{t \to \infty} t^{-1} (K(t + s, Q) - K(t - s, Q)) S(s, A)^* x^* = 0$ for all $x^* \in E$ and $s > 0$.

Then

(i) $Q_s^1 \subseteq Q_w^1 \subseteq Q_{w^*}$ are projectors, where $\| Q_{w^*}^1 \| \leqslant \varlimsup_{t \to \infty} t^{-1} \| S(t, A) \|$, while $D(Q_s^1) \subseteq D(Q_w^1)$ and $D(Q_{w^*}^1)$ are strongly closed;

(ii) $\mathcal{R}(Q_s^1) = \mathcal{R}(Q_w^1) = \mathcal{R}(Q_{w^*}^1) = \mathcal{N}(Q)$, $\mathcal{N}(Q_s^1) \subseteq \mathcal{N}(Q_w^1) \subseteq \overline{\mathcal{R}(Q)}$ and $III := \overline{\operatorname{span} \{ \mathcal{R}(K(t, Q) - I^*) : t > 0 \}} \subseteq \mathcal{N}(Q_{w^*}^1) \subseteq w^* - \mathrm{cl}(\mathcal{R}(Q))$. If condition b) is replaced by the stronger condition b′) $s - \lim_{t \to \infty} t^{-1} (K(t + s, Q) -$

$K(t-s, Q))\,S(s, A)^* = O$ for all s > 0, then $\mathcal{U} \subset \mathcal{N}(Q_S^1)$.

If we translate Def. 7.1.4 into the language of COF's and write $P := s-\lim_{t\to\infty} t^{-1}S(t, A)$, we can say that a COF C(t,A) is strongly (C,1)-ergodic if D(P) = E and uniformly (C,1)-ergodic if $\| t^{-1}S(t,A) - P \| \to 0$ as $t \to \infty$.

9.2.12. [371] Assume that conditions a) and b) of 2.11 are satisfied and E is a Grothendieck space. Then $\mathcal{R}(Q_{w^*}^1) = \mathcal{N}(Q)$, $\mathcal{N}(Q_{w^*}^1) = \overline{\mathcal{R}(Q)}$ and $D(Q_{w^*}^1) = \mathcal{N}(Q) \ominus \overline{\mathcal{R}(Q)} = E^*$. If, however, condition b') is satisfied, then K(t,Q) is strongly (C,1)-ergodic, i.e., $D(Q_s^1) = E$.

9.2.13. [371] A COF C(t,A) defined on a Grothendieck space E is strongly (C,1)-ergodic if and only if the following conditions are satisfied:

$$\text{(i) } \|S(t, A)\| = O(t) \text{ for } t \to \infty;$$
$$\text{(ii) } s\text{-}\lim_{t\to\infty} t^{-1}C(t, A)\,S(s, A) = O \text{ for all } s > 0;$$
$$\text{(iii) } w^*\text{-cl}(\mathcal{R}(A^*)) = \overline{\mathcal{R}(A^*)}.$$

9.2.14. [371] Under the assumptions of 2.13 with a space E that also has the Dunford—Pettis property, the COF C(t,A) is uniformly (C,1)-ergodic if and only if condition (i) of 2.13 is satisfied, $\| C(t,A)S(t,A) \| = O(t)$ for all s > 0 as $t \to \infty$, and, finally, $w^*\text{-cl}(\mathcal{R}(A^*)) = \overline{\mathcal{R}A^*}$.

9.2.15. [371] Let $\| T(t,A) \| = O(t^2)$ as $t \to \infty$ and $s\text{-}\lim_{t\to\infty} t^{-2}K(t, Q) = O$ for all $x^* \in D(Q)$. Then $Q_S^2 = Q_w^2 \subseteq Q_{w^*}^2$ are bounded projectors (see [371] for the definition) such that $\| Q_{w^*}^2 \| \leqslant \varlimsup_{t\to\infty} 2t^{-2}\| T(t, A)\|$, $\mathcal{R}(Q_s^2) = \mathcal{R}(Q_{w^*}^2) = \mathcal{N}(Q)$, $\mathcal{R}(Q) = \mathcal{N}(Q_s^2) \subseteq \mathcal{N}(Q_{w^*}^2) \subseteq w^*\text{-cl}(\mathcal{R}(Q))$. The subspaces $D(Q_s^2)$ and $D(Q_w^*)$ are strongly closed in E^* and $D(Q_s^2) = \mathcal{N}(Q) \oplus \overline{\mathcal{R}(Q)} = \{x^* \in E^* : \exists t_n \to \infty : \lim_{n\to\infty} 2t_t^{-2}T(t_n)x^* \text{ exists}\}$.

9.2.16. [371] Let E be a Grothendieck space and let K(t,Q) be a w^*-continuous COF. If $\| T(t,A) \| = O(t^2)$ as $t \to \infty$ and $w^*\text{-}\lim_{t\to\infty} t^{-2}K(t, Q)T(s)^* x^* = 0$ for all $x^* \in E^*$ and s > 0, then $\mathcal{R}(Q_{w^*}^2) = \mathcal{N}(Q)$, $\mathcal{N}(Q_{w^*}^2) = \mathcal{R}(Q)$, $D(Q_{w^*}^2) = E^*$. Moreover, if $s\text{-}\lim_{t\to\infty} t^{-2}K(t, Q)T(s)^* = 0$ for all x > 0, then the COF K(t,Q) is strongly (C,2)-ergodic.

9.2.17. [371] A COF C(t,A) on a Grothendieck space E is strongly (C,2)-ergodic if and only if

$$\text{(i) } \|T(t, A)\| = O(t^2) \text{ for } t \to \infty;$$
$$\text{(ii) } s\text{-}\lim_{t\to\infty} t^{-2}C(t, A)T(s, A) = O \text{ for all } s > 0;$$
$$\text{(iii) } w^*\text{-cl}(\mathcal{R}(A^*)) = \overline{\mathcal{R}(A^*)}.$$

9.2.18. [371] Assume that under the assumptions of 2.17 the space E has the Dunford—Pettis property. Then the COF C(t,A) is uniformly (C,2)-ergodic if and only if $\| T(t) \| = O(t^2)$, $\| C(t,A)T(s,A) \| = O(t^2)$ for S > 0 as $t \to \infty$, and $s > 0$, $\text{cl}(\mathcal{R}(A^*)) = \mathcal{R}(A^*)$.

9.2.19. [371] Let $Gx^* := s\text{-}\lim_{t\to 0+} t^{-1}S(t, A)^* x^*$ for those $x^* \in E^*$ for which the limit exists. The for the w^*-continuous COF K(t,Q) we have $D(Q) = \bigcup_{t>0} \overline{\mathcal{R}(S(t, A)^*)} = \{x^* \in E^* : \exists t_n \to \infty : w^*\text{-}\lim_{n\to\infty} t_n^{-1}S(t_n, A)^* x^* \text{ exists}\}$. In this case $Q = I_{D(G)}$.

9.3. Supplementary Comments. Note that it follows from 1.2 that the adjoint OS $\exp(tA)^*$, $t \in \bar{\mathbf{R}}_+$, is a C_0^*-semigroup, i.e., all of the properties of C_0-semigroups hold in the sense of w^* properties. However, if we consider arbitrary w^*-semigroups in E^*, it turns out that different w^*-semigroups may have the same w^*-generating operator. The paper [171] considers the situations in which it is possible to obtain Hille—Phillips type theorems for w^*-semigroups, and this paper introduces the notion of integrable w^*-generating operators (in this connection, see [172, 173, 183, 260]).

There is an extensive discussion of σ-continuous semigroups in the book [10]. We have given only some of the results presented in this book.

Concerning adjoint OS's and COF's, see also Props. 4.1.8, 8.1.36-43, and 8.2.17.

NOTATION

E — Banach space 1.1
$L(E)$ — the set of linear operators defined on E 1.1
$B(E)$ — Banach algebra of continuous linear operators defined on E 1.1
$D(A)$ — domain of operator A 1.1
$\mathscr{R}(A)$ — range of operator A 1.1
$|||\cdot|||$ — graphic norm 1.1
$\mathscr{D}(A)$ — Banach space of $D(A)$ with graphic norm 1.1
$\rho(A)$ — resolvent set of operator A 1.1
$\sigma(A)$ — spectrum of A 1.1
$\mathscr{C}(E)$ — set of closed operators with $\overline{D(A)}=E$ and $\rho(A)\neq\emptyset$ 1.1
$\mathbf{R}$ — the real line $(-\infty,+\infty)$ 1.1
$C(J;E)$ — the set of continuous functions mapping $j\subseteq\mathbf{R}$ into E 1.1
$C^k(J;E)$ — the set of k-times continuously differentiable functions mapping J into E 1.1
$\mathbf{R}_+$ — the non-negative reals $[0,\infty)$ 1.2
$u^n(t)$ — n-th derivative of $u(t)$, $d^nu(t)/dt^n$ 1.2
D — a set dense in E for which the Cauchy problem (1.2.1)-(1.2.2) is solvable 1.2
$\mathscr{K}$ — arbitrary compact subset of $\mathbf{R}$ 1.2
$\mathscr{F}_j(t)$ — propagators for Cauchy problem (1.2.1)-(1.2.2) 1.2
δ_{ij} — the Kronecker delta 1.2
$U(t)$ — semigroup of operators 1.3
C_0 — the class of strongly continuous semigroups (OS's) 1.3
$\mathscr{B}$ — Banach algebra with identity $\tilde{e}$ 1.4
$C(t)$ — cosine operator function (COF) 1.4
$C(t,A)$ — COF with generating operator A 1.4
$S(t)$ — sine operator function (SOF) 1.4
$S(t,A)$ — SOF associated with $C(t,A)$ 1.4
E^k — linear manifold $\{x\in E : C(t,A)x\in C^k(\mathbf{R};E)\}$ 1.4
$\exp(tA)$ — OS with generating operator A 2.1
$\omega(A)$ — type of OS $\exp(tA)$ 2.1
$R(\lambda,A)=(\lambda I-A)^{-1}$ — resolvent of the operator A 2.1
$\mathscr{G}(\mathbf{M},\omega)$ — set of generating operators for OS's 2.1
$\mathbf{C}$ — the complex numbers 2.1
$A_{\tilde{D}}$ — restriction of operator A to the set $\tilde{D}$ 2.1

$D(A^\infty)=\cap_{k\in\mathbf{N}}D(A^k)$ 2.1
$\mathscr{D}$ — core of an operator 2.1
$\tilde{E}_\alpha=\mathscr{R}(\exp(\alpha A))$ — range of an OS 2.1
$\tilde{\mathbf{E}}_\alpha=\cup_{\alpha>0}\tilde{e}_\alpha$ 2.1
$\mathbf{N}$ — the natural numbers 2.1
$\omega_b(\delta,F(\cdot))=\sup\{\|f(t)-f(s)\| : t-s<\delta,\ t,s\in[0,b]\}$ 2.1
$\mathscr{A}=(\exp(\alpha A)-I)/\alpha$ 2.1

$\Delta_\tau^m=\sum_{k=0}^{m}(-1)^{m-k}C_m{}^k\exp(k\tau A)$ 2.1

$C_m{}^k$ — binomial coefficient 2.1
$\psi(t,x)=\|(\exp(tA)-I)x\|$ 2.1

$\tilde{B}_t^m=(m!/t^m)(\exp(tA)-\sum_{k=0}^{m}t^kA^k/k!)$ 2.1

$\tilde{C}_t^m = t^{-m}\left\{\sum_{k=0}^{m}(-1)^{m-k}C_m^k \exp(ktA)\,x\right\}$ 2.1

$T_m(t) = [I \min (I \min \exp(ta))^m$ 2.1

C_1 — Cesaro descrete limit 2.1

$\mathcal{GR}(M,\omega)$ — set of generating operators for a group 2.2

$D(A,\eta) = \{x \in D(A^\infty) : \exists c_{k,\eta} = c > 0,\ \|A^k x\| < c\eta^k,\ k \in \mathbf{N}\}$ 2.2

$EXP(A) - \cup_{\eta>0} D(A,\eta)$ 2.2

$\mathfrak{A}(A)$ — set of analytic vectors of A, i.e., $\{x \in D(A^\infty) : \forall a > 0 \exists c > 0 : \|A^k x\| \leqslant c a^k k^k,\ k \in \mathbf{N}\}$ 2.2

$\mathfrak{A}_c(A)$ — set of integral vectors of A, i.e., $\{x \in D(A^\infty) : \forall a > 0 \exists c > 0 : \|A^k x\| \leqslant c a^k k^k,\ k \in \mathbf{N}\}$ 2.2

$\mathcal{H}(\theta,\omega)$ set of generators of analytic OS's 2.4

$\Sigma(\theta)$ — sector in the complex plane 2.4

$\mathcal{N}(A)$ — the kernel of A 2.4

$\partial(\Omega)$ — boundary of Ω 2.4

$\mathrm{nul}\,(A) = \dim \mathcal{N}(A)$ — dimension of $\mathcal{N}(A)$ 2.5

$\mathrm{def}\,(A) = \mathrm{codim}\,\mathcal{R}(A)$ — codimension of range 2.5

ind(A) = (nul(A) — def(A)) — index of operator A 2.5

$\mathrm{dist}(x,\hat{E}E) = \inf\{\|x - y\| : y \in \hat{E}\}$ 2.5

$\gamma(A)$ — reduced minimal modulus of operator A 2.5

SF(A) — half plane of semi-Fredholmness of resolvent 2.5

$\langle x,y^*\rangle$ value of functional y^* at x 2.5

E^* — adjoint of E 2.5

W(A) — numerical range of operator A 2.5

$P\sigma(A)$ — characteristic values of A 2.6

$R\sigma(A)$ — residual spectrum of A 2.6

$C\sigma(A)$ — continuous spectrum of A 2.6

$\Omega F(A)$ — region of Fredholmness of A 2.6

$E\sigma(A)$ — essential spectrum of A 2.6

Er(B) — essential spectral radius of B 2.6

$\alpha(\Omega)$ — Kuratowski measure of Ω 2.6

$\|\|B\|\|_\alpha$ — measure of noncompactness of $B \in B(E)$ 2.6

$E\omega(A)$ — essential growth factor of OS exp(tA) 2.6

$B\sigma(A)$ — boundary spectrum of $A \in \mathcal{G}(M,\omega)$ 2.6

$\omega(x,A)$ — bound of exponential growth of exp(tA)x 2.7

$\omega_1(A)$ — bound for exponential growth of solutions to Cauchy problem 2.7

$\gamma(x,A)$ — left bound of ω such that the function $R(\lambda,A)x$ is holomorphic when $\mathrm{Re}\,\lambda > \omega$ 2.7

$\gamma_0(x,A)$ — left bound of ω such that the function $R(\lambda,A)x$ is bounded when $\mathrm{Re}\,\lambda > \omega$ 2.7

$\gamma_1(x,A)$ — left bound of ω such that $R(\lambda,A)x$ converges to zero as $\mathrm{im}\,\lambda \to \infty$, $\mathrm{Re}\,\lambda > \omega$ 2.7

H^∞ — set of functions f(z) that is bounded and analytic when Re z > 0 ..... 2.7

$b(x) = \inf\left\{\mathrm{Re}\,\lambda : \int_0^\infty e^{-\lambda t}\exp(tA)\,x\,dx \quad \text{exists}\right\}$ 2.7

$E_{st} = \{x \in E : \lim_{t\to\infty} \|\exp(tA)x\| = 0\}$ 2.7

$E_{ap} = \mathrm{span}\{x \in D(A) : Ax = \lambda x,\ \lambda \in iR\}$ 2.7

E_+ — cone in Banach space E 2.8

$\leqslant$ — sign of partial ordering 2.8

sup(S) — supremum of S in the sense of $\leqslant$ 2.8

$x^+(x^-)$ — positive (negative) part of x 2.8

$|x|$ — modulus of member of a Banach structure $(x^+ + x^-)$ 2.8

E_+^* — set of positive functionals 2.8

$D(A)_+ = D(A) \cap E_+$ 2.8
J_x — principal ideal corresponding to x 2.8
$S(A) = \sup\{\mathrm{Re}\ \lambda : \lambda \in \sigma(A)\}$ 2.8
$p_+(x)$ — canonical functional 2.8
sign_x — linear operator of Kato inequality 2.8
$\mathrm{int}\ E_+$ — interior of cone E_+ 2.8
$\mathbf{Z}$ — the integers 2.8
$\mathrm{sign}(\alpha)$ — the sign of α 2.8
A_H — generator of the modulus of an OS 2.8
$\mathscr{C}(\mathbf{M}, \omega)$ — set of generating operators of a COF 3.1
$\omega^c(A)$ — type of COF C(t,A) 3.1
St(A) — Stieltjes vectors 3.1
$\mathfrak{A}_p(A)$ — semianalytic vectors 3.1

$$\mathfrak{A}_{pp}(A)=\left\{x\in E:\sum_{k=0}^{\infty}\|A^kx\|t^k/(2k)!<\infty \quad \text{for all } t>0\right\}. \quad \ldots\ldots\ 3.1$$

$\mathbf{NBI}_0(A) = \{t > 0$: the operator C(t,A) has no bounded inverse$\}$ 3.1
$\mathbf{NBI}_1(A) = \{t > 0$: the operator S(t,A) has no bounded inverse$\}$ 3.1
$J(f,\varepsilon)$ — set of all ε-periods of the function f(t) 5.0
$a(\lambda,f)$ — Bohr transform of f(t) 5.0
$\mathscr{R}_{f(t)}$ — complete range of a function, i.e., $\{x \in E : x = f(t), t \in \mathbf{R}\}$ 5.0
$P(\lambda)$ — Riesz projector onto root subspace corresponding to point of spectrum $\lambda \in \sigma(A)$ 6.0
$\mathrm{Com}(A) = \{t > 0 : \exp(tA) - I$ is a compact operator$\}$ 6.1
$\theta(x)$ — adjoint set 7.0
$\varphi(x)$ — dual mapping 7.0
$[x,y]$ — semiscalar product 7.0
(C,α)-lim — Cesaro limit 7.1
A-lim — Abel limit 7.1
$\mathscr{E}(t)$ — either an OS or a COF 8.1
$\mathfrak{M}(\mathscr{E})$ — the class of Miyadera perturbations 8.1
$B(E,F)$ — set of linear continuous operators mapping E into F 8.1
A^* — adjoint of operator A 9.1
x^{**} — characteristic mapping of $E \to E^*$ 9.1
$\exp(tA)^{\odot}$ — Phillips adjoint of a semigroup 9.1
$E^{\odot}$ — subspace in E^* on which the adjoint semigroup $\exp(tA)^*$ is strongly continuous 9.1

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